

**QUESTION BANK
QUESTION MODEL**

Q.P. CODE: 122010

RegNo:.....

**First Semester B.Sc Nursing Degree Regular Examinations
August 2022**

Paper II – Applied Anatomy and Applied Physiology

Time: 3 Hrs

Max. Marks: 37

- Answer all questions to the point neatly and legibly • Do not leave any blank pages between answers • Indicate the question number correctly for the answer in the margin space • Answer all parts of a single question together • Leave sufficient space between answers • Draw diagrams wherever necessary

Section A – Applied Anatomy

Multiple Choice Questions

(6x1=6)

(choose the most appropriate answer from the options given)

1. Renal artery is a branch of"
a) Coeliac trunk b) Superior mesenteric c) Inferior mesenteric d) Abdominal aorta
2. Elbow joint is an example of _____ type of joint.
a) Ball & Socket b) Hinge c) Ellipsoid d) Pivot.
3. An example for fibrocartilage is
a) Costal cartilage b) Pinna of ear c) Intervertebral disc d) Epiglottis
4. Coronary sinus opens into
a) Density. b) Left atrium c) Right ventricle d) Left ventricle
5. The distal attachment of Biceps brachii muscle is to
a) Ulnar tuberosity b) Radial tuberosity c) Olecranon process d) Styloid process of radius
6. Motor area of Brain is located in _____ lobe.
a) Frontal b) Parietal c) Temporal d) Central

Essay

(1x10=10)

7. Mention the parts of Respiratory system. Describe Lungs in detail. (2+8)

Short notes

(3x5=15)

8. Thyroid gland
9. Ventricles of brain
10. Kidney

Differentiate between

(3x2=6)

11. Artery and Vein
12. Small intestine and Large intestine
13. Testis and Ovary.

UNIT 1

INTRODUCTION TO ANATOMICAL TERMS

MCQ

1. Holoenzyme is made of
a. Apoenzyme and Zymogen b. Apoenzyme and Co-enzyme c. Co-enzyme and Prosthetic group d. Prosthetic group and Co-factor
2. Which of the following organelle is called 'Suicidal Bag'
a. Mitochondria b. Endoplasmic reticulum c. Lysosome d. Ribosome
3. Most abundant blood cells in the human body are
a. WBCs b. RBCs c. Platelets d. Plasma Cells
4. Number of iron atoms in one haemoglobin molecule are

a. 1 b. 3 c. 4 d. 8

5. Which of the following is not a co-enzyme

a. NAD b. NADP c. FAD d. Mn⁺⁺

6. Activity of allosteric enzymes are influenced by

a. Allosteric modulators b. Allosteric site c. Catalytic site d. None of the above

7. In competitive inhibition, inhibitors bears a close structural similarity with the

a. Co-enzyme b. Co-factor c. Prosthetic group d. Substrate

8. Enzyme acts best at a particular temperature called

a. Catalytic Temperature b. At normal Body temperature c. Optimum temperature d. None of the above

9. Lock and Key model is also known as

a. Template model b. Induced fit model c. Khosland's Model d. Enzyme-substrate interaction model

10. Which bond is not associated with Enzyme-substrate interaction –

a. Hydrogen bonds b. Ionic bonds c. Di-sulfide bonds d. Van der Waal's force of attraction

11. Which of the following statement is incorrect

a. Enzymes are protein in nature b. Enzymes are colloidal in nature c. Enzymes are thermolabile d. Enzymes are inorganic catalyst

12. Apoenzymes dissociates from co-enzymes due to

a. Change in pH b. Change in temperature c. Change in substrate concentration d. Change in inhibitor concentration

13. Which of the following enzyme inhibitions shows decreased Km Value ?

a. Competitive inhibition b. Un-competitive inhibition c. Non-competitive inhibition d. Feed back inhibition

14. Amount of total blood volume in an individual is approximately

a. 50 ml/Kg body weight b. 60 ml/Kg body weight c. 90 ml/Kg body weight d. 80 ml/Kg body weight

15. Normal blood pH is

a. 7.3 b. 7.2 c. 7.4 d. 8.4

16. Haematocrit value is the ratio of

a. WBC to plasma b. Platelets to plasma c. RBCs to plasma d. Total blood cells to plasma

17. Plasma represents _____ percent of total blood volume
a. 35 b. 45 c. 55 d. 5

18. Normal amount of plasma protein ranges from

a. 2.2-4.3 gm% b. 4.4-6.3 gm% c. 6.4-8.3 gm% d. 8.4-10.2 gm%

19. Which component of protein contribute to maximum percentage to total plasma protein

a. Albumin b. Globulin c. Fibrinogen d. Prothrombin

20. Serum does not contain

a. Calcium b. Prothrombin c. Factor VIII d. Factor-X

21. Combination of haem with O₂ is called

a. Oxyhaemoglobin b. Oxidation c. Oxygenation d. Oxidized haem

22. Adult haemoglobin contains _____ polypeptide chains

a. 2 α , 2 γ b. 2 α , 2 β c. 2 α , 2 δ d. 2 β , 2 γ

23. Each haemoglobin molecules carries _____ number of O₂ molecules

a. 2 b. 4 c. 1 d. 8

24. Each gram% of haemoglobin, when fully saturated, can carry _____ ml of O₂
a. 1.34 ml b. 3.14 ml c. 4.13 ml d. 5ml
25. In Sick cell anaemia, the defect lies in which polypeptide
a. Alpha chain b. Beta chain c. Gamma chain d. Delta chain
26. Average mean corpuscular diameter is _____ µm
a. 5.1 b. 6.3 c. 7.3 d. 8.5
27. Increase in RBC count beyond 10 million per cu mm is known as
a. Anisocytosis b. Poikilocytosis c. Polycythemia d. Leucocytosis
28. During erythropoiesis haemoglobin first appears in
a. Early normoblast b. Intermediate normoblast c. Late normoblast d. Pronormoblast
29. During hypoxia Kidney releases
a. Renin b. Renal Erythropoietic factor c. Erythropoietin d. None of the above
30. Intrinsic factor is secreted by a. Liver b. Chief cells of stomach c. Parietal cells of stomach d. Beta cells of pancreas
31. Which of the following extrinsic factor is required for maturation of RBCs
a. Vit B12 b. Folic acid c. Iron d. Both (a) and (b)
32. Largest WBCs in peripheral blood is
a. Neutrophil b. Large lymphocyte c. Monocyte d. Eosinophil
33. The process by which WBCs squeeze through pores in capillary wall is
a. Chemotaxis b. Pinocytosis c. Opsonization d. Diapedesis
34. Smallest blood cell is
a. Small lymphocyte b. Platelet c. RBC d. Neutrophil
35. Commonest anaemia in India is
a. Pernicious anaemia b. Sick cell anaemia c. Iron deficiency anaemia d. None of the above
36. The term ER was coined by
a. Camillo Golgi b. Porter c. Robert Brown d. Benda
37. Which of the following organelle has a continuous connection with nuclear membrane
a. Golgi apparatus b. Lysosome c. RER d. SER
38. In RER, ribosomes are located on
a. the cytoplasmic side b. on the luminal side c. both (a) and (b) d. all throughout
39. Which of the following statements were true regarding ER
a. ER provides structural framework to the cell b. ER acts as intra cellular transporting system c. SER is involved in the synthesis of lipid d. All of the above
40. Which of the following statements are correct regarding Golgi apparatus
a. sorting and packaging b. exocytosis of melanin granules c. exocytosis of thyroxine hormone d. all of the above
41. The term Golgi apparatus was coined by
a. Camillo Golgi b. Robert Brown c. Robert Hook d. Benda
42. F₀-F₁ Particles are located on
a. Thylakoids b. inner mitochondrial membrane c. Golgian vacuoles d. None of the above
43. In mitochondria cristae act as sites for
a. protein synthesis b. phosphorylation of flavoproteins c. breakdown of macromolecules d. Oxidation–reduction reaction
44. Mitochondrial inner membrane is rich in which phospholipid
a. Phosphatidyl inositol b. Phosphatidyl serine c. Cardiolipin d. Phosphatidyl choline
45. Which of the following is NOT a function of mitochondrion
a. electron transport and associated ATP production b. Fatty acid breakdown c. non-shivering thermogenesis d. glycolysis and associated ATP production
46. Who coined the term mitochondria
a. Kolliker b. Benda c. Fleenming d. Robert Brown
47. Nucleus was first discovered by
a. Robert Hook b. Strasburger c. Robert Brown d. None of the above
48. Nuclear membrane is in continuous connection with
a. SER b. RER c. Golgi apparatus d. Lysosomes
49. The number of nuclear pores depends on
a. Size of cells b. Transcriptional activity of the cell c. DNA content of the cell d. all of the above
50. The DNA Protein ratio in chromatin is
a. 3:1 b. 2:1 c. 1:1 d. 4:1
51. The function of nucleolus is
a. RNA synthesis b. DNA synthesis c. Histone synthesis d. Ribosomal subunit synthesis
52. The basic protein of the nucleus are
a. nucleohistones b. nuceoprotamines c. both (a) and (b) d. none of these
53. Lysosomes are present in all except
a. muscle cells b. acinar cells c. erythrocytes d. hepatocytes
54. Which of the following is the function of lysosomes
a. autophagy b. autolysis c. digestion d. all of the above
55. Lysosomes are involved in
a. Extracellular digestion b. Intracellular digestion c. both (a) and (b) d. none of the above
56. Who identified lysosome
a. Novikoff b. Claude c. Palade d. none of the above
57. All the following has ribosomes except
a. nucleus b. mitochondrion c. chloroplast d. cytoplasm
58. In 70S ribosome 'S' stands for
a. S.I unit b. Solubility factor c. Svedberg unit d. None of the above
59. 80S ribosomes are found in
a. Eukaryotes b. Prokaryotes c. Both eukaryotes and Prokaryotes d. Eukaryotic plant cells
60. The subunits of 80S ribosomes include
a. 40S and 50S b. 30S and 50S c. 40S and 60S d. 20S and 60S
61. The subunits of 70S ribosomes include
a. 40S and 50S b. 30S and 40S c. 30S and 50S d. 20S and 50S
62. 70S ribosomes occur in
a. Viruses b. prokaryotes c. eukaryotic plant cells d. eukaryotic animal cells
63. Ribosomes are made up of
a. RNA only b. RNA and Proteins c. RNA, DNA and Proteins d. nucleic acids, proteins and lipids
64. The rough ER is specially well developed in cells actively engaged in
a. Protein synthesis b. Nucleotide synthesis c. Lipid synthesis d. Secretory functions
65. The nucleus contains

a. Mitochondria b. Golgi apparatus c. Chromosomes d. Lysosomes

66. Plasma membrane is

a. Permeable b. Selectively permeable c. Impermeable d. Semi-permeable

67. Most accepted structural model of plasma membrane is

a. Sandwich model b. Unit membrane model c. Lamellar model d. Fluid-mosaic model

68. Plasma membrane is composed of

a. Glycoproteins b. Lipoproteins c. Chromoproteins d. Lipids

69. Ribosomes contain maximum amount of

a. Steroids b. Lipids c. RNA d. DNA

70. Which structure is present in animal cell but is absent from plant cell ?

a. Centrioles b. Golgi apparatus c. Mitochondria d.

Endoplasmic reticulum

71. A unit membrane is about :

a. 50-60 Å thick b. 60-75 Å thick c. 75-100 Å thick d. 100-120 Å thick

72. The enzymes which break up starch into sugar are called a.

Hydrolases b. Amylases c. Lipases d. Nucleases

73. Apoenzyme is a

a. Protein b. Carbohydrate c. Vitamin d. Amino acid

74. Coenzyme is :

a. Always a protein b. Often a metal c. Always an inorganic compound d. Often a vitamin

75. Enzymes are named after their substrates by adding suffix :

a. -in b. -ase c. -ose d. -sin

76. Enzyme exist in the cells as

a. Solid b. Crystals c. Colloid d. None of the above

77. An enzyme brings about :

a. Reduction in activation energy b. Increase in reaction time c. Increase in activation energy d. All the above

78. Which of the following statement is "NOT" correct

a. All enzymes are thermolabile b. All enzymes are biocatalysts c. All enzymes are proteins d. All proteins are enzymes

79. Who discovered blood groups

a. F. Galton b. Carl Linnaeus c. Edward Jenner d. C. Landsteiner

80. Enzymes bringing about hydrolysis of esters and peptides are :

a. Transferases b. Lyases c. Hydrolases d. All of the above

81. Aerobic respiration is performed by :

a. Glyoxisomes b. Mitochondria c. Lysosomes d. Chloroplast

82. Bile reduces the surface tension and causes

a. Emulsification of fat b. Digestion of fat c. Absorption of fat d. All of the above

83. Dialysis causes

a. Separation of colloids from crystalloids b. Purification of colloids c. Precipitation of colloids d. None of the above

84. Ultrafiltration helps in

a. Formation of Glomerular filtrate b. Formation of urine c. Accumulation of proteins d. Separation of vitamins

85. pH of RBC is lower due to

a. Na^+ -K⁺ Pump action b. Gibb's Donnan Effect c. Efflux of OH⁻ from RBC d. None of the above

86. Gibb's Donnan effect creates a. Diffusion b. Surface tension c. Osmotic pressure d. None of the above

87. Transmembrane potential results due to

a. Donnan Effect b. Influx and efflux of oppositely charged ions c. Retention of anions inside the membrane d. All the above

88. Rate of diffusion of a substance depends on

a. Presence of semi-permeable membrane b. Concentration gradient of solute c. Concentration of solvent d. Concentration of ions

89. Homeostasis means

a. Control of internal environment of the body b. Adaptation with the environment c. Constant environment of the body d. All of the above

90. Diffusion is more rapid in

a. Solid b. Liquid c. Gas d. Mixture of liquid and gas

91. In Osmosis, movement of _____ occurs through the semi-permeable membrane

a. Solvent b. Solute c. Both (a) and (b) d. All the above

92. Viscosity of blood increases with rise in

a. Albumin b. Globulin c. Fibrinogen d. Prothrombin

93. Osmotic pressure across the capillary wall is exerted by

a. Size of the molecule b. Shape of the molecule c. Concentration of the molecule d. All the above

94. Effect of change of temperature on viscosity involves

a. Increase in viscosity b. Decrease in viscosity c. No change d. Both (a) and (b)

95. pH means

a. $-\log [\text{H}^+]$ b. $-\log_{10} [\text{H}^+]$ c. $-\log [\text{H}]$ d. $\log [\text{H}^+]$

96. Microcytic anaemia develops in

a. Vit B12 deficiency b. Folic acid deficiency c. Both (a) and (b) d. None of the above

97. Extrinsic system of blood clotting is initiated by

a. Factor-III b. Factor-VIII c. Factor-II d. Factor-I

98. One of the following is NOT an anticoagulant 1

a. Heparin b. Protein-C c. Antithrombin-III d. Thrombin

99. Following are the membrane bound cell organelles except

a. Endoplasmic reticulum b. Lysosome c. Ribosomes d. Peroxisome

100. The intrinsic protein present in the cell membrane mainly functions as

a. Enzymes b. Carrier c. Pores d. Channels

ANSWERS 1.(b) 2.(c) 3.(b) 4.(c) 5.(d) 6.(a) 7.(d) 8.(c) 9.(a)

10.(c) 11.(d) 12.(a) 13.(a) 14.(d) 15.(c) 16.(c) 17.(c) 18.(a)

19.(a) 20.(a) 21.(a) 22.(b) 23.(d) 24.(b) 25.(b) 26.(c) 27.(c)

28.(b) 29.(b) 30.(c) 31. (d) 32.(c) 33.(d) 34.(b) 35.(c) 36.(b)

37.(c) 38.(a) 39.(d) 40.(d) 41.(a) 42.(b) 43.(d) 44.(c) 45.(d) 46.

(b) 47.(c) 48.(b) 49.(b) 50.(c) 51.(d) 52.(c) 53.(c) 54.(d) 55.(c)

56.(a) 57.(a) 58.(c) 59.(a) 60.(c) 61.(c) 62.(b) 63.(b) 64.(a)

65.(c) 66.(b) 67.(d) 68.(b) 69.(c) 70.(a) 71.(c) 72.(b) 73.(a)

74.(d) 75.(b) 76.(c) 77.(a) 78.(d) 79.(d) 80.(c) 81.(b) 82.(d)

83.(a) 84.(a) 85.(b) 86.(c) 87.(a) 88.(b) 89.(c) 90.(c) 91.(a)

92.(a) 93.(c) 94.(d) 95.(b) 96.(c) 97.(a) 98.(d) 99.(c) 100.(a)

LONG ESSAY

1. Classify Epithelium with examples for each. Describe in detail the stratified squamous epithelium with diagram

- Describe in detail about the human cell structure and enumerate the nucleus with diagrammatic representation.
- Define tissue. Enlist the types of tissues. Explain about the epithelial tissue
- Enumerate the muscular tissue, nervous tissue and excitable cells and describe the structure and classification of membranes and glands.

SHORT ESSAY

- Classify connective tissue.
- Explain the organization of human body and organization of body cells.
- Explain the anatomical structure of a human cell.
- Explain in detail classification of connective tissues.
- Explain tumor. Explain the types of effects of tumor.
- Briefly narrate the body cavities.
- Describe in detail the stratified squamous epithelium with diagram.
- Draw the labeled diagram of typical cell
- Describe transitional epithelium with a labeled diagram. Give an examples
- Describe pseudo stratified columnar epithelium with a diagram

SHORT NOTES

- Structure of cell.
- Homeostasis
- Diffusion
- Meiosis
- Biopsy
- Simple squamous epithelium
- Histology of cartilage
- Parts of Mitochondria

ANSWER BRIEFLY

- Classification of tissues
- Enlist the functions of lysosomes
- Characteristics of tissues
- Compound epithelium
- Stratified squamous epithelium
- Types of connective tissue
- Meiosis and Mitosis
- Skeletal muscle and cardiac muscle
- Cardiac muscle and smooth muscle
- Membranes and glands
- Bone and cartilage
- Epithelial tissue and connective tissue
- Muscle tissue and Nervous tissue
- Squamous epithelium and columnar epithelium

ANSWER KEY

UNIT 1

INTRODUCTION TO ANATOMICAL TERMS

LONG ESSAY

- Classify Epithelium with examples for each. Describe in detail the stratified squamous epithelium with diagram

INTRODUCTION

DEFINITION

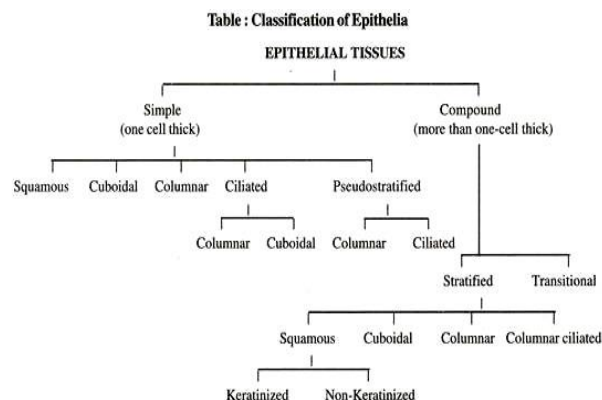
EPITHELIUM- epithelium covers body surfaces, lines body cavities and constitute glands.

Epithelium is an avascular tissue composed of cells that cover the exterior body surfaces and line internal closed cavities (including the vascular system) and body tubes that communicate with the exterior (the alimentary, respiratory, and genitourinary tracts).

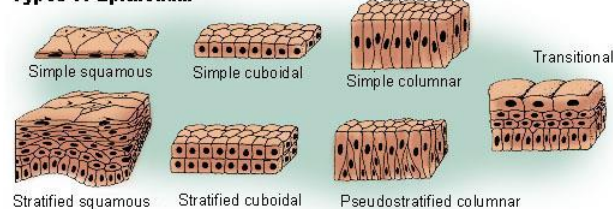
Epithelium also forms the secretory portion (parenchyma) of glands and their ducts.

In addition, specialized epithelial cells function as receptors for the special senses.

CLASSIFICATION



Types of Epithelium

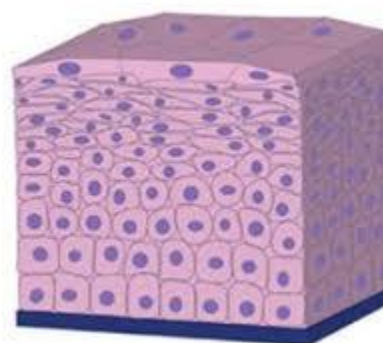


STRATIFIED SQUAMOUS EPITHELIUM

Basal layer of cuboidal or columnar cells which become flatter and more typically squamous as they move progressively toward the surface

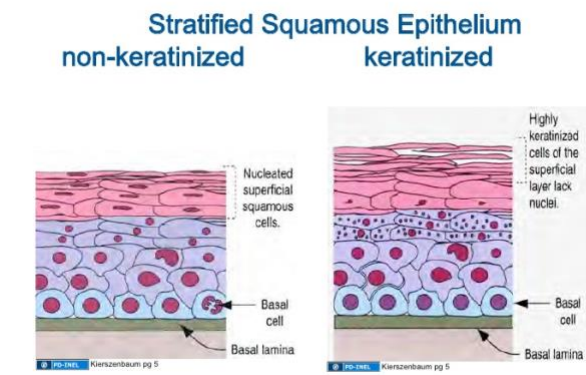
Found in locations subject to friction or attrition: – Skin – Mouth – Esophagus – Anal Canal – Vagina

Stratified Squamous Epithelium -To differentiate stratified squamous epithelium from transitional epithelium (it's "look-alike"), examine the cells at the surface. Stratified squamous epithelium will exhibit flattened squamous cells whereas the surface cells of transitional epithelium are more rounded and dome-shaped.

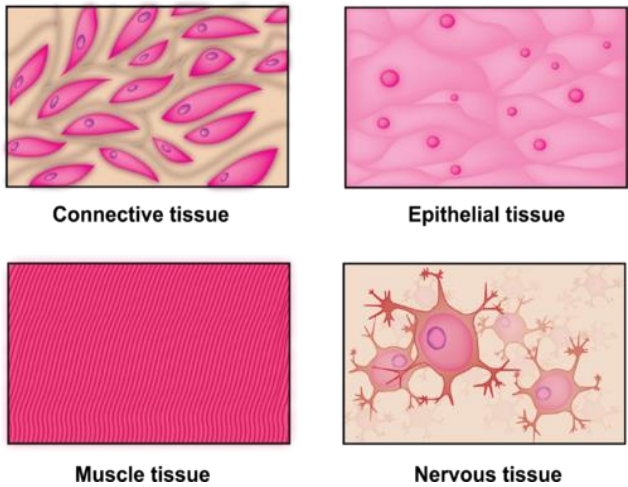


Stratified Squamous

CLASSIFICATION :- Keratinised and non-keratinised epithelium



Four Types of Tissues

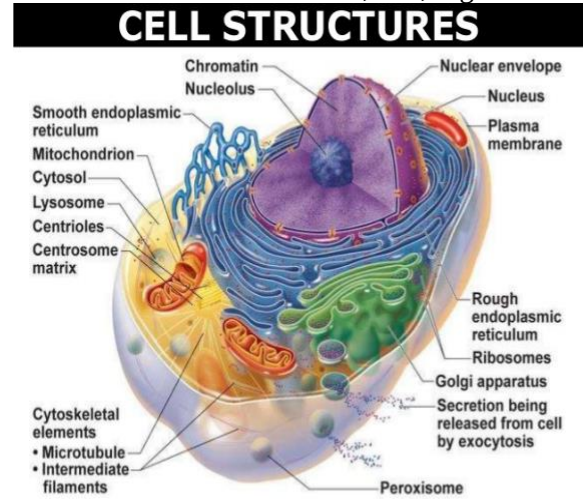


2. Describe in detail about the human cell structure and enumerate the nucleus with diagrammatic representation.

INTRODUCTION

CELL.STRUCTURE AND FUNCTIONS

- ❖ Plasma membrane
- ❖ Organelles
- Nucleus
- Mitochondria
- Ribosomes
- Endoplasmic reticulum
- Golgi apparatus
- Lysosomes
- Cytoskeleton
- Microfilaments
- Microtubules
- Centrosomes
- Cell extensions like microvilli,cilia,flagella etc.



3. Define tissue. Enlist the types of tissues. Explain about the epithelial tissue

INTRODUCTION

DEFINITION

Tissues are group of cells

TYPES

[Also refer ans-1]

4. Enumerate the muscular tissue,nervous tissue and exictable cells and describe the structure and classification of membranes and glands.
- MUSCULAR TISSUES –consists of fibres that are specialized for contraction
- It provides motion, maintainence of posture ,heat production and protection.
 - Myocytes can use ATP to generate force.
 - 3 main types :-Skeltal-attached to bones and tendons and facilitates motion.
- Cardiac :- found in the heart wall and enables blood pumbing to all parts of body.
- Smooth :-found in iris ,blood vessels,lungs ,stomach etc.
- NERVOUS TISSUE- consists of neurons and neuroglia.
- Its function is toexhibit sensitivity to various types of stimuli,converts them to other neurons.
 - Nueroglia do not conduct or generate nerve impulses.

EXCITABLE CELLS-

Neurons and muscle fibres exhibit electrical excitability,they are considered as excitable cells.

MEMBRANES-

Membranes are sheets of epithelial tissues and their supporting connective tissue that cover or line internal structures or cavities.

Types

- ❖ Mucous
- ❖ Serous
- ❖ Cutaneous
- ❖ Synovial membrane (explain each)

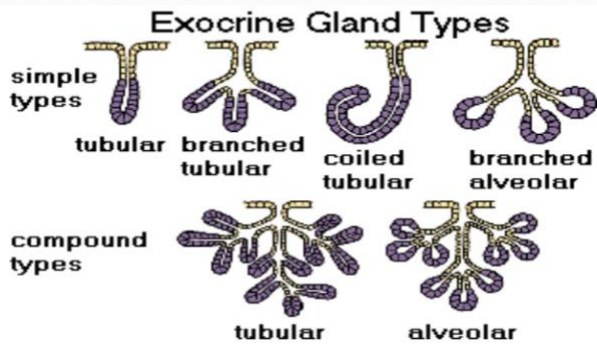
GLANDS-

Glands are group of epithelial cells which produce specialized secretions

Types

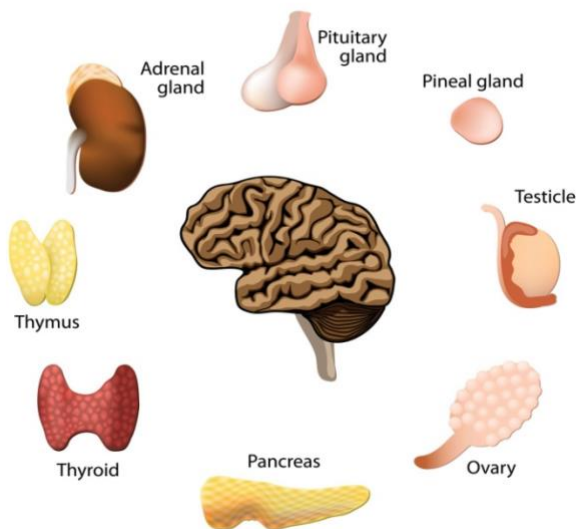
- ❖ Exocrine glands- will release secretion directly into ducts (sweat ,sebaceous,salivary and mammary)

Exocrine Gland Type :-



- ❖ Endocrine glands-will release secretions directly into blood and body secretions. Secretions are called hormones. (thyroid, pineal)

ENDOCRINE SYSTEM



SHORT ESSAY

- Classify connective tissue.

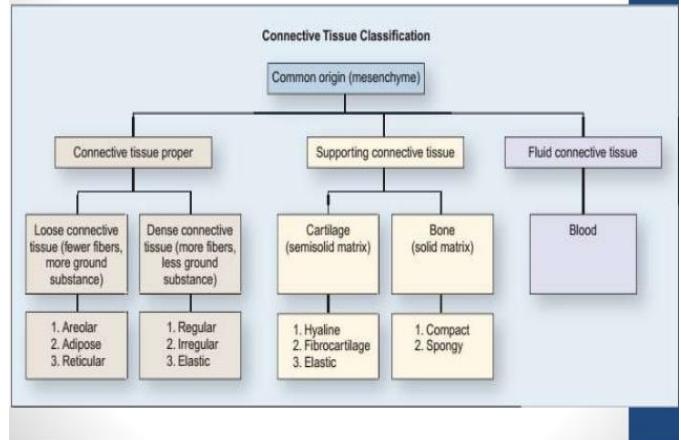
CONNECTIVE TISSUE- a tissue of mesodermal origin that consists of various cells (such as fibroblasts and macrophages) and interlacing protein fibers (as of collagen) embedded in a chiefly carbohydrate ground substance, that supports, ensheathes, and binds together other tissues, and that includes loose and dense forms (such as adipose tissue, tendons, ligaments, and aponeuroses) and specialized forms (such as cartilage and bone)

TYPES OF CONNECTIVE TISSUE CELLS

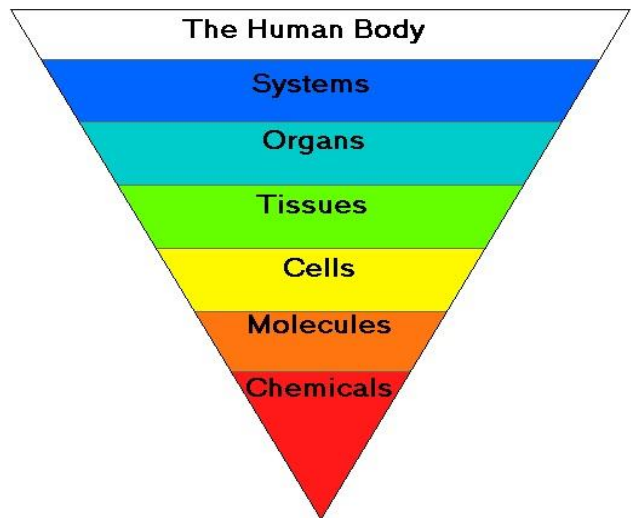
- Fibroblasts
- Macrophages
- Plasma cells
- Mast cells
- Adipocytes
- White blood cells\

TYPES OF CONNECTIVE TISSUE

Classification of connective tissue



- Explain the organization of human body and organization of body cells.



- Explain the anatomical structure of a human cell.
- Explain in detail classification of connective tissues.
- Explain tumor. Explain the types of effects of tumor.
A tumor or neoplasm is a mass of tissue that grows faster than normal in an uncoordinated manner and continues to grow after the initial stimulus has ceased.
TYPES
 - BENIGN
 - MALIGNANT**CAUSES**
 - ✓ Genetic
 - ✓ Environmental
 - ✓ Carcinogens**CELL DIFFERENTIATION**
METASTASIS
EFFECTS OF TUMORS
 - ❖ Pressure effects
 - ❖ Hormonal effects
 - ❖ Cachexia
- Briefly narrate the body cavities.

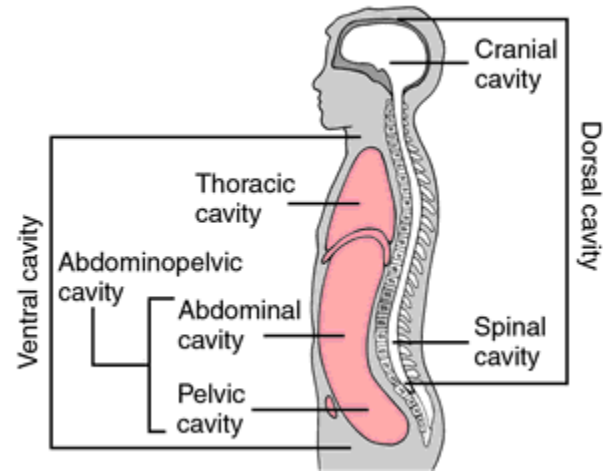
DEFINITION

Vital organs are suspended in internal body chambers called cavities.(coelem)

FUNCTIONS

- Protect delicate body organs
- Permits significant changes in size and shape of visceral organs.

TYPES



OTHER CAVITIES

- ❖ Oral and digestive cavities
- ❖ Nasal cavity
- ❖ Orbital cavity
- ❖ Middle ear cavity
- ❖ Synovial cavity

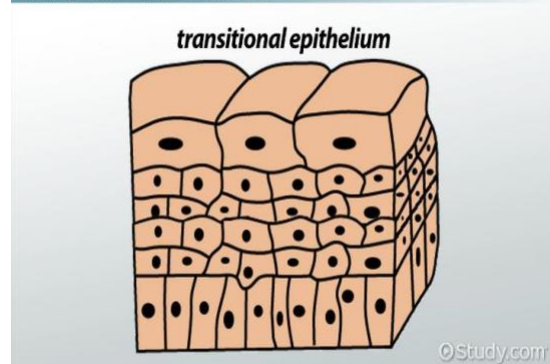
17. Describe in detail the stratified squamous epithelium with diagram.

18. Draw the labeled diagram of typical cell

19. Describe transitional epithelium with a labeled diagram. Give an examples

Transitional epithelium is a type of stratified epithelium – tissue consisting of multiple layers of epithelial cells which can contract and expand as needed. It is so named because of this function in the transition of degree of distension. It has a basement membrane. This tissue structure type is found in urothelium, including that of the renal pelvis, urinary bladder, the ureters, the superior urethra, and the prostatic and ejaculatory ducts of the prostate.

WHAT IS TRANSITIONAL EPITHELIUM?



20. Describe pseudo stratified columnar epithelium with a diagram

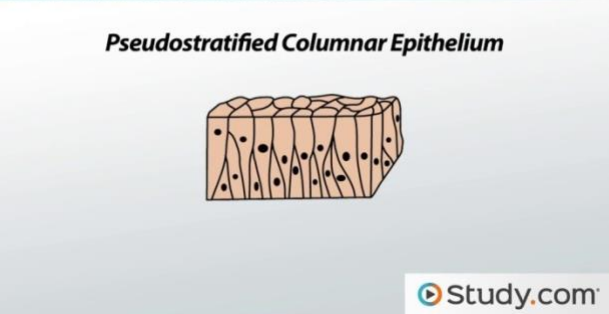
A **pseudostratified epithelium** is a type of epithelium that, though comprising only a single layer of cells, has its **cell nuclei** positioned in a manner suggestive of stratified epithelia.

As it rarely occurs as **squamous** or **cuboidal epithelia**, it is usually considered synonymous with the term **pseudostratified columnar epithelium**.

The term *pseudostratified* is derived from the appearance of this epithelium in section which conveys the erroneous (*pseudo* means almost or approaching) impression that there is more than one layer of cells, when in fact this is a true simple epithelium since all the cells rest on the **basal lamina**.

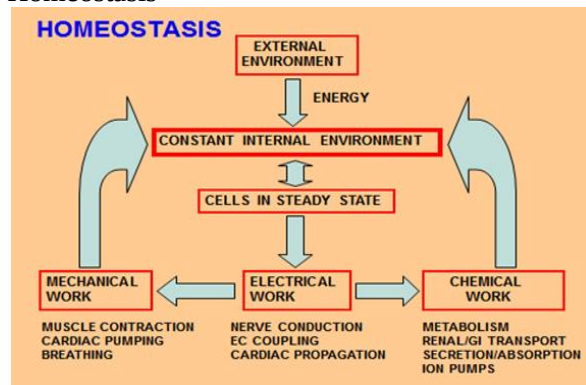
Pseudostratified epithelia function in secretion or absorption. If a specimen looks stratified but has **cilia**, then it is a pseudostratified ciliated epithelium, since stratified epithelia do not have cilia.

PSUEDOSTRATIFIED COLUMNAR EPITHELIUM



SHORT NOTES

9. Structure of cell.
10. Homeostasis



11. Diffusion

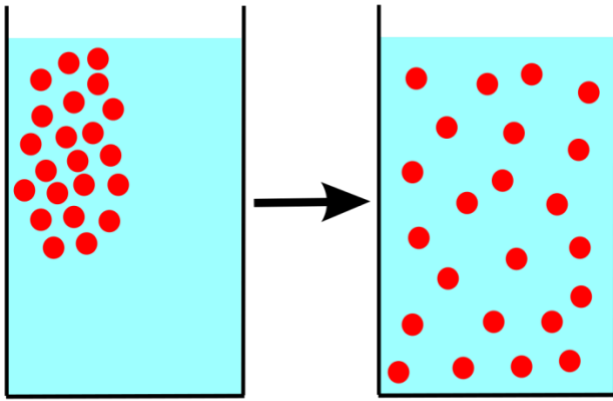
Diffusion is the net movement of molecules or atoms from a region of high concentration (or high chemical potential) to a region of low concentration (or low chemical potential) as a result of random motion of the molecules or atoms. Diffusion is driven by a gradient in chemical potential of the diffusing species.

A **gradient** is the change in the value of a quantity e.g. concentration, **pressure**, or **temperature** with the change in another variable, usually **distance**. A change in concentration over a distance is called a concentration gradient, a change in pressure over a distance is called a pressure gradient, and a change in temperature over a distance is called a temperature gradient.

The word **diffusion** derives from the Latin word, *diffundere*, which means "to spread way out".

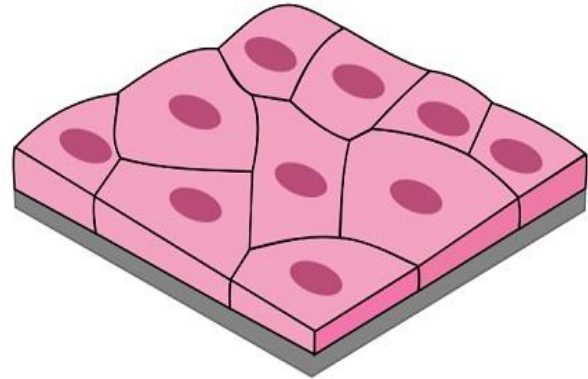
A distinguishing feature of diffusion is that it depends on particle random walk, and results in mixing or mass transport without requiring directed bulk motion. Bulk motion, or bulk

flow, is the characteristic of advection. The term convection is used to describe the combination of both transport phenomena.



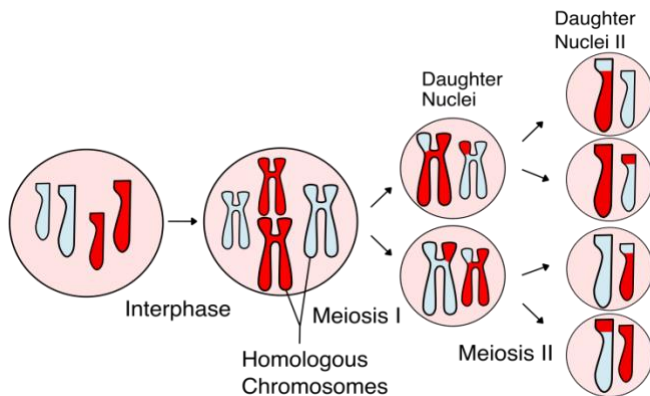
This type of epithelium is often permeable and occurs where small molecules need to pass quickly through membranes via filtration or diffusion.

- Simple squamous epithelia are found in [capillaries](#), [alveoli](#), [glomeruli](#), outer layer of skin and other tissues where rapid diffusion is required. Cells are flat with flattened and oblong nuclei.
- It is also called **pavement epithelium** due to its tile-like appearance.
- This tissue is extremely thin, and forms a delicate lining. It offers very little protection.



12. Meiosis

Meiosis is a specialized type of cell division that reduces the chromosome number by half, creating four haploid cells, each genetically distinct from the parent cell that gave rise to them



13. Biopsy

A biopsy is a medical test commonly performed by a surgeon, interventional radiologist, or an interventional cardiologist involving extraction of sample cells or tissues for examination to determine the presence or extent of a disease.

PURPOSES

TYPES

- Bone marrow biopsy
- Endoscopic biopsy
- Needle biopsies
- Core needle biopsies
- Fine needle biopsies
- Image-guided biopsies
- Vacuum-assisted biopsies
- Skin biopsy
- Surgical biopsy

INDICATIONS

PRE-PROCEDURAL CARE

INTRA-PROCEDURAL CARE

POST-PROCEDURAL CARE

COMPLICATIONS

14. Simple squamous epithelium

- A **simple squamous epithelium** is a single layer of flat cells in contact with the basal lamina (one of the two layers of the basement membrane) of the epithelium.

15. Histology of cartilage

Cartilage is a resilient and smooth elastic tissue, a rubber-like padding that covers and protects the ends of long bones at the joints, and is a structural component of the rib cage, the ear, the nose, the bronchial tubes, the intervertebral discs, and many other body components. It is not as hard and rigid as bone, but it is much stiffer and much less flexible than muscle. The matrix of cartilage is made up of chondrin.

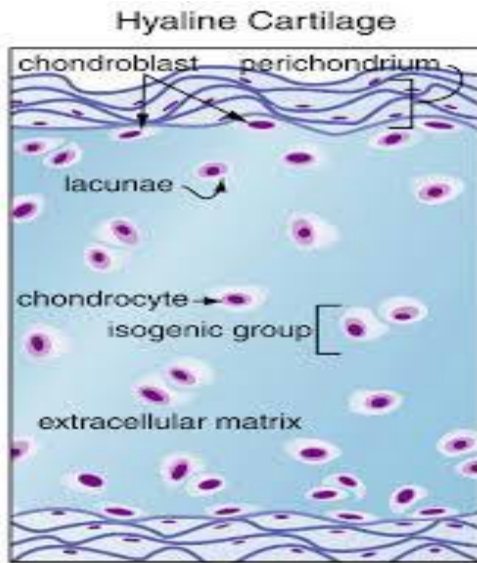
GENERAL CHARACTERISTICS

COMPONENTS

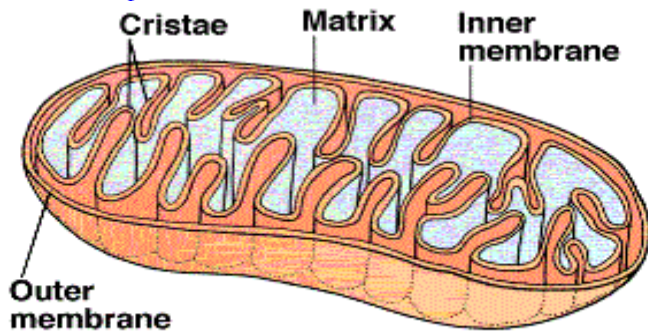
Cells- chondroblasts and chondrocytes
Fibres-collagen and elastic
Ground substance- mucopolysaccharides (chondroitin sulphate, Keratan sulphate and hyaluronic acid)

TYPES

- HYALINE CARTILAGE
- ELASTIC CARTILAGE
- FIBRO CARTILAGE



16. Parts of Mitochondria
The mitochondrion is a double-membrane-bound organelle found in most eukaryotic organisms.
- STRUCTURE**
1. the outer mitochondrial membrane,
 2. the intermembrane space (the space between the outer and inner membranes),
 3. the inner mitochondrial membrane,
 4. the cristae space (formed by infoldings of the inner membrane),
 5. the matrix (space within the inner membrane).
- Mitochondria stripped of their outer membrane are called [mitoplasts](#).



QUESTIONS UNIT II RESPIRATORY SYSTEM

MCQ

1. Which of the following is correct for the partial pressure of oxygen in alveoli?
(a) less than carbon dioxide
(b) less than the blood
(c) more than the blood
(d) equal to that of the blood
Answer: (c)
2. Which of the following is a chronic respiratory disorder caused by smoking?
(a) Asthma
(b) Emphysema
(c) Respiratory alkalosis

- (d) Respiratory acidosis
Answer: (b)
3. Most of the carbon dioxide produced in the tissues is transported to the lungs as
(a) carbonates
(b) bicarbonates
(c) dissolved in the blood
(d) attached to haemoglobin
Answer: (b)

4. The gaseous exchange in alveoli is a type of
(a) simple diffusion
(b) osmosis
(c) active transport
(d) passive transport
Answer: (a)

5. The linings that separate air contained in lungs from venous blood
(a) squamous epithelium and tunica media
(b) squamous epithelium and endothelium
(c) transitional epithelium and tunica externa
(d) none of the above
Answer: (b)

6. Which of the following organs only receives oxygenated blood?
(a) Lungs
(b) Gills
(c) Liver
(d) Spleen
Answer: (d)

7. Which of the following parts of the brain regulates the respiratory process?
(a) The vagus nerve
(b) Cerebral peduncle
(c) Medulla oblongata
(d) Cerebellum
Answer: (c)

8. Which of the following membranes encloses lungs?
(a) Pleural membrane
(b) Pericardium
(c) Perichondrium
(d) Periosteum
Answer: (a)

9. RBCs present in the blood of a person residing at higher altitudes
(a) decrease in size
(b) increase in size
(c) decrease in the number
(d) increase in the number
Answer: (d)

10. The alveolar epithelium of lungs is
(a) ciliated squamous
(b) nonciliated squamous
(c) nonciliated columnar
(d) ciliated columnar
Answer: (b)

11. In which part of the respiratory system, gaseous exchange takes place?
(a) Alveoli
(b) Pharynx
(c) Larynx

(d) Trachea

Sol. (a) Alveoli.

12. _____ is located between two pleural sacs and is the central compartment of the thoracic cavity?

(a) Hilum

(b) Pleura

(c) Mediastinum

(d) Thoracic cage

Sol. (c) Mediastinum.

13. Which of the following statements is true about involuntary breathing?

(a) It is controlled by the bronchioles

(b) It is controlled by the pulmonary arterioles

(c) It is controlled by the alveolar-capillary network

(d) It is controlled by the neurons, located in the medulla and pons

Sol. (d) It is controlled by the neurons, located in the medulla and pons.

14. Which of the following are parts of the human respiratory system?

(a) Trachea

(b) Diaphragm

(c) The lungs

(d) All of the above

Sol. (d) All of the above.

15. Which of the following gas is released out during the process of respiration?

(a) Oxygen

(b) Hydrogen

(c) Carbon dioxide

(d) None of the above

Sol. (c) Carbon dioxide.

16. The tiny air sacs present in human lungs is called_____.

(a) Alveoli

(b) Bronchus

(c) Bronchioles

(d) All of the above

Sol. (a) Alveoli.

17. Which of the following functions by filtering and keeping the mucus and dirt away from our lungs?

(a) Cilia

(b) Bronchioles

(c) Hairs in the lungs

(d) All of the above

Sol. (a) Cilia.

18. The total number of alveoli present in the human lungs is estimated to be around _____.

(a) 1 billion

(b) 800 million

(c) 500 million

(d) 1500 million

Sol: (c) 500 million.

19. The exchange of gases between the external environment and the lungs_____.

(a) Respiration

(b) External respiration

(c) Cellular respiration

(d) None of the above

Sol: (b) External respiration

20. Which one of the following statements is false about the trachea?

(a) Has C-shaped rings

(b) It is covered by epiglottis

(c) It splits into the right and left lungs

(d) None of the above

Sol: (b) It is covered by epiglottis

21. The maximum volume of air contained in the lung by a full forced inhalation is called _____.

(a) Tidal volume

(b) Vital capacity

(c) Ventilation rate

(d) Total lung capacity

Sol: (d) Total lung capacity.

22. Which one of the following is correct regarding larynx?

(a) It houses the vocal cords

(b) It prevents the invading pathogens into the trachea

(c) It is an organ made of cartilage and connects the pharynx to the trachea

(d) All of the above.

Sol: (d) All of the above.

23. Which of the following is the function of the trachea?

(a) Gaseous Exchange

(b) Filters the air we breathe

(c) Exhales the air from the body

(d) All of the above

Sol: (b) Filters the air we breathe.

24. Which of these statements is true about internal respiration?

(a) Production of ATP

(b) Exchange of gases between the bloodstream and tissue cells

(c) Exchange of gases between alveoli and the bloodstream

(d) Breathing between the atmosphere and the alveoli

Sol: (b) Exchange of gases between the bloodstream and tissue cells.

25. Which of the following organs functions as an air conditioner?

(a) Larynx

(b) Pharynx

(c) Nasal chambers

(d) All of the above

Sol: (c) Nasal chambers.

26. The normal breathing process is controlled by _____.

(a) Lungs

(b) Ventral respiratory group

(c) Dorsal respiratory group

(d) Both (b) and (c)

Sol: (d) Both (b) and (c).

27. In Aves, the exchange of gases occurs within the _____.

(a) Lungs

(b) Air sacs

(c) Air sacs and Lungs

(d) None of the above

Sol: (a) Lungs.

28. Which of the following statements is true about the entry of air into the lungs?

(a) Air enters the body and travels to the lungs through the mouth and the nose

- (b) Air enters the body and travels to the lungs through the esophagus and gullet
 (c) Air enters the body and travels to the lungs through the windpipe and the pores
 (d) Air enters the body and travels to the lungs through the nose and the nervous system.

Sol: (a) Air enters the body and travels to the lungs through the mouth and the nose.

29. The windpipe is also called the _____.

- (a) Larynx
 (b) Lungs
 (c) Trachea
 (d) Esophagus

Sol: (c) Trachea.

LONG ESSAYS

1. Mention the different parts of respiratory system.
2. Describe the right lung under the following headings: _ Surfaces and borders _ Relations _ Broncho-pulmonary segments _ Applied anatomy
3. Describe the structure and blood supply of larynx and explain structure of the bronchi and smaller air passages.

SHORT ESSAYS

4. Name the parts of respiratory tract and describe lungs in detail with applied aspects
5. Name the parts of the respiratory system. Describe the right lung in detail. Mention the differences from left lung
6. Describe the structure of lungs in detail with a diagram.
7. Describe the location of Pharynx and its parts
8. Describe the location of the trachea and outline its structure.

SHORT ANSWER

9. Respiratory muscles

ANSWER BRIEFLY

10. Name the air sinuses opening into middle meatus
11. List the various respiratory diseases
12. List the various respiratory organs
13. Name the para nasal air sinuses
14. Name the openings in the lateral wall of nose
15. Name the bones forming nasal septum
16. Name 4 cartilages of larynx
17. Name 4 cartilages of larynx
18. Name any four muscles of pharynx
19. Name two pleural recesses
20. Mention 4 structures passing through the diaphragm
21. Mention the action of intercostals muscle
22. Medial wall of nasal cavity

DIFFERENTIATE BETWEEN

23. Frontal sinus and maxillary sinus
24. Right lung and left lung
25. Pharynx and larynx
26. Right lobe and left lobe of lung
27. Trachea and bronchi

ANSWER KEY

UNIT II RESPIRATORY SYSTEM

LONG ESSAYS

1. Mention the different parts of respiratory system.

INTRODUCTION

ORGANS OF RESPIRATORY SYSTEM

- Nose
- Pharynx
- Larynx
- Trachea
- Bronchi
- Bronchioles and smaller air passages
- Two lungs their covering pleura
- Respiratory muscles

POSITION OF LUNGS AND ASSOCIATED

STRUCTURES(diagram)

VASCULAR ,NERVOUS SUPPLY AND LYMPHATIC SUPPLY

APPLIED ANATOMY

2. Describe the right lung under the following headings: _ Surfaces and borders _ Relations _ Broncho-pulmonary segments _ Applied anatomy

INTRODUCTION

FEATURES (DIAGRAM)

PARTS

- ❖ Right lung-3 LOBES
- ❖ Left lung-2 LOBES
- FISSURES AND LOBES OF LUNGS
- RIGHT LUNG
- 3 lobes
- FISSURES-2
- Oblique
- Longitudinal

BOUNDARIES

VASCULAR ,NERVOUS AND LYMPHATIC SUPPLY

3. Describe the structure and blood supply of larynx and explain structure of the bronchi and smaller air passages.

INTRODUCTION

POSITION

STRUCTURES ASSOCIATED WITH LARYNX

STRUCTURE

- Cartilages
- 1 thyroid cartilage
- 1 cricoid cartilage
- 2 arytenoid cartilage
- 1 epiglottis

INTERIOR OF LARYNX

LIGAMENTS AND MEMBRANES

BLOOD AND NERVE SUPPLY

APPLIED ANATOMY

SHORT ESSAYS

4. Name the parts of respiratory tract and describe lungs in detail with applied aspects

REFER QUE1

5. Name the parts of the respiratory system. Describe the right lung in detail. Mention the differences from left lung

REFER QUE 1

6. Describe the structure of lungs in detail with a diagram

REFER QUE 2.

7. Describe the location of Pharynx and its parts

INTRODUCTION

POSITION

BOUNDARIES ASSOCIATED WITH PHARYNX

PARTS –Nasopharynx

-Oropharynx

-Laryngopharynx

STRUCTURE OF PHARYNX

❖ Mucus membrane lining

❖ Fibrous tissue

❖ Muscle tissue

VASCULAR ,NERVOUS AND LYMPHATIC

SUPPLY

APPLIED ANATOMY

8. Describe the location of the trachea and outline its structure

INTRODUCTION

POSITION

BOUNDARIES OF TRACHEA

STRUCTURE OF TRACHEA(diagram)

○ Outer layer

○ Middle layer

○ Inner lining

VASSCULAR ,NERVOUS AND LYMPHATIC SUPPLY

APPLIED ANATOMY

SHORT ANSWER

9. Respiratory muscles

INTRODUCTION

LAYERS OF MUSCLE

❖ External internal muscle fibres

❖ Internal intercostal muscle fibres

❖ Diaphragm

QUESTIONS

UNIT III

DIGESTIVE SYSTEM

MCQ

1. These cells of ‘Crypts of Lieberkühn’ secrete lysozyme

(a) Argentaffin cells

(b) Kupffer cells

(c) Zymogen cells

(d) Paneth cells

Answer: (d)

2. The opening of the hepatopancreatic duct into the duodenum is guarded by

(a) Ileocecal valve

(b) Semilunar valve

(c) Sphincter of Oddi

(d) Pyloric sphincter

Answer: (c)

3. Which of the following cells secretes gastric acid in the stomach?

(a) Parietal cells

(b) Chief cells

(c) Acidic cells

(d) Peptic cells

Answer: (a)

4. Which part of the digestive system contains symbiotic microorganisms?

(a) Duodenum

(b) Appendix and rectum

(c) Oral cavity and tongue

(d) Caecum

Answer: (d)

5. Which of the following is present on the surface of the epithelial lining of the intestine and help in the absorption of digested food?

(a) Phagocytic vesicles

(b) Zymogen granules

(c) Microvilli

(d) Pinocytic vesicles

Answer: (c)

6. Which of the following parts of the digestive system contains Brunner’s glands?

(a) Duodenum

(b) Ileum

(c) esophagus

(d) Stomach

Answer: (a)

7. Which part of the digestive system contains lacteals?

(a) Ischium

(b) Ileum

(c) Pharynx

(d) esophagus

Answer: (b)

8. The cells which secrete tooth enamel are known as

(a) ameloblast

(b) dentoblast

(c) odontoblast

(d) osteoblast

Answer: (a)

9. Kupffer’s cells are found in

(a) liver

(b) brain

(c) kidney

(d) spleen

Answer: (a)

10. Chief cells are present in

(a) duodenum

(b) pyloric region of stomach

(c) fundic region of stomach

(d) cardiac region of stomach

Answer: (c)

11. The enzymes present in pancreatic juice are

(a) Amylase, Trypsinogen, Peptidase, Rennin

(b) Trypsinogen, Lipase, Amylase, Procarboxypeptidase

(c) Peptidase, Pepsin, Amylase, Rennin

(d) Maltase, Amylase, Trypsinogen, Pepsin

Answer: (b)

12. Which of the following hormones stimulates the production of pancreatic juice and bicarbonate?

(a) Insulin and glucagon

(b) Cholecystokinin and secretin

(c) Gastrin and insulin

(d) Angiotensin and epinephrine

Answer: (b)

13. Infants' gastric juice contains

- (a) nuclease, pepsinogen, lipase
- (b) maltase, pepsinogen, rennin
- (c) amylase, rennin, pepsinogen
- (d) pepsinogen, lipase, rennin

Answer: (d)

14. The absorption of fructose by intestinal mucosa is

- (a) co-transport mechanism
- (b) simple diffusion
- (c) facilitated transport
- (d) active transport

Answer: (c)

15. Carboxypeptidase requires _____ for its activity

- (a) copper
- (b) niacin
- (c) iron
- (d) Zn

Answer: (d)

16. Na⁺ and other carrier ions facilitate absorption of

- (a) amino acids and fructose
- (b) fatty acids and glycerol
- (c) fatty acids and glucose
- (d) amino acids and glucose

Answer: (d)

17. Secretin and cholecystokinin are secreted in

- (a) pyloric region
- (b) Ileum
- (c) duodenum
- (d) Oesophagus

Answer: (c)

18. Which of the following hydrolytic enzymes act in low pH?

- (a) Peroxidases
- (b) Hydrolases
- (c) Amylases
- (d) Proteases

Answer: (d)

19. Which of the following is synthesized and stored in the liver cells?

- (a) Galactose
- (b) Lactose
- (c) Glycogen
- (d) Arabinose

Answer: (c)

10. The disease caused due to deficiency of nicotinic acid is

- (a) anaemia
- (b) osteomalacia
- (c) xerophthalmia
- (d) pellagra

Answer: (d)

LONG ESSAYS

1. Describe the location, blood supply, nerve supply and applied anatomy of rectum
2. Describe the parts, features, blood supply and histology of stomach
3. Explain the various parts of alimentary tract
4. Mention different parts of digestive system. Describe the esophagus in detail
5. Name the salivary glands. Where do they open?
6. Describe the parotid gland in detail

SHORT ESSAYS

7. Name the parts of extra hepatic biliary apparatus. Describe the anatomy of the liver
8. Describe the pharynx
9. Describe the duodenum
10. Name the parts of digestive system. And describe stomach in detail.
11. Name the location, surfaces, borders, blood supply and applied anatomy of liver.

SHORT NOTES

1. Parotid gland
2. Submandibular gland
3. Oesophagus
4. Histology of oesophagus
5. Describe the blood supply of stomach
6. Stomach bed
7. Soft palate
8. Duodenum
9. Spleen
10. Gall bladder
11. Parts and relations of pancreas
12. Peritoneum and its function
13. Vermiform appendix
14. Digestive system disorders

ANSWER BRIEFLY

15. Name the blood vessels supplying the rectum
16. Describe the interior of anal canal
17. Inferior/ visceral surface of liver
18. Mention the components of saliva
19. List the structure of salivary glands
20. List the properties and composition of bile
21. Name the peritoneal folds of liver
22. Name different lobes of liver
23. Nerve supply of small intestine
24. Name the Papillae of tongue.
25. List the parts of digestive system.
26. Name the parts of large intestine.
27. Name the salivary glands
28. Name the parts of extra hepatic biliary apparatus.

DIFFERENTIATE BETWEEN

29. Mention the differences between small and large intestine.
30. Difference between jejunum and ileum

ANSWER KEY

UNIT III

DIGESTIVE SYSTEM

LONG ESSAYS

1. Describe the location, blood supply, nerve supply and applied anatomy of rectum

INTRODUCTION

RECTUM

STRUCTURE (DIAGRAM)

DIMENSIONS

COURSE AND DIRECTION RELATIONS

- ❖ Peritoneal relations
- ❖ Visceral relations

ARTERIAL SUPPLY

VENOUS SUPPLY

NERVOUS SUPPLY

LYMPHATIC SUPPLY

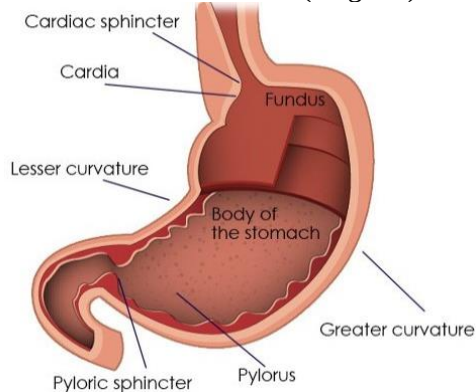
CLINICAL ANATOMY

2. Describe the parts, features, blood supply and histology of stomach

INTRODUCTION

LOCATION

EXTERNAL FEATURES (diagram)



TWO ORIFICES

- ❖ Cardiac orifice
- ❖ Pyloric orifice

TWO CURVATURES

- Lesser curvature
- Greater curvature

TWO SURFACES

- Anterior surfaces
- Posterior surfaces

DIMENSIONS

BOUNDARIES

PARTS

CARDIAC

- Fundus
- Body

PYLORIC

- Pyloric antrum
- Pyloric canal

RELATIONS OF STOMACH

BLOOD SUPPLY

LYMPHATIC DRAINAGE

NERVOUS SUPPLY

APPLIED ANATOMY

3. Explain the various parts of alimentary tract

INTRODUCTION

ORGANS OF ALIMENTARY CANAL

- Mouth
- Pharynx
- Oesophagus
- Stomach
- Small intestine

- Large intestine
- Rectum
- Anal canal

ACCESSORY ORGANS

- 3 pairs of salivary glands
- Pancreas
- Liver
- Biliary duct

BASIC STRUCTURE OF ALIMENTARY TRACT (diagram)

LAYERS

- ❖ Adventitia
- ❖ Muscle layer
- ❖ Submucosal layer
- ❖ Mucosa lining

VASCULAR SUPPLY

NERVOUS SUPPLY

LYMPHATIC SUPPLY

APPLIED ANATOMY

4. Mention different parts of digestive system. Describe the esophagus in detail

REFER QUE 3

ESOPHAGUS- The word *esophagus* is the Greek word *oisophagos*, meaning "gullet **food pipe** or **gullet (gut)**", an organ in vertebrates through which food passes, aided by peristaltic contractions, from the pharynx to the stomach.

DIMENSIONS

The esophagus is a fibromuscular tube, about 25 centimetres long in adults, which travels behind the trachea and heart, passes through the diaphragm and empties into the uppermost region of the stomach. "

STRUCTURE

- Sphincters
- Nerve supply
- Gastro-esophageal l junction
- Microanatomy

• DEVELOPMENT

• FUNCTION

- Swallowing
- reducing gastric reflux

• VASCULAR,NERVOUS AND LYMPHATIC SUPPLY

• CLINICAL SIGNIFICANCE

- Inflammation
- Barrett's esophagus
- Cancer
- Varices
- Motility disorders
- Malformations

5. Describe the parotid gland in detail

INTRODUCTION

The **parotid gland** is a major salivary gland in many animals. In humans, the two parotid glands are present on either side of the mouth and in front of both ears. They are the largest of the salivary glands.

Each parotid is wrapped around the mandibular ramus, and secretes serous saliva through the parotid duct into the mouth, to facilitate mastication and swallowing and to begin the digestion of starches.

The word *parotid* (paraotic) literally means "beside the ear".

□ STRUCTURE

- LOCATION
- BOUNDARIES
- BLOOD SUPPLY
- LYMPHATIC DRAINAGE
- NERVE SUPPLY
- HISTOLOGY
- DEVELOPMENT
- VASCULAR,NERVOUS AND LYMPHATIC SUPPLY
- CLINICAL SIGNIFICANCE
- PAROTITIS
- FIBROUS REACTIONS
- SALIVARY STONES
- INJURY
- CANCER AND TUMOURS

SHORT ESSAYS

6. Describe the anatomy of the liver

INTRODUCTION

ORGANS ASSOCIATED WITH LIVER

PORTAL FISSURE

STRUCTURE(diagram)

DIMENSIONS

BOUNDARIES

VASCULAR SUPPLY,NERVOUS SUPPLY AND

LYMPHATIC SUPPLY

CLINICAL SIGNIFICANCE

7. Describe the pharynx

INTRODUCTION

DIMENSIONS

BOUNDARIES

PARTS

❖ Nasopharynx

❖ Oropharynx

❖ Laryngopharynx

STRUCTURE OF PHARYNX(diagram)

ORIGIN AND INSERTION OF PHARYNX MUSCLES

VASCULAR,NERVOUS AND LYMPHATIC SUPPLY

CLINICAL SIGNIFICANCE

8. Describe the duodenum

INTRODUCTION

- STRUCTURE(diagram)
 - FIRST PART
 - SECOND PART
 - THIRD PART
 - FOURTH PART
 - BLOOD SUPPLY
 - LYMPHATIC DRAINAGE
 - HISTOLOGY
 - VARIATION
- FUNCTION

- CLINICAL SIGNIFICANCE
 - ULCERATION
 - CELIAC DISEASE
- 9. Name the parts of digestive system. And describe stomach in detail.

REFER QUE 2 AND 3

10. Name the location, surfaces, borders, blood supply and applied anatomy of liver.

REFER QUE 6

SHORT NOTES

11. Parotid gland

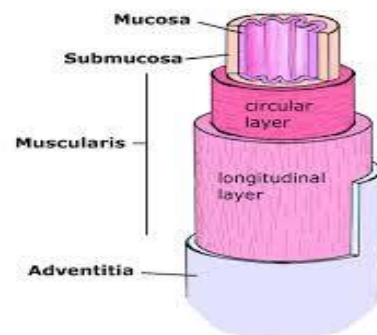
REFER QUE 5

12. Submandibular gland

INTRODUCTION

- STRUCTURE
 - HISTOLOGY
 - BLOOD SUPPLY
 - LYMPHATIC DRAINAGE
 - INNERVATION
 - DEVELOPMENT
 - CLINICAL SIGNIFICANCE
 - 13. Oesophagus
- REFER QUE 4
14. Histology of oesophagus
- INTRODUCTION

LAYERS OF THE ESOPHAGEAL WALL



15. Describe the blood supply of stomach

INTRODUCTION

STOMACH

ARTERIAL SUPPLY

- ❖ LEFT GASTRIC ARTERY
- ❖ RIGHT GASTRIC ARTERY
- ❖ GASTROEPIPLOIC ARTERY
- ❖ SPLENIC ARTERY

VENOUS SUPPLY

- ❖ RIGHT GASTRIC VEIN
 - ❖ LEFT GASTRIC VEIN
 - ❖ MESENTRIC VEIN
 - ❖ SPLENIC VEIN
- CLINICAL SIGNIFICANCE

16. Stomach bed

INTRODUCTION

STOMACH

STOMACH BED

Boundaries

The **stomach bed** refers to the structures upon which the stomach rests in mammals

These include the Pancreas, Spleen, Left Kidney, Left Suprarenal gland, Transverse colon and its mesocolon, and the diaphragm.

The term was introduced around 1896 by Mr Philip Polson of the Catholic University School of Medicine, Dublin.

STRUCTURES

- Left Crus Of Thoracic diaphragm
- Tortuous Splenic Artery
- Gastric Surface of Spleen
- Transverse Mesocolon
- Left Colic Flexure
- Anterior Surface Of Left Kidney
- Left Suprarenal Gland
- Body of the Pancreas

ARTERIAL SUPPLY

VENOUS SUPPLY

17. Soft palate

PALATE

SOFT PALATE

MUSCLE OF SOFT PALATE

- ❖ Tensor palate
- ❖ Levator palate
- ❖ Musculus uvulae
- ❖ Palatoglossus
- ❖ Palatopharyngeus

BLOOD,NERVE AND LYMPHACTIC SUPPLY

CLINICAL SIGNIFICANCE

18. Duodenum

REFER QUE-6

19. Spleen

INTRODUCTION

LOCATION

EXTERNAL FEATURES

Two ends

Two borders

Two surfaces

RELATIONS

Peritoneal relations

Visceral relations

DEVELOPMENT

FUNCTIONS

BOUNDARIES

DIMENSIONS

BLOOD,NERVE AND LYMPHACTIC SUPPLY

CLINICAL SIGNIFICANCE

20. Gall bladder

INTRODUCTION

DIMENSIONS

BOUNDARIES

LAYERS

DUCTS OF GALL BLADDER

FUNCTIONS

BLOOD SUPPLY,NERVE SUPPLY AND

LYMPHACTIC SUPPLY

CLINICAL SIGNIFICANCE

21. Parts and relations of pancreas

DESCRIPTION

LOCATION

STRUCTURE(diagram)

SHAPE

HEAD OF PANCREAS

Features

NECK OF PANCREAS

BODY OF PANCREAS

TAIL OF PANCREAS

DUCTS OF PANCREAS

DIMENSIONS

BOUNDARIES

BLOOD ,NERVE ,LYMPHACTIC SUPPLY

CLINICAL SIGNIFICANCE

22. Peritoneum and its function

Peritoneum is in the form of a closed sac which is invaginated by a number of viscera.

LAYERS

- ❖ Outer or parietal layer
- ❖ Inner or visceral layer

Functions

DEVELOPMENT

PERITONEAL FOLDS

BLOOD ,NERVE ,LYMPHACTIC SUPPLY

CLINICAL ANATOMY

23. Vermiform appendix

DIMENSIONS

POSITIONS

- ❖ Paracolic
- ❖ Retrocolic
- ❖ Splenic
- ❖ Promonteric
- ❖ Pelvic
- ❖ Subcaecal

APPENDICULAR ORIFICE

BLOOD SUPPLY

CLINICAL SIGNIFICANCE

24. Digestive system disorders

INTRODUCTION

- DENTAL CARIES
- PERIDONTAL DISEASE
- PEPTIC ULCER DISEASE
- DIVERTICULAR DISEASE
- COLORECTAL CANCER
- HERNIA

QUESTIONS

UNIT IV

CIRCULATORY & LYMPHATIC SYSTEM

MCQ

1. The lymphatic system consists of all the following except:
A) **Blood**
B) lymph nodes
C) lymphatic vessels
D) lymph

2. Which of the following applies to the thoracic duct?
 - A) it drains the entire body above the diaphragm
 - B) **it empties its contents into the subclavian vein**
 - C) it carries blood into the lymphatic system
 - D) it arises in the vessels of the brain
3. Lymphatic capillaries resemble blood capillaries because lymphatic capillaries
 - A) have the same permeability as blood capillaries
 - B) lead to the vena cava
 - C) **have a lining of endothelium**
 - D) are thick-walled tubes
4. The fluid that passes through the lymphatic vessels
 - A) Flows toward the lungs
 - B) Passes from the lymphatic vessels into the arteries
 - C) Enters the left ventricle of the heart through the right thoracic duct
 - D) **Moves in a single direction toward the heart**
5. The t-lymphocytes and B-lymphocytes are the major cells of the
 - A) **Lymph nodes**
 - B) Lymphatic vessels
 - C) Adrenal gland
 - D) Thymus
6. Lymph nodes may be located in the human body in the tissues of the
 - A) stomach and brain
 - B) **groin and neck**
 - C) Ventricle and atrium
 - D) thyroid gland and adrenal gland
7. The nodules of lymphoid tissue found in the wall of the intestinal tract are known as
 - A) Hashimoto's nodes
 - B) grave's region
 - C) di-george's nodes
 - D) **Peyer's patches**
8. The movement of fluid through the lymphatic vessels is assisted by
 - A) pressure from the right ventricle
 - B) **pressure of contracting skeletal muscles**
 - C) movement of phagocytes such as macrophages
 - D) movement of red blood cells
9. Those lymphatic vessels that are rich in fat
 - A) are found only in the brain
 - B) **are known as lacteals**
 - C) enter the left atrium of the heart
 - D) are found only within the spleen
10. Which of the following is not likely to be found in the lymph?
 - A) **red blood cells**
 - B) protein molecules
 - C) Microorganisms
 - D) macrophages
11. All the following functions are associated with the spleen except
 - A) it provides a filtration system for blood
 - B) it is the site of red blood cell breakdown
 - C) it is a storage depot for blood
 - D) **it is the major site of white blood cell formation**
12. The lymphokines secreted by cytotoxic t-lymphocytes increase the activity of
 - A) red blood cells
 - B) brain cells
 - C) **macrophages**
 - D) b-lymphocytes
13. Where is the thymus gland located?
 - A. **Lower part of the mediastinum**
 - B. Upper part of the mediastinum
 - C. Behind the mediastinum
 - D. In front of the mediastinum
14. Where is the spleen situated?
 - A. **Abdominal cavity**
 - B. Pelvic cavity
 - C. In the neck
 - D. Git
15. What is the function of the lymphatic system
 - A. Immunity
 - B. **Transports lipids from GIT to the blood**
 - C. Drains excess interstitial fluid
 - D. All of the above
16. Why lymphatic nodules are not known as lymphatic organs?
 - A. Because they are less in number
 - B. Due to their small size
 - C Lacks the surrounding capsule
 - D. **Both b and c**
17. Submandibular lymph glands are present under.....
 - (a) Axilla
 - (b) **Chin**
 - (c) Chest
 - (d) Groins
18. One of the following is not a lymphatic organ
 - (a) **Pancreas**
 - (b) Spleen
 - (c) Thymus
 - (d) Tonsils
19. The lymphatic fluid returning from the small intestine is rich in
 - (a) Proteins
 - (b) Calcium
 - (c) **Fats**
 - (d) Phosphorus
20. Spleen is called the "graveyard" of the...
 - (a) **Erythrocytes**
 - (b) Leucocytes
 - (c) Lymphocytes
21. Lymphatic fluid is rich in
 - (a) Iron
 - (b) **Antibodies**
 - (c) Neutrophils
 - (d) Basophils
22. The lymphatic fluid is
 - (a) Brownish
 - (b) Pinkish

(c) **Colorless**

(d) Bluish

23. The thoracic duct of the lymphatic system pours its contents into the

(a) Aorta

(b) **Subclavian vein**

(c) Internal jugular veins

(d) Right auricle

24. The lymphatic fluid is

(a) **Alkaline**

(b) Acidic

(c) Neutral

(d) Partially acidic and partially alkaline

25. Trabeculae in the lymph nodes are formed of

(a) Lymphocytes

(b) **Fibrous tissue**

(c) Blood vessels

(d) Lymphatic vessels

26. The structures of the lymphatic system present in the intestine are called

(a) Mucosa

(b) Villi

(c) Chyle

(d) **Lacteals**

27. What is the circulatory system?

- ☒ The body's breathing system
- ☐ The body's system of nerves
- ☐ The body's food-processing system
- ☐ The body's blood-transporting system

28. From what source do cells get their food?

- ☒ Blood
- ☐ Oxygen
- ☐ Other cells
- ☐ Carbon dioxide

29. Why is oxygen important to blood and to the cells?

- ☒ ➤ Oxygen helps the blood to clot.
- ☐ ➤ Oxygen brings food to the cells.
- ☐ ➤ Oxygen is necessary for cell growth and energy.
- ☐ ➤ Oxygen is not important -- carbon dioxide is the most important substance to the body.

30. Which type of blood vessels carries blood away from the heart?

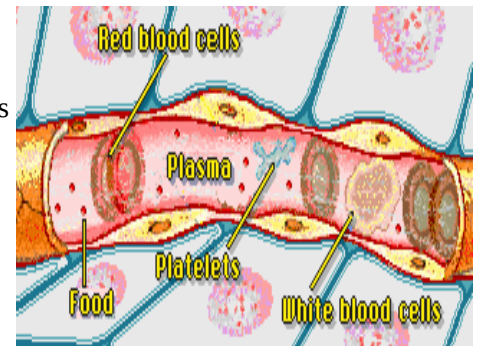
- ☐ Veins
- ☐ Arteries
- ☐ Capillaries
- ☐ Arteries, veins and capillaries

31. Why is blood that flows from the lungs to the heart bright red rather than dark red?

- ☐ ➤ Oxygen makes it red.
- ☐ ➤ Carbon dioxide makes it red.
- ☐ ➤ Gastric juices produce the red color of the blood.
- ☐ ➤ The lungs add a pigment (dye) to blood as it flows through them.

32. What part of the blood carries minerals, vitamins, sugar, and other foods to the body's cells?

- ☐ Plasma
- ☐ Platelets
- ☐ Red corpuscles
- ☐ White corpuscles



33. What is the main job of the red corpuscles in the blood?

- ☐ ➤ To clot blood
- ☐ ➤ To fight disease
- ☐ ➤ To transport oxygen to the body's cells and carry away carbon dioxide from the cells
- ☐ ➤ To transport carbon dioxide to the body's cells and carry away oxygen from the cells

34. Which of the following can best be compared to soldiers?

- ☐ Lungs
- ☐ Capillaries
- ☐ Red blood cells
- ☐ White blood cells

35. Which element in the blood is round and colorless?

- ☐ Plasma
- ☐ Platelets
- ☐ Red blood cells
- ☐ White blood cells

36. What would happen to people who have an open wound and whose blood did not clot naturally?

- ☐ They may bleed to death.
- ☐ Nothing. Clotting is not important.
- ☐ They would have to take regular doses of plasma.
- ☐ They would have to take regular doses of platelets.

37. What happens when a clot occurred in an undamaged blood vessel?

- ☐ ➤ You would bleed to death.
- ☒ ➤ A scab will form on the skin surface.
- ☐ ➤ Platelets stick to the edges of the cut and to one another, forming a plug.
- ☐ ➤ The flow of blood to tissues beyond the clot may be cut off.

38. What happens to blood when it is pumped into the thin-walled blood vessels of the lungs?

- ☒ ➤ Platelets are exchanged for plasma.
- ☐ ➤ Carbon dioxide is replaced with oxygen.
- ☐ ➤ Blood fills the lungs and causes coughing.
- ☐ ➤ Nothing -- the lungs are just a place blood goes through on its way back to the heart.

39. What is the function of the blood vessels and capillaries?

- ☐ ➤ They pump blood to the heart.
- ☐ ➤ They filter impurities from the blood.
- ☐ ➤ They carry blood to all parts of the body.

- ☐ ➤ They carry messages from the brain to the muscles.

40. Why does blood turn dark red as it circulates through the body?

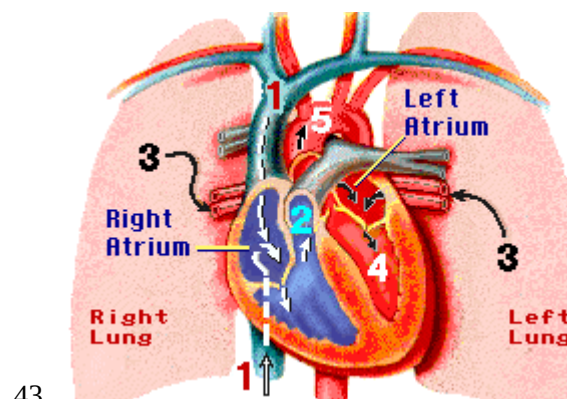
- ☐ It starts to clot.
- ☐ It gets old and dirty flowing through the body.
- ☐ The oxygen in it is replaced with carbon dioxide.
- ☐ The farther blood is from the heart, the more dark red it is.

41. How many major types of blood have scientists discovered?

- ☐ One: Type "O"
- ☐ Two: white cells and red cells
- ☐ Three: white cells, red cells, and plasma
- ☐ Four: Types A, B, AB, and O

42. What is the organ that pumps blood all throughout the human body?

- ☐ The lungs
- ☐ The heart
- ☐ The kidneys
- ☐ The blood vessels and capillaries



43. ~ How the heart works ~

1. The heart receives oxygen-deficient blood (see the white arrows) from the body into the right upper atrium.
2. When the heart contracts, the right lower ventricle will pump the blood into the lungs, where the carbon dioxide is exchanged for oxygen.

3. After the exchange, the blood containing fresh oxygen flows into the left upper atrium.
4. Oxygen-rich blood (see the black arrows) flows from the left upper atrium into the left lower ventricle.
5. When the heart contracts, the left lower ventricle will force the blood out to the body through a network of arteries.

LONG ESSAYS

1. Describe the heart under the following headings: parts _ relations _ blood supply _ applied anatomy
2. Draw and label the lymphatic organs. Describe the microscopic structure of one of them.
3. Describe the gross anatomy of heart in detail
4. Name the chambers of the heart with a brief detail of each chamber and the valves in between them
5. Describe the right atrium in detail

SHORT ESSAYS

6. Describe the ventricles of heart. Write a note on blood supply of heart
7. Draw and label the main arteries of the upper limb. Describe the structure of blood vessels.
8. Explain spleen in detail location, external features, blood supply, histology and functions
9. Describe blood. Explain functions and composition of blood.

SHORT ANSWERS

9. Waldeyer's ring
10. Mention the formation and termination of thoracic duct
11. Apex beat
12. Chambers of heart
13. Conducting system of heart

ANSWER BRIEFLY

- 14.. Name 4 groups of cervical lymph nodes
15. Branches of the right coronary artery
16. Name 3 layers of wall of blood vessels
17. List the four important functions of spleen

DIFFERENTIATE BETWEEN

18. Large and medium sized arteries
19. SA node and AV node
20. Structure of large artery and large vein.
21. Erythrocytes and leucocytes
22. Lymphocytes and leucocytes

ANSWER KEY

UNIT IV

CIRCULATORY & LYMPHATIC SYSTEM

LONG ESSAYS

1. Describe the heart under the following headings: parts _ relations _ blood supply _ applied anatomy

INTRODUCTION

HEART

STRUCTURE(diagram)

PARTS

- ❖ Apex
- ❖ Base
- ❖ 2 Lateral surfaces

CHAMBERS

- Right atrium
- Left atrium
- Right ventricle
- Left ventricle

RELATIONS

VALVES OF HEART

- AV valves
- Semilunar valves

CONDUCTING SYSTEM

- ❖ SA node
- ❖ AV node
- ❖ Bundle of His
- ❖ Purkinje fibres

ARTERIAL SUPPLY

VENOUS SUPPLY

NERVOUS SUPPLY

APPLIED ANATOMY

2. Draw and label the lymphatic organs. Describe the microscopic structure of one of them.

3. Describe the gross anatomy of heart in detail

REFER QUE-1

4. Name the chambers of the heart with a brief detail of each chamber and the valves in between them

REFER QUE-1

5. Describe the right atrium in detail

INTRODUCTION

FEATURES (DIAGRAM)

PARTS

TRIBUTARIES OR INLETS

RIGHT ATRIOVENTRICULAR ORIFICE

INTERNAL FEATURES

- ❖ Smooth posterior part- Sinus venarum
- ❖ Rough anterior part-
- ❖ Interatrial septum

VASCULAR AND LYMPHATIC SUPPLY

CLINICAL SIGNIFICANCE

SHORT ESSAYS

6. Describe the ventricles of heart. Write a note on blood supply of heart

INTRODUCTION

HEART-CHAMBERS(diagram)

VENTRICLES OF HEART

FEATURES

Right ventricle

Left ventricle

- External surface-Anterior and Sternocostal surface
- Internal surface-Trabaculæ and Infundibulum

VALVES

ARTERIAL SUPPLY

VENOUS SUPPLY

NERVOUS SUPPLY

LYMPHATIC SUPPLY

CLINICAL SIGNIFICANCE

7. Draw and label the main arteries of the upper limb.
8. Describe the structure of blood vessels.
9. Explain spleen in detail location, external features, blood supply, histology and functions
10. Describe blood. Explain functions and composition of blood.

INTRODUCTION

DEFINITION

COMPOSITION OF BLOOD

Plasma

- ✓ Plasma proteins
- ✓ Albumins
- ✓ Globulins
- ✓ Clotting factors
- ✓ Inorganic salts
- ✓ Nutrients
- ✓ Organic wasteproducts
- ✓ Hormones
- ✓ Gases

Cellular components

- ✓ Leukocytes
- Granulocytes
 - Neutrophils
 - Eosinophils
 - Basophils
- Agranulocytes
 - Monocytes
 - Lymphocytes
- ✓ Erythrocytes
- ✓ Thrombocytes

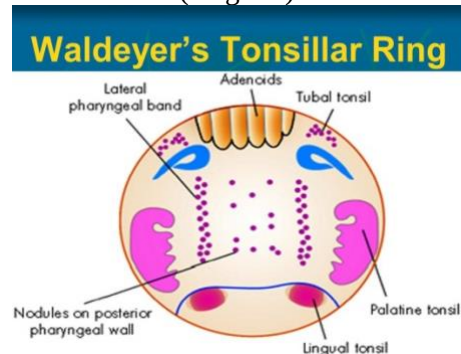
SHORT ANSWERS

1. Waldeyer's ring

TONSILLAR RING or PHARYNGEAL RING or

LYMPHOID RING

STRUCTURE(diagram)



- ❖ 1 Pharyngeal tonsil
- ❖ 2 Tubal tonsil
- ❖ 2 palatine tonsil
- ❖ 1 Lingual tonsil

10. Mention the formation and termination of thoracic duct

FEATURES

- ✓ In adults, the thoracic duct is typically 38–45 cm in length and has an average diameter of about 5 mm.
- ✓ The vessel usually starts from the level of the twelfth thoracic vertebrae (T12) and extends to the root of the neck.
- ✓ It drains into the systemic (blood) circulation at the angle of the left subclavian and internal jugular veins as a single trunk, at the commencement of the brachiocephalic vein.
- ✓ It also collects most of the lymph in the body other than from the right thorax, arm, head, and neck which are drained by the right lymphatic duct.

COURSE

TRIBUTARIES

APPLIED ANATOMY

11. Apex beat

The **apex beat**, also called the **apical impulse** is the pulse felt at the **point of maximum impulse (PMI)**, which is the point on the precordium farthest outwards (laterally) and downwards (inferiorly) from the sternum at which the cardiac impulse can be felt.

The cardiac impulse is the vibration resulting from the heart rotating, moving forward and striking against the chest wall during systole.

INTERPRETATION

VARIATION

APPLIED ANATOMY

12. Chambers of heart

REFER QUE 2

13. Conducting system of heart

REFER QUE 2

QUESTIONS

UNIT V

ENDOCRINE SYSTEM

MCQ

1. Which of these is true of the endocrine system?
 - A. secretes hormones that are transported to target cells by blood
 - B. causes changes in metabolic activities
 - C. effects are prolonged
 - D. All of above are true.
2. Name the gland that is located at the base of the throat, just inferior to the laryngeal prominence (Adam's apple) .
 - A. Pituitary.
 - B. Pineal gland.
 - C. Hypothalamus.
 - D. Thyroid .
3. What gland is located just superior to the kidneys?
 - A. Pituitary.
 - B. Adrenal.
 - C. Pancreas.
 - D. ovaries
4. In the pancreas, which are the cells that secrete insulin, decrease the blood levels of glucose.

- A. delta.
- B. alpha.
- C. beta.
- D. beta.

5. The endocrine gland responsible for the body's circadian rhythm is the:

- A. Thymus gland.
- B. Pineal gland.
- C. Parathyroid gland.
- D. Pituitary gland.

6. Endocrine glands differ from exocrine glands in that:

- A. Endocrine glands are ductless and exocrine glands release secretions at the body's surface or into ducts.
- B. Endocrine glands release hormones, whereas exocrine glands release waste.
- C. Endocrine glands are formed by epithelial tissue, but exocrine glands are primarily connective tissue.
- D. Endocrine glands are all interconnected; whereas exocrine glands act completely independently.

7. Which of these is not an endocrine property?

- A. hormones reach targets through the blood
- B. effects are slow and cyclic
- C. rapid acting effects
- D. effects caused by chemicals.

8. The primary target of the releasing and inhibiting hormones of the hypothalamus is the:

- A. Liver and adipose tissue
- B. Gonads
- C. Anterior pituitary
- D. Bone marrow

9. Which of these is not an endocrine gland?

- A. Pancreas.
- B. Testes.
- C. Salivary gland.
- D. Parathyroid.

10. Which is not a function of the hypothalamus?

- A. Affect heart rate
- B. Control temperature
- C. Affect water balance
- D. Secrete FSH

11. Which of the following gland which can be classified as an endocrine and an exocrine gland?

- A. Thyroid.
- B. Thymus.
- C. Pancreas.
- D. Pituitary.

12. Which of these hormones is made by the posterior pituitary?

- A. FSH

- B. LH
- C. ACTH
- D. ADH

13. Which gland controls basal metabolic rate (BMR)?

- A. Thyroid.
- B. Parathyroid.
- C. Testes.
- D. Pancreas.

14. How do hormones from the thyroid and parathyroid regulate the calcium concentration of the blood?

- A. Calcitonin lowers blood calcium; parathyroid hormone raises blood calcium.
- B. Parathyroid hormone lowers blood calcium; calcitonin raises blood calcium.
- C. Thyroxine and triiodothyronine together regulate calcium levels, as needs dictate.
- D. Both parathyroid hormone and the three thyroid hormones function to regulate blood calcium levels.

15. The posterior pituitary stores and releases:

- A. Growth hormone and prolactin.
- B. Prolactin and oxytocin.
- C. Oxytocin and antidiuretic hormone (ADH).
- D. ADH and growth hormone.

16. Which of the following hormones are responsible for the "fight-or-flight" response?

- A. Epinephrine and norepinephrine.
- B. Insulin and glucagon.
- C. Estradiol and progesterone.
- D. Thyroxine and melatonin.

17. The pituitary hormone that stimulates the male testes to produce sperm and stimulates the development of the follicle in the female on a monthly cycle is:

- A. growth hormone
- B. luteinizing hormone
- C. prolactin
- D. follicle-stimulating hormone

18. Which hormones of the adrenal glands supplement the sex hormones from the gonads?

- A. Mineralocorticoids, such as aldosterone
- B. Glucocorticoids, such as cortisol
- C. Gonadocorticoids, such as the androgens
- D. Epinephrine and norepinephrine.

19. The clusters of cells in the pancreas that produce hormones are the:

- A. Nodules.
- B. Islets of Langerhans.
- C. Pancreatic medulla.
- D. Pancreatic cortex.

20. The Glucagon is:

- A. accelerates the conversion of glycogen into glucose.
- B. slows down glucose formation from lactic acid.
- C. decreases the conversion of glycogen into glucose.
- D. speeds up protein synthesis within cells.

21. Most hormones of the endocrine system are regulated by a:

- A. Negative feedback mechanism.
- B. Positive feedback mechanism.
- C. Hormone-receptor complex.
- D. Hormone-gene complex.

22. Calcitonin is a hormone of which of the following:

- A. Adrenal cortex.
- B. Thyroid gland.
- C. Pituitary gland.
- D. Thymus gland.
- E. None of the above.

23. Calcium level in the blood is regulated by the:

- A. Thyroid.
- B. Parathyroid.
- C. Posterior pituitary.
- D. Adrenal medulla.
- E. A and B.

24. All of the following are hormones of the anterior pituitary except:

- A. Human growth hormone (GH).
- B. Follicle-stimulating hormone (FSH).
- C. Parathyroid hormone (PTH).
- D. Thyroid-stimulating hormone (TSH).

25. The secretions from which of these glands differ between males and females?

- A. Adrenal.
- B. Parathyroid.
- C. Gonadal.
- D. Pancreas.

ANSWER KEY

1. The correct answer is D.
2. The correct answer is D.
3. The correct answer is B.
4. The correct answer is D.
5. The correct answer is B.
6. The correct answer is A.
7. The correct answer is C.
8. The correct answer is C.
9. The correct answer is C.
10. The correct answer is D.
11. The correct answer is C.
12. The correct answer is D.
13. The correct answer is A.
14. The correct answer is A.
15. The correct answer is C.
16. The correct answer is A.
17. The correct answer is D.
18. The correct answer is C.

19. The correct answer is B.

20. The correct answer is A.

21. The correct answer is A.

22. The correct answer is B.

23. The correct answer is E.

24. The correct answer is C.

25. The correct answer is C.

SHORT ESSAYS

1. Describe the pituitary gland in detail
2. Describe the location, histology of anterior and posterior pituitary gland.
3. Name the endocrine glands. Describe the thyroid gland in detail
4. Name the endocrine glands. Describe the pancreas
5. Describe pancreas under following heading- location, parts, importance relations and histology

SHORT ESSAYS

6. Describe the location, histology of adrenal gland
7. Describe the location, histology of pancreatic cells

SHORT NOTES

8. Thyroid gland
9. Functions of adrenal gland
10. Parathyroid glands
11. Relations of thyroid gland

ANSWER BRIEFLY

1. Name the endocrine glands
2. Mention the parts of pituitary gland
3. Mention the types of cells found in anterior pituitary
4. Name 2 hormones produced by neurohypophysis
5. Name any 4 hormones produced by pituitary gland
6. Name 2 arteries supplying the thyroid gland
7. Name 2 ovarian hormones
8. Name the hormones secreted by thyroid gland
9. Name the hormones secreted by placenta
10. Mention the blood supply of pancreas
11. Describe islets of Langerhans
12. Name the disorders of thyroid gland.
13. Parts of thyroid gland.

DIFFERENTIATE BETWEEN

14. Anterior pituitary and posterior pituitary
15. Adrenal cortex and medulla

ANSWER KEY

UNIT V

ENDOCRINE SYSTEM

SHORT ESSAYS

1. Describe the pituitary gland in detail
INTRODUCTION
PITUITARY GLAND

Anterior pituitary
Hypophyseal portal system
(TSH,FSH,LH,Prolactin,ACTH,MSH)
Posterior pituitary(Prolactin,ADH)
HORMONES ACTION

LOCATION

VASCULAR SUPPLY

NERVE SUPPLY

APPLIED ANATOMY

2. Describe the location ,histology of anterior and posterior pituitary gland.

3. REFER QUE 1

4. Name the endocrine glands. Describe the thyroid gland in detail

INTRODUCTION

ENDOCRINE GLANDS

THYROID GLAND

Microscopic structure of thyroid gland(diagram)

❖ Follicular cells

❖ Para follicular cells

Hormones

DIMENSIONS

BORDERS

VASCULAR SUPPLY

NERVE SUPPLY

APPLIED ANATOMY

5. Name the endocrine glands. Describe the pancreas

REFER QUE 13

6. Describe pancreas under following heading- location, parts, importance relations and histology

REFER QUE 13

SHORT ESSAYS

12. Describe the location ,histology of adrenal gland

INTRODUCTION

ADRENAL GLAND

ADRENAL CORTEX

ADRENAL MEDULLA

HORMONES

BLOOD SUPPLY

NERVE SUPPLY

APPLIED ANATOMY

13. Describe the location ,histology of pancreatic cells

PANCREAS

FEATURES

CELL TYPES

Alpha cells

Beta cells

Delta cells

F cells

APPLIED ANATOMY

SHORT NOTES

14. Thyroid gland

REFER QUE 4

15. Functions of adrenal gland

16. Parathyroid glands

PARATHYROID GLAND

FEATURES

LOCATION

DIMENSIONS

HISTOLOGY

HORMONES AND ACTIONS

VASCULAR SUPPLY,NERVE

SUPPLY,LYMPHATIC SUPPLY

CLINICAL ANATOMY

17. Relations of thyroid gland

THYROID GLAND

RELATIONS

❖ Lower larynx

❖ Trachea

❖ Parathyroid gland

VASULAR SUPPLY

CLINICAL SIGNIFICANCE

QUESTIONS

UNIT VI

SENSORY ORGANS

MCQ

1. The change in an environment to which a body responds is called

- ☐ A. A stimulus
- ☐ B. A response
- ☐ C. A retina
- ☐ D. A ligament

2. The reaction to a stimulus is called

- ☐ A. A response
- ☐ B. A choroid
- ☐ C. A function
- ☐ D. A reply

3. Which one of the following is not a sense organ?

- ☐ A. Liver
- ☐ B. Eyes
- ☐ C. Ears
- ☐ D. Skin

4. Which one of the following is NOT one of the human senses?

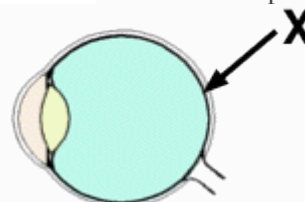
- ☐ A. Nerve
- ☐ B. Taste
- ☐ C. Touch
- ☐ D. Sight

5. Which one of the following is a sense organ?

- ☐ A. Tongue
- ☐ B. Sight
- ☐ C. Smell
- ☐ D. Touch

6. Where are the sense cells found in the eye?

- ☐ A. Retina
- ☐ B. Iris
- ☐ C. Lens
- ☐ D. Blind spot



7.

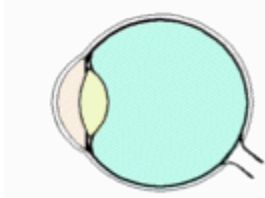
The part labelled X in the diagram is called the

- ☐ A. Retina
- ☐ B. Blind spot
- ☐ C. Lens
- ☐ D. Cornea

8. What is the function of the iris in the human eye?

- ☐ A. It controls the amount of light entering the eye.
- ☐ B. It focuses light onto the retina.
- ☐ C. It changes the shape of the lens.
- ☐ D. It allows light enter the eye.

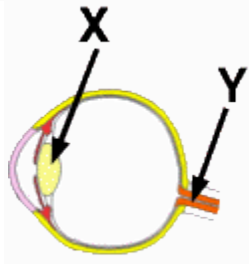
9.



Where is the sclera located?

- ☐ A. On the outer layer of the eye.
- ☐ B. On the retina.
- ☐ C. On the fovea.
- ☐ D. In the vitreous humour

10.



Name the parts labelled X and Y in the diagram

- ☐ A. X = lens, Y = optic nerve
- ☐ B. X = lens, Y = retina
- ☐ C. X = iris, Y = optic nerve
- ☐ D. X = pupil, Y = retina

11. The transparent front part of the eye is called the

- ☐ A. Cornea
- ☐ B. Lens
- ☐ C. Pupil
- ☐ D. Iris

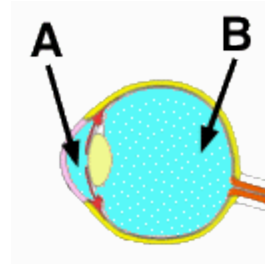
12.



The coloured part of the eye is called the

- ☐ A. Iris
- ☐ B. Retina
- ☐ C. Choroid
- ☐ D. Pupil

13.



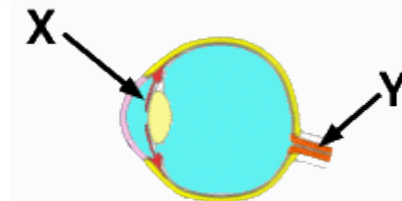
Name the jelly-like parts of the eye labelled in the diagram

- ☐ A. A = aqueous humour, B = vitreous humour
- ☐ B. A = vitreous humour, B = aqueous humour
- ☐ C. A = aqueous humour, B = lens
- ☐ D. A = vitreous humour, B = pupil

14. What part of the eye focusses light onto the retina?

- ☐ A. Lens
- ☐ B. Pupil
- ☐ C. Iris
- ☐ D. Retina

15.



Name the parts labelled X and Y in the diagram

- ☐ A. X = iris, Y = optic nerve
- ☐ B. X = iris, Y = fovea
- ☐ C. X = optic nerve, Y = iris
- ☐ D. X = pupil, Y = blind spot

16. The outermost transparent layer of the eye is called the

- ☐ A. Sclera
- ☐ B. Iris
- ☐ C. Fovea
- ☐ D. Pupil

17. The middle part of the eye containing blood vessels is called the

- ☐ A. Choroid
- ☐ B. Pupil
- ☐ C. Vitreous humour
- ☐ D. Optic nerve

18. The coloured part of the eye that controls the amount of light entering is called the

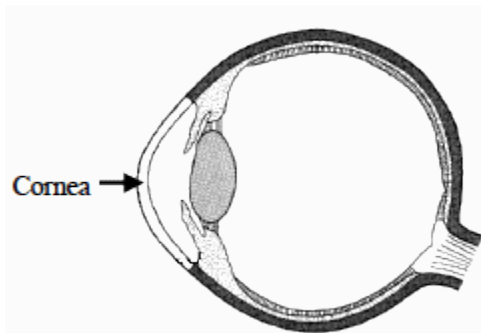
- ☐ A. Iris
- ☐ B. Pupil
- ☐ C. Lens
- ☐ D. Sclera

19. The lens focusses light onto the retina by changing its shape. What causes it to change shape?

- ☐ A. Ciliary muscles
- ☐ B. Fovea
- ☐ C. Iris

- D. Cornea
20. The light receptor cells in the eye are located in the
- A. Retina
 - B. Blind spot
 - C. Pupil
 - D. Cornea

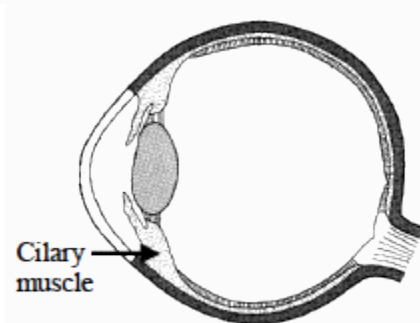
21.



The diagram shows the human eye. Give the function of the cornea

- A. To allow light into the eye.
- B. To carry nerve impulses to the brain.
- C. To change the shape of the lens.
- D. To detect light.

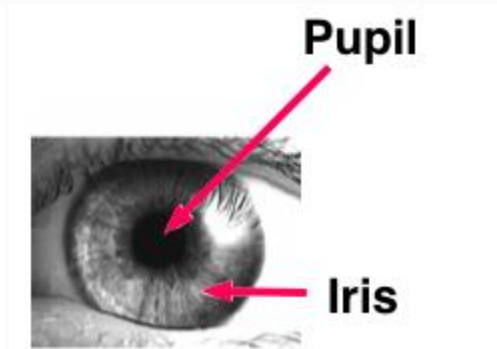
22.



The diagram shows the human eye. What is the function of the ciliary muscle?

- A. To change the shape of the lens.
- B. To allow light into the eye.
- C. To carry nerve impulses to the brain.
- D. To detect light

23.



What is the function of the pupil?

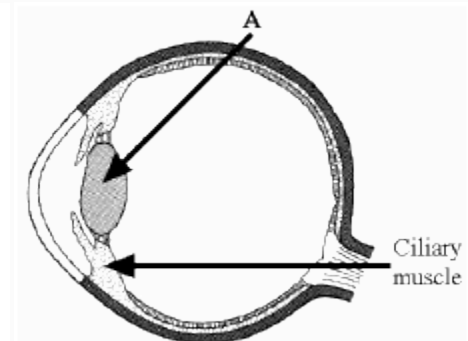
- A. To allow light enter the eye.
- B. To control the amount of light entering the eye.

- C. To change the shape of the lens.
- D. To focus light onto the retina.

24. The pupil of the human eye is transparent. Why does the pupil appear to be black in most situations?

- A. Because most of the light entering the eye is absorbed.
- B. Because light cannot travel through the vitreous humour.
- C. Because the lens does not let light pass back out.
- D. Because the light does not have enough energy.

25.



The diagram is of the human eye. Name the part labelled A.

- A. Lens
- B. Iris
- C. Fovea
- D. Blind spot

26. The inner most layer of human eye is

- a) Sclera
- b) Cornea
- c) Choroid
- d) Retina

25. Color blindness is due to defect in _____

- a) Cones
- b) Rods
- c) Rods and cons
- d) Rhodopsin

26. The purplish red pigment rhodopsin contained in rods type of photoreceptor cell is a derivative of _____

- a) Vitamin B1
- b) Vitamin C
- c) Vitamin D
- d) Vitamin A

27. Color perception in man is due to _____

- a) Rhodopsin present in rod cells
- b) Iodopsin present in cone cells
- c) Rhodopsin present in cone cells
- d) Iodopsin present in rod cells

28. Organ of Corti helps in _____

- a) Balancing
- b) Maintaining equilibrium
- c) Hearing
- d) Formation of wax

29. Which is thickened to form organ of Corti?
 - a) Basilar membrane
 - b) Reissner's membrane
 - c) Tectorial membrane
 - d) Epithelial cells
30. A person goes to ophthalmologist. He has a problem of reading books because he is not able to contract his
 - a) Iris
 - b) Suspensory ligament
 - c) Pupil
 - d) Ciliary muscles
31. Eustachian tube connects _____
 - a) Pharynx with middle ear
 - b) Middle ear with external ear
 - c) Middle ear with external ear
 - d) External ear with internal ear
32. Aqueous and vitreous humours are divided by ____
 - a) Lens
 - b) Iris
 - c) Retina
 - d) Optic nerve
33. Organ of Ruffini is sensitive to _____
 - a) Cold
 - b) Heat
 - c) Pressure
 - d) Touch
34. Which of the following prevents internal reflection of light inside the eye?
 - a) Cornea
 - b) Sclera
 - c) Choroid
 - d) Retina
35. In which of the following only cone cells are found?
 - a) Fovea centralis
 - b) Retina
 - c) Fossa ovalis
 - d) Blind spot
36. Ora serrata is _____
 - a) Present in ear
 - b) Oral cavity of protochordates
 - c) Gland present in oral cavity
 - d) A part of third wall in retina of eye
37. Sense of smell is perceived by _____
 - a) Pituitary
 - b) Hypothalamus
 - c) Olfactory lobe
 - d) Cerebrum

LONG ESSAY

1. Describe the structure of skin with diagrammatic representation
2. Describe the anatomy of structure in ear
3. Draw a diagram of eyeball and briefly explain its parts

SHORT ESSAY

4. Describe the layers of skin with diagram
5. Describe the extra ocular muscles
6. Explain the parts of internal ear.

SHORT ANSWERS

7. Tympanic membrane
8. Tongue
9. Dorsum of tongue
10. Anatomy of eye ball
11. Visual pathway.
12. Gustatory receptors
13. Auditory tube
14. Epidermis
15. Structure of eyeball
16. Structure of tongue
17. Structure of skin
18. Nasal sinuses.
19. Nerve supply to tongue.

ANSWER BRIEFLY

20. Layers Of Eye Ball
 21. Lens of eye
 22. Name The Ear Ossicles
 23. Any Four Functions Of Skin
 24. Name any four functions of skin
 25. Name the papillae of tongue
 26. List the auditory apparatus
 27. Name 3 parts of bony labyrinth
 28. Mention the functions of eyelid
 29. Give the nerve supply of extra-ocular muscles
 30. Mention 4 contents of middle ear
 31. List the accessory structures of eye
 32. List the disorders of the skin
 33. List the refractory errors
 34. Explain organ of corti
 35. List the functions of vestibular apparatus.
 36. Enlist the tastes and taste buds
- DIFFERENTIATE BETWEEN**
37. Superior and Inferior oblique muscle of eye
 38. Membranous labyrinth and bony labyrinth
 39. Inner ear and middle ear

ANSWER KEY UNIT VI SENSORY ORGANS

LONG ESSAY

11. Describe the structure of skin with diagrammatic representation

INTRODUCTION

SKIN

FUNCTIONS

STRUCTURE

Epidermis

Dermis-Blood vessels

-Lymph vessels

-Sensory nerve endings

-Sweat glands and their ducts

-Hair, erector pili and sebaceous glands

- Nails

VASCULAR, NERVOUS AND LYMPHATIC SUPPLY

APPLIED ANATOMY

12. Describe the anatomy of structure in ear

INTRODUCTION

REGIONS

- External ear
- Middle ear
- Internal ear
- Bony labyrinth
- ❖ Semicircular canals
- ❖ Vestibule
- ❖ Cochlea
- Membranous labyrinth
- ❖ Utricle
- ❖ Sacule

VASCULAR,NERVOUS AND LYMPHACTIC SUPPLY

APPLIED ANATOMY

13. Draw a diagram of eyeball and briefly explain its parts

INTRODUCTION

3 LAYERS

- ❖ FIBROUS TUNIC
- ❖ VASCULAR TUNIC
- ❖ RETINA

VASCULAR ,NERVOUS AND LYMPHACTIC SUPPLY

APPLIED ANATOMY

SHORT ESSAY

14. Describe the layers of skin with diagram

REFER QUE-10

15. Describe the extra ocular muscles

REFER UNIT III QUE-2

INTRODUCTION

STRUCTURE

- ORIGINS AND INSERTIONS
- BLOOD SUPPLY
- NERVE SUPPLY
- DEVELOPMENT
- FUNCTION
- MOVEMENTS
- MOVEMENT COORDINATION
- CLINICAL SIGNIFICANCE
- EXAMINATION

16. Explain the parts of internal ear.

INTRODUCTION

REFER QUE-2

SHORT ANSWERS

17. Tympanic membrane

INTRODUCTION

STRUCTURE

NERVE SUPPLY

VASCULAR SUPPLY

CLINICAL SIGNIFICANCE

18. Tongue

INTRODUCTION

EXTERNAL FEATURES(diagram)

PAPILLAE OF THE TONGUE

- Vallate or Circumvallate papillae

- Fungiform papillae
- Filiform papillae or conical papillae

DEVELOPMENT OF TONGUE

MUSCLES OF TONGUE

- INTRINSIC MUSCLE
- ❖ Superior longitudinal
- ❖ Inferior longitudinal
- ❖ Transverse
- ❖ Vertical
- EXTRINSIC MUSCLE
- ❖ Genioglossus
- ❖ Hypoglossus
- ❖ Styloglossus
- ❖ Palatoglossus

ARTERIAL SUPPLY OF TONGUE

LYMPHATIC DRAINAGE

NERVE SUPPLY

CLINICAL ANATOMY

19. Dorsum of tongue

INTRODUCTION

TONGUE(diagram)

PARTS- CURVED UPPER SURFACE(DORSUM)

-INFERIOR SURFACE

DORSUM

2 Parts

- ✓ Oral part
- ✓ Pharyngeal part(lymphoid part)

Papillae of the tongue

CLINICAL SIGNIFICANCE

20. Anatomy of eye ball

INTRODUCTION

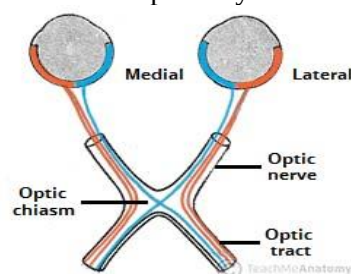
LAYERS OF EYEBALL(diagram)

- ❖ Fibrous tunic
- ❖ Vascular tunic
- ❖ Retina

VASCULAR,NERVOUS AND LYMPHACTIC SUPPLY

APPLIED ANATOMY

21. Visual pathway.



22. Gustatory receptors

INTRODUCTION

5 PRIMARY TASTES

- ❖ Sour
- ❖ Sweet
- ❖ Bitter
- ❖ Salty
- ❖ Umami (meaty)

ANATOMY OF TASTE BUDS AND PAPILLAE (diagram)

23. Auditory tube

REFER QUE 7

24. Epidermis

REFER QUE 1

25. Structure of eyeball

REFER QUE-10

26. Structure of tongue

REFER QUE -8

27. Structure of skin

REFER QUE-1

28. Nasal sinuses.

INTRODUCTION

TYPES OF SINUS

❖ MAXILLARY SINUSES

❖ FRONTAL SINUSES

❖ ETHMOIDAL SINUSES

❖ SPHENOID SINUSES

VASCULAR, NERVE AND LYMPHATIC SUPPLY

CLINICAL SIGNIFICANCE

29. Nerve supply to tongue.

Innervation of the tongue consists of motor fibers, special sensory fibers for taste, and general sensory fibers for sensation

- Motor supply for all intrinsic and extrinsic muscles of the tongue is supplied by efferent motor nerve fibers from the hypoglossal nerve (CN XII), with the exception of the palatoglossus, which is innervated by the vagus nerve (CN X).

Innervation of taste and sensation is different for the anterior and posterior part of the tongue because they are derived from different embryological structures (pharyngeal arch 1 and pharyngeal arches 3 and 4, respectively)

- Anterior two thirds of tongue (anterior to the vallate papillae):
 - Taste: chorda tympani branch of the facial nerve (CN VII) via special visceral afferent fibers
 - Sensation: lingual branch of the mandibular (V3) division of the trigeminal nerve (CN V) via general visceral afferent fibers
- Posterior one third of tongue:
 - Taste and sensation: glossopharyngeal nerve (CN IX) via a mixture of special and general visceral afferent fibers
 - Base of tongue
 - Taste and sensation: internal branch of the superior laryngeal nerve (itself a branch of the vagus nerve, CN X)

QUESTIONS

UNIT VII

MUSCULO SKELETAL SYSTEM

MCQ

1. Which of the following is the most superior vertebra of the spine?

Possible Answers:

The dens

The axis

The coccyx

The atlas

Correct answer:

The atlas

Explanation:

The most superior vertebra of the spine is the atlas, or C1. This bone sits on the axis, or C2, over a bony process called the dens. The coccyx, or "tailbone," is part of the sacrum.

2. How many cervical vertebrae are in the human spine?

Possible Answers:

12

5

6

7

Correct answer:

7

Explanation:

There are 7 cervical vertebrae, 12 thoracic vertebrae, and 5 lumbar vertebrae. The mnemonic for this is "breakfast at 7, lunch at 12, dinner at 5."

3. What bone forms the most posterior and inferior part of the skull?

Possible Answers:

The ethmoid bone

The occiput

The maxilla

The sphenoid bone

Correct answer:

The occiput

Explanation:

The occiput is the inferior and posterior bone of the skull. It wraps under the bottom of the skull and forms the base on which the head articulates with the spine. The ethmoid bone and sphenoid bone are housed primarily within the skull, and the maxilla forms the upper jaw and part of the orbit.

4. What is the name for the bony protuberance(s) at which the occipital bone articulates with the atlas (C1)?

Possible Answers:

The external occipital protuberance

The external occipital crest

The occipital condyles

The inion

Correct answer:

The occipital condyles

Explanation:

The bony protuberance(s) at which the occipital bone articulates with the atlas (C1) are the occipital condyles. The external occipital protuberance, external occipital crest, and inion are all part of the occipital bone and serve as the attachment points for various muscles and ligaments.

5. A patient presents to the ER complaining of "pain in his heart." When the nurse asks him to point to the area that he feels the pain, he places his finger on the bony protuberance just below his rib cage. Seeing this, the nurse has reason to suspect that his pain may be gastrointestinal or musculoskeletal in origin as well as cardiac. What bony protuberance is this patient likely touching?

Possible Answers:

The costal cartilage
The manubrium
The xiphoid process
The body of the sternum
Correct answer:

The xiphoid process

Explanation:

The costal cartilage of the ribcage is joined to the sternum, which is made up of three bony structures: the most superior is the manubrium, which articulates not just with the costal cartilage of the first rib but also with the clavicles. The body of the sternum is inferior to the manubrium, and articulates with the second through fifth ribs. The xiphoid process is a small protuberance just below the body of the sternum, at the point where the bottom of the ribcage meets the sternum. It is an attachment point for several muscles, including the diaphragm, the rectus abdominis, and the transverse abdominis. This is a common location for referred pain from gallbladder disease, GERD, or pain due to musculoskeletal injury or irritation.

6. Which of the following is the name of the bony landmark that forms the ridge of the brow?

Possible Answers:

The superciliary arch
The glabella
The coronoid process
The orbit

Correct answer:

The superciliary arch

Explanation:

The bony ridge of the brow is formed by the superciliary processes (superciliary means *super*, or above, *cilia*, the lens of the eye). The orbit refers to the bony socket in which the eye sits. The glabella is the small depression between the eyebrows where the two superciliary arches meet. The coronoid process is part of the mandible, or jawbone, and unrelated to the brow.

7. What structures of the skull form the cheek bones?

Possible Answers:

The mental protuberance
The styloid processes
The pterion
The zygomatic arches

Correct answer:

The zygomatic arches

Explanation:

The cheek bones are formed by the zygomatic arches, which are part of the temporal bones of the face. The mental protuberance forms the chin, the styloid processes are just under the ears and serve as an attachment point for several muscles of the mouth and throat, and the pterion is the suture where the frontal, parietal, temporal, and sphenoid bones come together.

8. Which of the following is the name for the foramen that forms the ear canal?

Possible Answers:

The foramen ovale
The external auditory meatus

The internal auditory meatus
The foramen rotundum

Correct answer:

The external auditory meatus

Explanation:

The foramen that forms the ear canal is called the external auditory meatus. This travels through the temporal bone and joins the inner ear to the outer ear. The internal auditory meatus passes through the temporal bone between the posterior cranial fossa and the inner ear, and is the track through which the vestibulocochlear nerve, the facial nerve, and the labyrinthine artery travel from the inner ear toward the CNS. The foramen ovale and foramen rotundum are both openings in the sphenoid bones and have to connection to the ear.

9. What is the name of the large opening through which the spinal cord exits the skull?

Possible Answers:

The foramen spinosum
The jugular foramen
The foramen magnum
The foramen cecum

Correct answer:

The foramen magnum

LONG ESSAY

1. Classify synovial joint with examples. Describe typical synovial joint under the following headings: _ parts _ movements _ blood supply _ applied anatomy
2. Classify bones and describe the different types. Draw and label both the macroscopic and microscopic structure of long bone
3. Classify the bones with examples. Describe the histology of compact bone.

SHORT ESSAYS

4. Describe shoulder joint in detail
5. Describe knee joint under following headings
 - a) type and articular surfaces
 - b) arterial supply
6. What is patella? Mention its importance
7. Classify the skeletal system. explain in detail about the skull

Adult human skelton consist of 206 bones

SHORT NOTE

8. Maxillae
9. Mandible
10. Scapula
11. Femur
12. Bones of adult skeletal system
13. Bones of cranium
14. Facial bones
15. Vertebral column
16. Bones in thorax
17. Clavicle
18. Humerus
19. Skelton of hand
20. Pelvic girdle

21. Haversian system
22. Arterial supply of hip joint
23. Bones of the lower limb
24. Draw a labeled diagram of synovial joint.
- ANSWER BRIEFLY**
25. Name the carpal bones
26. Classification of joints
27. Name the bones forming the shoulder joint
28. Name the bones forming elbow joint
29. List the tarsal bones
30. Name any 4 cranial bones
31. Name the parts of sternum
32. List 4 features of lumbar vertebrae
33. Muscles causing pronation and supination
34. Name the parts of hip bone.
35. Parts of developing long bones.
36. List the bones of the upper limb.

DIFFERENTIATE BETWEEN

37. Ulna and radius
38. Fetal skull and adult skull
39. Tibia and fibula
40. Femur and humerus
41. Axial and appendicular skeletal system
42. Maxilla and mandible
43. Knee joint and elbow joint
44. Metacarpal and metatarsal bones

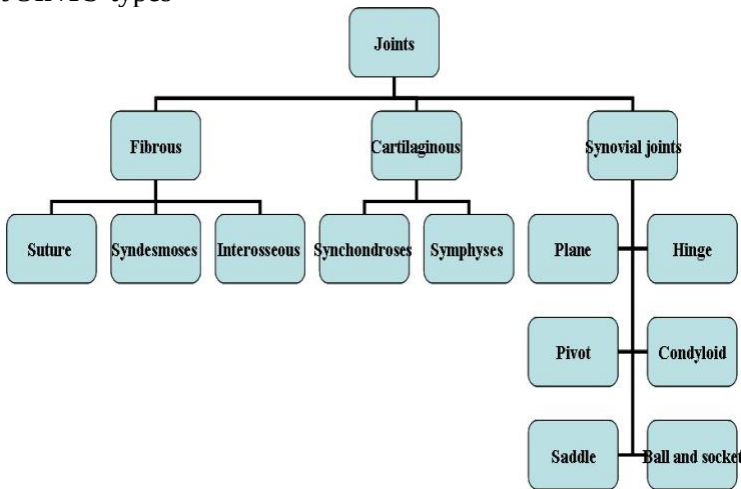
**ANSWER KEY
UNIT VII
SKELETAL SYSTEM**

LONG ESSAY

1. Classify synovial joint with examples. Describe typical synovial joint under the following headings: _ parts _ movements _ blood supply _ applied anatomy

INTRODUCTION

JOINTS-types

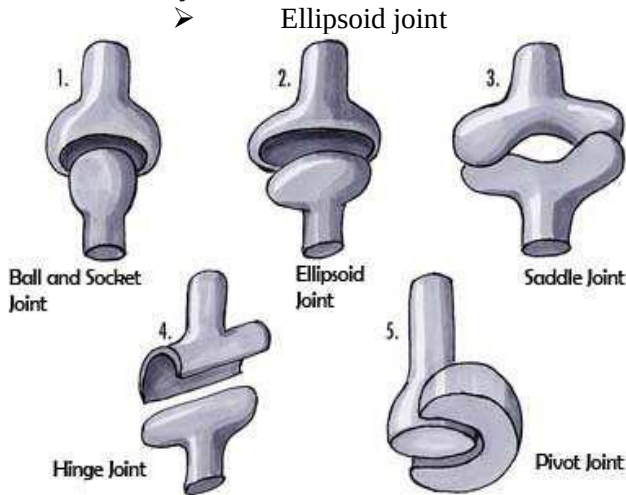


SYNOVIAL JOINT

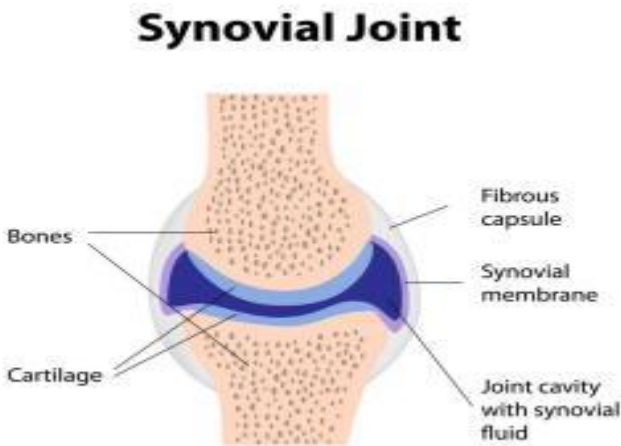
CLASSIFICATION OF SYNOVIAL JOINT

- Planar joint
- Hinge joint
- Pivot joint

- Condylloid joint
- Saddle joint
- Ball and socket joint



STRUCTURE OF A TYPICAL SYNOVIAL JOINT



FUNCTION

The movements possible with synovial joints are

- abduction: movement away from the mid-line of the body
- adduction: movement toward the mid-line of the body
- extension: straightening limbs at a joint
- flexion: bending the limbs at a joint
- rotation: a circular movement around a fixed point

BLOOD SUPPLY

Periarticular arterial plexuses– circulus articularis vasculosus

- Articular cartilage: avascular
- Fibrous capsule & ligaments: poor blood supply
- Synovial membrane: rich blood supply

NERVE SUPPLY

- HILTON'S LAW

The nerves supplying the joint capsule also supply the muscles regulating the movement of the jt & skin over the joint.

APPLIED ANATOMY

Clinical significance

The *joint space* equals the distance between the involved bones of the joint.

A *joint space narrowing* is a sign of either (or both) [osteoarthritis](#) and inflammatory degeneration

The normal joint space is at least 2 mm in the [hip](#) (at the superior [acetabulum](#)), at least 3 mm in the [knee](#), and 4–5 mm in the [shoulder joint](#).

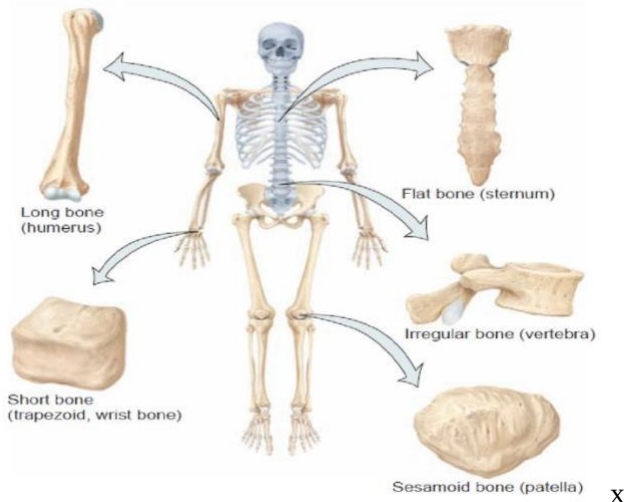
For the [temporomandibular joint](#), a joint space of between 1.5 and 4 mm is regarded as normal. Joint space narrowing is therefore a component of several [radiographic classifications of osteoarthritis](#).

2. Classify bones and describe the different types. Draw and label both the macroscopic and microscopic structure of long bone

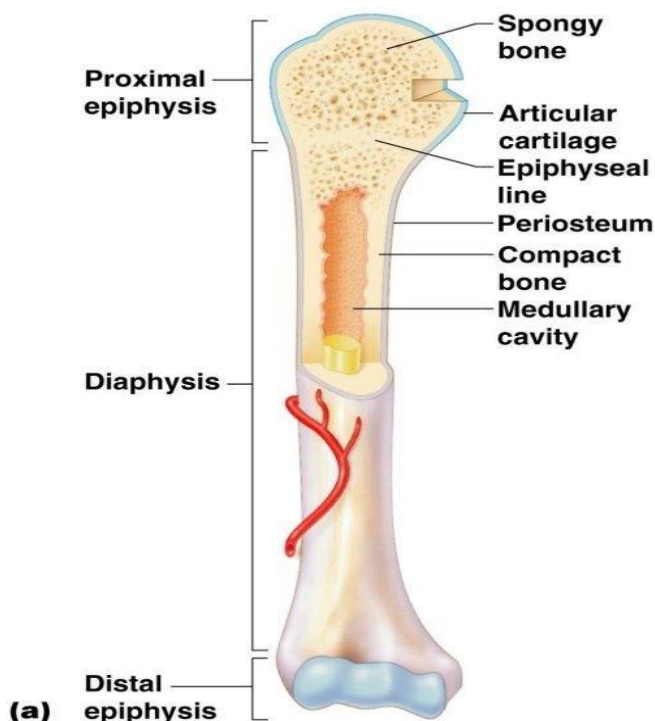
INTRODUCTION

BONES

TYPES OF BONES

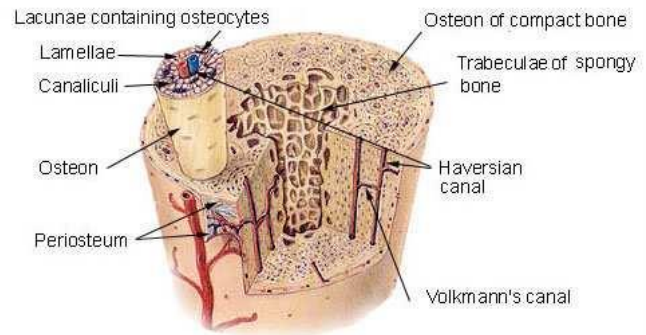


MACROSCOPIC STRUCTURE OF LONG BONE



MICROSCOPIC STRUCTURE OF BONE

Compact Bone & Spongy (Cancellous Bone)



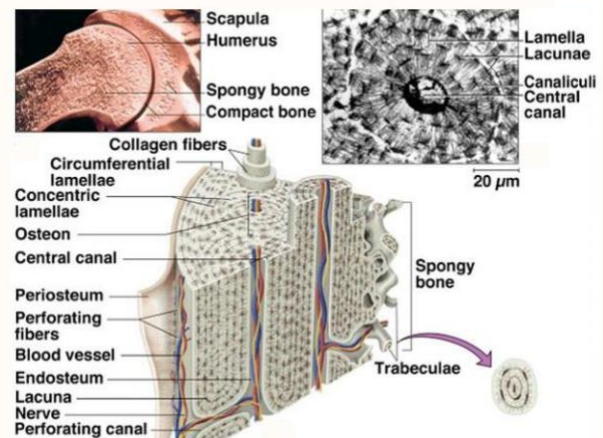
3. Classify the bones with examples. Describe the histology of compact bone.

INTRODUCTION

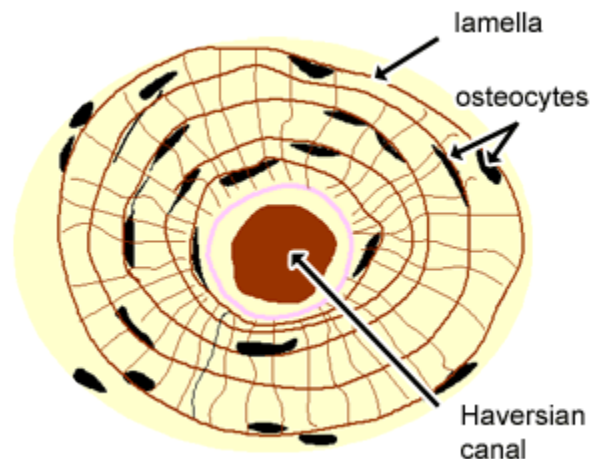
BONES

HISTOLOGY OF COMPACT BONE

Histology of Compact Bone



HAVERSIAN SYSTEM



SHORT ESSAYS

4. Describe shoulder joint in detail

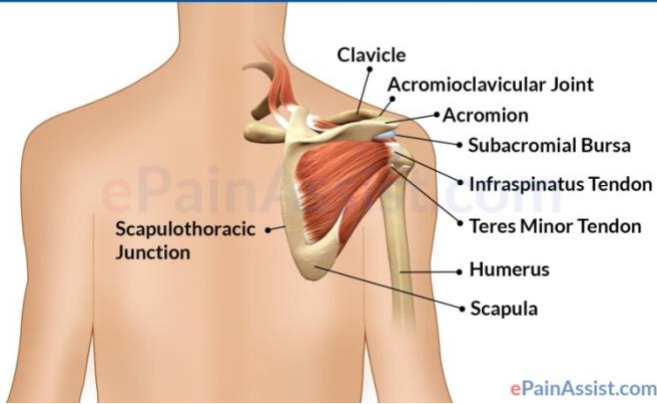
INTRODUCTION

JOINT

TYPES

SHOULDER JOINT

Shoulder Joint Anatomy

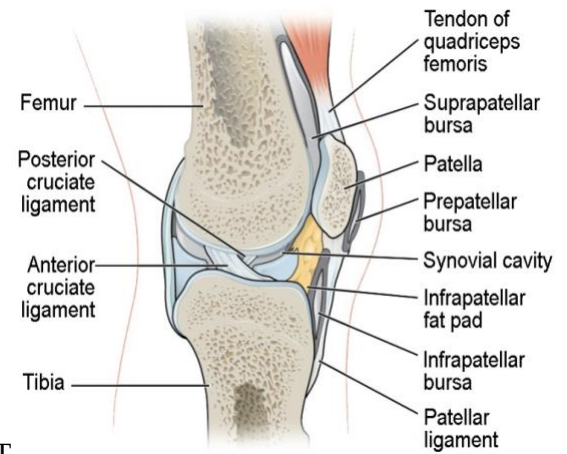


- The shoulder complex is made up of three bones, which are connected by muscles, ligaments, and tendons.
- The large bone in the upper arm is called the **humerus**.
- The shoulder blade is called the **scapula** and the collarbone is called the **clavicle**.
- The top of the humerus is shaped like a ball.
- This ball sits in a socket on the end of the scapula.
- The ball is called the **head of the humerus** and the socket is called the **glenoid fossa**, hence the term "**glenohumeral**" joint.
- The glenoid fossa has a rim of tissue around it called the **glenoid labrum**.
- The glenoid labrum makes the glenoid fossa deeper.
- The glenohumeral joint is the most mobile joint in the body.
- Articular cartilage is a smooth shiny material that covers the humeral head and the glenoid fossa of the glenohumeral joint.
- **Articular cartilage** allows these bones to slide easily over each other as the arm moves.
- The glenohumeral joint is just one of the joints in the shoulder complex.
- The other two joints are the **sternoclavicular joint** and the **acromioclavicular joint**.
- The sternoclavicular joint allows a small amount of movement to occur between the inner (medial) part of the clavicle and the breastbone (sternum).
- The acromioclavicular joint allows a small amount of movement to occur between the outer (lateral) part of clavicle and a projection on the top of the scapula called the acromion process.
- The scapula sits on the back of the ribs and moves as the arm moves.

5. Describe knee joint under following headings

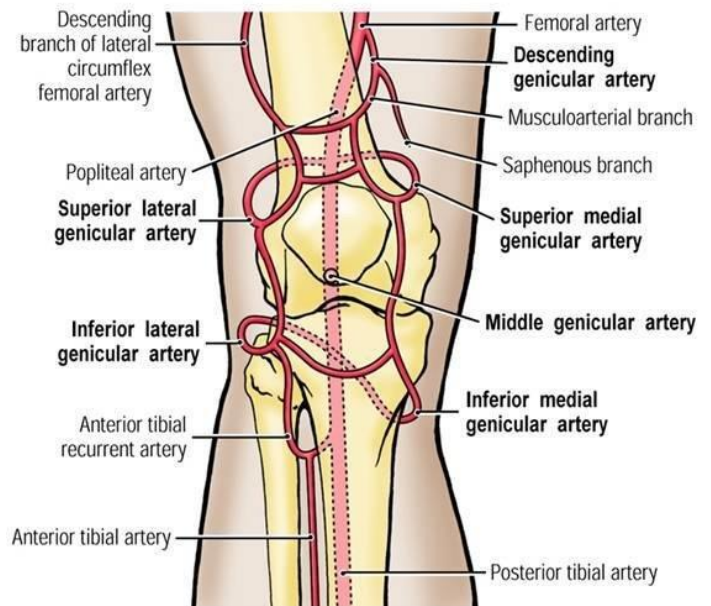
- a) type and articular surfaces
- b) arterial supply

INTRODUCTION



KNEE JOINT TYPES

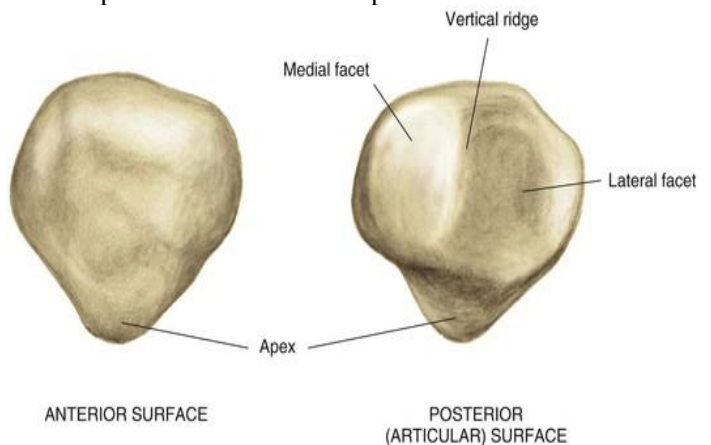
ARTERIAL SUPPLY OF KNEE JOINT



A. Anterior View

Copyright © 2009 Wolters Kluwer Health | Lippincott Williams & Wilkins

6. What is patella? Mention its importance



- ❖ Located on the anterior of the distal **femur**, the **patella** is constructed of a triangular sesamoid **bone** of significant size.
- ❖ Tension and stress on the patella tendon determine the response and action of the patella.

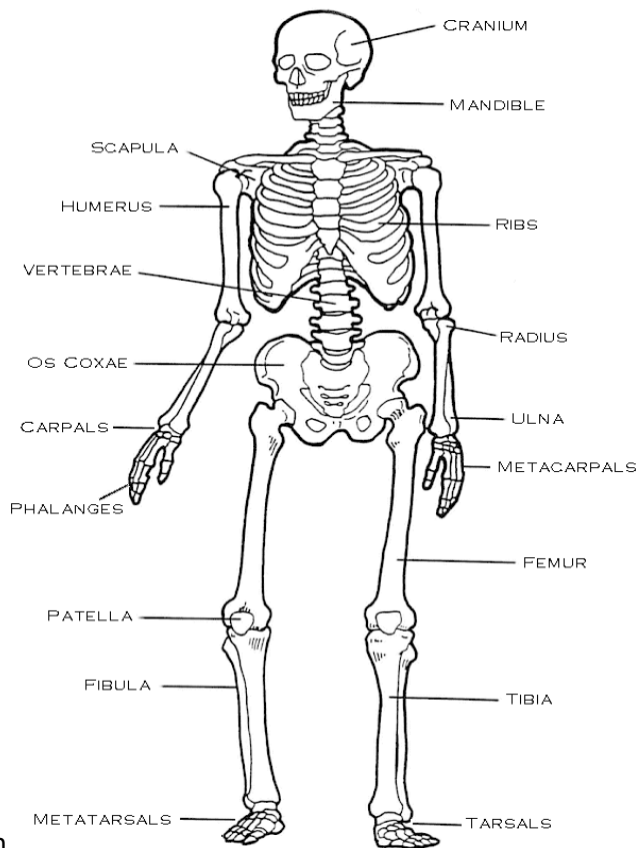
- ❖ The base is rather broad and its inferior apex finalizes in a near point.
- ❖ The medial and lateral femoral condyles interface with the articular facets of the same surface in order to provoke response.
- ❖ The quadriceps relies on the patella for leverage concerning the femoris muscle when extending into a straight [leg](#) position.

FUNCTIONS

- ❖ The primary responsibility of the patella includes protection of the knee joint
- ❖ It also serves to help in strengthening the patella tendon.

INTRODUCTION

ADULT HUMAN SKELTON



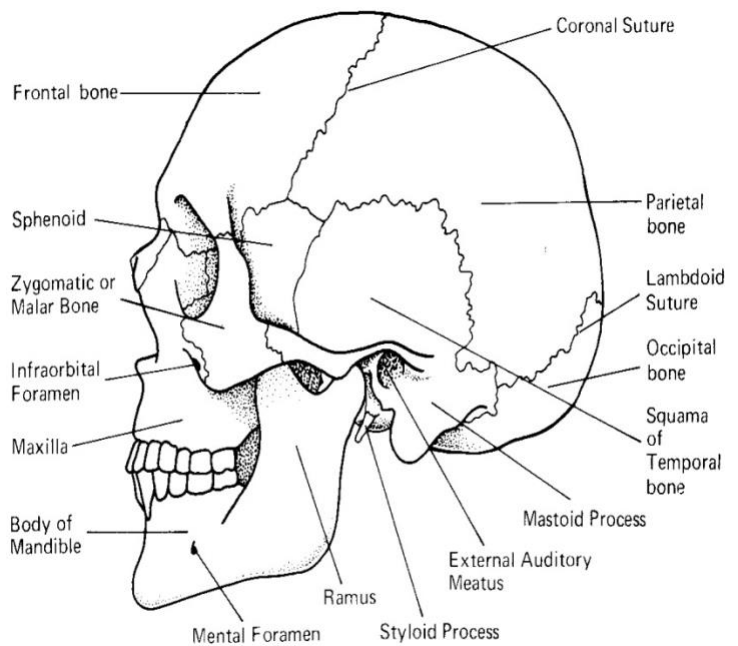
on

Adult human skelton consist of 206 bones

CLASSIFICATION

- AXIAL SKELTAL SYSTEM-skull,vertebral column and ribs.
- APPENDICULAR SKELTAL SYSTEM-shoulder girdle and pelvic girdle

SKULL



The skull—lateral view.

BONES OF THE SKULL

28 BONES

2 parts

❖ Cranium

Bones

- ✓ 1 Frontal bone
- ✓ 2 Parietal bones
- ✓ 2 Temporal bones
- ✓ 1 Occipital bones
- ✓ 1 Sphenoid bone
- ✓ 1 Ethmoid bone
- ✓ 3 Paired ear ossicles (malleus ,Incus and Stapes)
- ❖ Face- formed by 14 bones
- ✓ 2 Zygomatic or cheek bones
- ✓ 1 Maxilla
- ✓ 2 Nasal bones
- ✓ 2 Lacrimal bones
- ✓ 1 Vomer
- ✓ 2 Palatine bones
- ✓ 2 inferior Conchae
- ✓ 1 Mandible

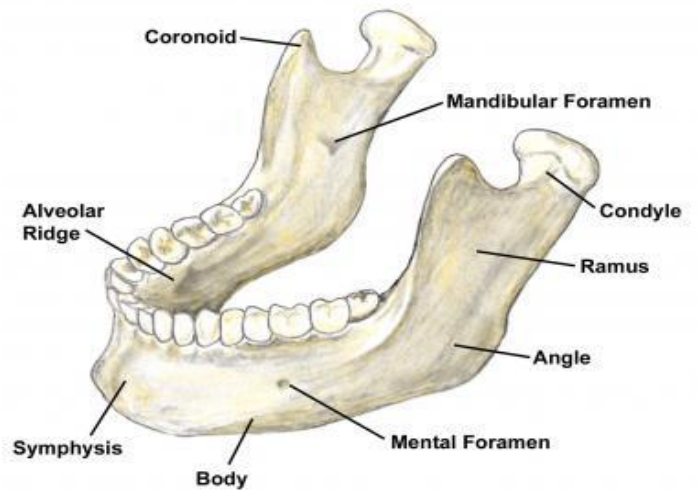
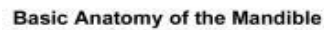
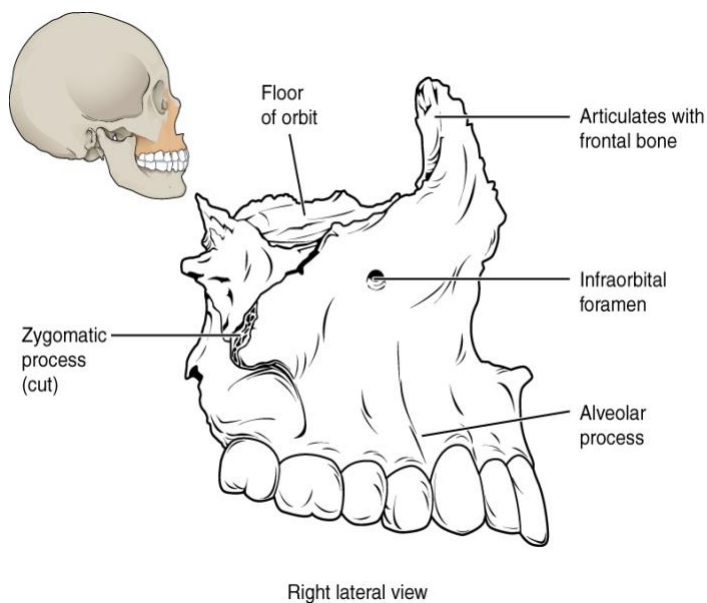
SKULL JOINTS

ANATOMICAL POSITION OF SKULL

PECULIARITIES OF SKULL BONES

SHORT NOTE

8. Maxillae



- ❖ The **maxillary bone** , often referred to simply as the maxilla (plural = maxillae), is one of a pair that together form the upper jaw, much of the hard palate, the medial floor of the orbit, and the lateral base of the nose .
- ❖ The curved, inferior margin of the maxillary bone that forms the upper jaw and contains the upper teeth is the **alveolar process of the maxilla**.
- ❖ Each tooth is anchored into a deep socket called an alveolus.
- ❖ On the anterior maxilla, just below the orbit, is the infraorbital foramen.
- ❖ This is the point of exit for a sensory nerve that supplies the nose, upper lip, and anterior cheek.
- ❖ On the inferior skull, the **palatine process** from each maxillary bone can be seen joining together at the midline to form the anterior three-quarters of the hard palate
- ❖ The **hard palate** is the bony plate that forms the roof of the mouth and floor of the nasal cavity, separating the oral and nasal cavities.

9. Mandible INTRODUCTION

The **mandible, lower jaw** or **jawbone** is the largest, strongest and lowest bone in the human face. It forms the lower jaw and holds the lower teeth in place. The mandible sits beneath the maxilla. It is the only movable bone of the skull (discounting the ossicles of the middle ear).

DIAGRAM

FORAMINA AND RELATIONS TO NERVES AND VESSELS

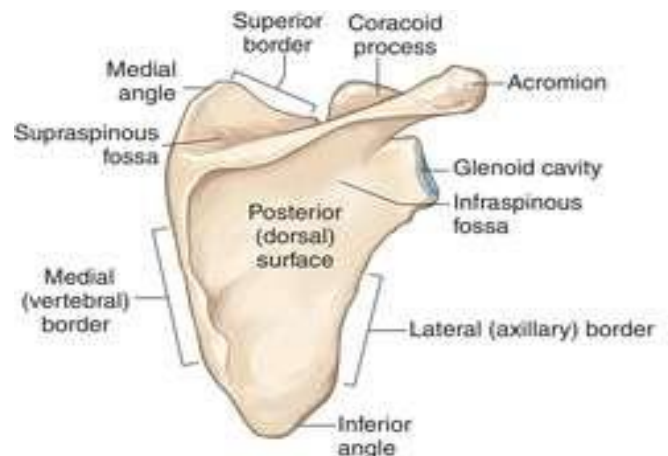
OSSIFICATION

SSTRUCTURES RELATED TO MANDIBLE

10. Scapula

INTRODUCTION

SURFACES



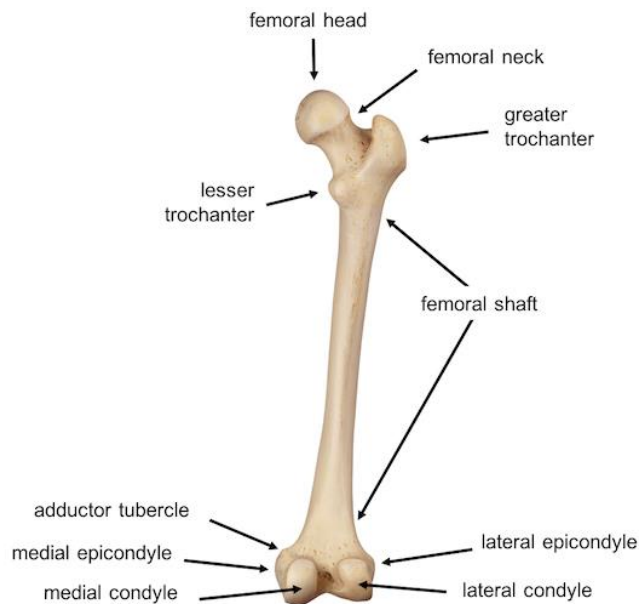
ANGLES
BORDERS
DEVELOPMENT
FUNCTION
CLINICAL SIGNIFICANCE

- Scapular fractures
- Impingement syndrome

11. Femur

INTRODUCTION

SIDE DETERMINATION



ANATOMICAL POSITION

Upper end comprises of head, neck and two trochanters ,greater and lesser trochanters.

- Head
- Shaft

BLOOD SUPPLY

APPLIED ANATOMY

12. Bones of adult skeletal system

INTRODUCTION

The **human skeleton** is the internal framework of the body. It is composed of around 270 [bones](#) at birth – this total decreases to around 206 bones by adulthood after some bones get fused together.



• SKELETAL DIVISIONS

- AXIAL SKELETON
- APPENDICULAR SKELETON

• FUNCTIONS

- SUPPORT
- MOVEMENT
- PROTECTION
- BLOOD CELL PRODUCTION
- STORAGE
- ENDOCRINE REGULATION

• GENDER DIFFERENCES

- SKULL
- DENTITION
- LONG BONES
- PELVIS

• BLOOD SUPPLY AND NERVOUS SUPPLY

• CLINICAL SIGNIFICANCE

- ARTHRITIS
- OSTEOPOROSIS

13. Bones of cranium

REFER QUES-7

14. Facial bones

REFER QUE-7

15. Vertebral column

INTRODUCTION

STRUCTURE

- ❖ Body
- ❖ Pedicles
- ❖ Lamina
- ❖ Vertebral foramen
- ❖ Spinous process
- ❖ Transverse process
- ❖ Superior articular process
- ❖ Inferior articular process
- ❖ Vertebral notch
- REGIONAL VARIATION
 - 7 CERVICAL VERTEBRAE
 - 12 THORACIC VERTEBRAE
 - 5 LUMBAR VERTEBRAE
 - 5 SACRUM
 - 5 COCCYX
- DEVELOPMENT
- FUNCTION
- BLOOD AND NERVOUS SUPPLY
- CLINICAL SIGNIFICANCE

16. Bones in thorax

INTRODUCTION

The bones of the thorax, called the "thoracic skeleton" is a component of the [axial skeleton](#).

It consists of the [RIBS](#) and [STERNUM](#).

The ribs of the thorax are numbered in ascending order from 1-12. 11 & 12 are known as floating ribs because they have no anterior attachment point in particular the cartilage attached to the sternum, as 1-7 are, and therefore are termed "floating".

Whereas ribs 8-10 are termed false ribs as their costal cartilage articulates with the costal cartilage of the rib above.

FEATURES OF TYPICAL RIB

2 END-Anterior end

-Posterior end

:-head

:-tubercle

ATTACHMENTS and BORDERS OF EACH RIBS

BLOOD,NERVOUS SUPPLY

STERNUM

Flat bone,17 cm long.

PARTS

- Manubrium
- Body
- Xiphoid process

ATTACHMENTS

ANATOMICAL LOCATION

BLOOD,NERVOUS SUPPLY

CLINICAL ANATOMY

17. Clavicle

INTRODUCTION

- STRUCTURE
 - MEDIAL END
 - LATERAL END
 - SHAFT

- MEDIAL TWO-THIRDS OF THE SHAFT
- LATERAL THIRD OF THE SHAFT
- DEVELOPMENT
- VARIATION
- FUNCTIONS
- MUSCLES
- CLINICAL SIGNIFICANCE
- COLLARBONE FRACTURE

18. Humerus

INTRODUCTION

The **humerus** ([/'hju:mərəs/](#), plural: *humeri*) is a long bone in the arm or forelimb that runs from the shoulder to the elbow.

CONNECTIONS

It connects the scapula and the two bones of the lower arm, the radius and ulna.

PARTS

It consists of three sections.

The **humeral upper extremity** consists of a rounded head, a narrow neck, and two short processes (tubercles, sometimes called tuberosities).

The body is cylindrical in its upper portion, and more prismatic below.

The lower extremity consists of 2 epicondyles, 2 processes (trochlea & capitulum), and 3 fossae (radial fossa, coronoid fossa, and olecranon fossa).

- ☐ STRUCTURE
- DIAGRAM
- ARTICULATIONS
- BLOOD SUPPLY AND NERVES
- FUNCTION
- MUSCULAR ATTACHMENT
- CLINICAL SIGNIFICANCE

19. Skelton of hand

INTRODUCTION

CLAVICLE

- ❖ Side determination
- ❖ Peculiarities
- ❖ Muscle attachment
- ❖ Border
- ❖ Processes

HUMERUS

- ❖ GENERAL FEATURES
- ❖ SIDE DETERMINATION
- ❖ ATTACHMENTS
- ❖ PARTS-HEAD

- GREATER TUBERCLE
- LESSER TUBERCLE
- SHAFT OF HUMERUS
- LOWER END

RADIUS

- ❖ PARTS –upper end,shaft and lower end
- ❖ BORDER
- ❖ GENERAL FEATURES
- ❖ ATTACHMENTS

ULNA

- ❖ PARTS-upper end, shaft and lower end.
- ❖ BORDERS
- ❖ PROCESSES

CARPEL BONES

- ❖ 8 CARPEL BONES
- ❖ 2 ROWS
- Proximal row:-Scaphoid, Lunate, Triquetral and Pisiform bones.
- Distal row:-Trapezium ,trapesoid, Capitate and hamate bone
- ❖ ATTACHMENTS

METACARPEL BONES- 5 Minature bones

- ❖ Charactersics of individual metacarpal bones.

- ❖ Attachments

PHALANGES-14 bones.3 of each finger,2 for thumb.

- Parts-Base ,a shaft, and ahead.
- Attachments

BLOOD AND NERVOUS SUPPLY

CLINICAL SIGNIFICANCE.

20. Pelvic girdle

INTRODUCTION

4 BONES

- ❖ 2 hip bones
- ❖ Sacrum
- ❖ Coccyx

PELVIC PLANES-False pelvis

-True pelvis

LESSER PELVIS

- Pelvic planes
- Bony walls
- Ligaments and membranes
- Muscles
- Pelvic inlet
- Pelvic outlet
- Pelvic floor
- Pelvic cavity
- Anatomical position
- Structures crossing pelvic brim

CLINICAL SIGNIFICANCE.

21. Haversian system

REFER QUES 2

22. Arterial supply of hip joint

INTRODUCTION

Blood supply

The hip joint is supplied with blood from the medial circumflex femoral and lateral circumflex femoral arteries, which are both usually branches of the deep artery of the thigh (profunda femoris), but there are numerous variations and one or both may also arise directly from the femoral artery. There is also a small contribution from the foveal artery, a small vessel in the ligament of the head of the femur which is a branch of the posterior division of the obturator artery, which becomes important to avoid avascular necrosis of the head of the femur when the blood supply from the medial and lateral circumflex arteries are disrupted (e.g. through fracture of the neck of the femur along their course).

The hip has two anatomically important [anastomoses](#), the [cruciate](#) and the [trochanteric anastomoses](#), the latter of which provides most of the blood to the head of the femur. These anastomoses exist between the femoral artery or profunda femoris and the gluteal vessels

23. Bones of the lower limb

INTRODUCTION

❖ HIP BONE

❖ FEMUR

❖ PATELLA

❖ TIBIA

❖ FIBULA

❖ BONES OF FOOT

➤ Talus

➤ Calcaneus

➤ Navicular bone

➤ Cuneiform bones

• Medial cuneiform

• Intermediate cuneiform

• Lateral cuneiform

• Cuboid

• Metatarsus-5 bones

• Phalanges

❖ FEATURES

❖ ATTACHMENTS

❖ BORDERS

❖ VASCULAR AND NERVOUS SUPPLY

❖ CLINICAL SIGNIFICANCE

24. Draw a labeled diagram of synovial joint.

REFER QUES-1

QUESTIONS UNIT VII(CONTD) MUSCULAR SYSTEM

LONG ESSAY

1. Explain the structure of three types of muscular tissue
2. Explain the distribution of muscles in the body.
Describe the muscles of face neck ,trunk ,abdomen and pelvic floor
3. Explain in detail about muscles of upper and lower limbs

SHORT ESSAY

4. Describe the insertion,action and innervations of extrinsic eye muscles
5. Describe the insertion,action and innervations of extrinsic tongue muscles
6. Describe the insertion,action and innervations of respiratory muscles
7. Explain the properties and narrate the types of muscle tissue
8. Name muscles of gluteal region. Describe gluteus maximus in detail
9. Describe rectus sheath and rectus abdominal muscle

SHORT ANSWERS

10. Muscular dystrophy
11. Fibromyalgia

12. Gluteus maximus
13. Myasthenia gravis
14. Femoral hernia
15. Muscles of mastication
16. Sternocleidomastoid muscle

ANSWER BRIEFLY

17. Name the muscles of mastication
18. Name 2 pectoral muscles and give its nerve supply
19. Name 2 muscles of forearm
20. Name the muscles of back of the thigh
21. Name the contents of femoral triangle
22. Name 4 hamstring muscles
23. Name 4 parts of quadriceps femoris muscles
24. Describe 4 recti muscles
25. Name 4 muscles in the gluteal region
26. Mention the contents of cubital fossa
27. Mention the contents of popliteal fossa
28. List the type of muscles.
29. Enlist the abnormal contractions of skeletal muscle
30. Deltoid muscle
31. Muscles used for intramuscular injections.
32. Trapezius muscle
33. Gluteus maximus
34. Origin, insertion and nerve supply of deltoid muscle.
35. Biceps brachii muscle
36. Pelvic Floor muscle

DIFFERENTIATE BETWEEN

37. Skeletal and smooth muscle
38. Cardiac and skeletal muscle
39. Actin and myosin

ANSWER KEY UNIT VII (CONTD) MUSCULAR SYSTEM

LONG ESSAY

1. Explain the structure of three types of muscular tissue

INTRODUCTION

MUSCLE TISSUE

TYPES

- Skeletal
- Cardiac
- Smooth

SKELTAL MUSCLE

Microscopic anatomy

Structure of skeletal muscle(with diagram)

Characteristics and functions

- ❖ Sarcolemma ,transverse tubules and sarcoplasm
- ❖ Filaments and sarcomere
- ❖ Muscle proteins
- ❖ Contractile proteins
- ❖ Regulatory proteins
- ❖ Structural proteins

CARDIAC MUSCLE

STRUCTURE(with diagram)

- ❖ Intercalated disc
- ❖ Nucleus

- ❖ Faint straitions
SMOOTH MUSCLE
- ❖ Features
- ❖ Types
 - Visceral Smooth muscle
 - Multi-unit smooth muscle
- ❖ Microscopic antomy
 - Myofibrils
 - Nucleus
 - Sarcoplasm
 - Plasma membrane

CLINICAL SIGNIFICANCE

2. Explain the distribution of muscles in the body.
Describe the muscles of face neck ,trunk ,abdomen and pelvic floor

INTRODUCTION

MUSCLES OF FACE

- Mouth –Buccinator, orbicularis oris
- Eye- Orbicularis oculi
Extrinsic muscle –lateral rectus, medial rectus, superior oblique, inferior oblique, superior rectus, inferior rectus
- Nose- Nasalis
- Scalp- Frontalis
- Muscle of mastication-Masseter,Temporalis
- Tongue-Lingual muscles
(Genioglossus,hypoglossus)

(Mention each of its movements and nerve supply)

MUSCLES OF NECK

- Sternocleidomastoid muscle
- Trapezius

MUSCLES OF TRUNK

- Trapezius
- Deltoid
- Latissimus dorsi
- Erector spinae

MUSCLES OF THORAX

- External intercostals
- Internal intercostals
- Pectoris major and minor
- Diaphragm

MUSCLES OF ABDOMEN

- External oblique
- Internal oblique
- Rectus abdominis
- Transverse abdominis

MUSCLES OF PELVIC FLOOR

- Levator ani
- Illiococcygeal

(Mention each of its movements and nerve supply)

3. Explain in detail about muscles of upper and lower limbs

INTRODUCTION

MUSCLES OF UPPER LIMB

- Deltoid
- Pectoralis major
- Coracobrachialis
- Biceps

- Brachialis
- Triceps
- Brachioradialis
- Pronator quadrates
- Pronator teres
- Supinator
- Flexor carpi radialis
- Flexor carpi ulnaris
- Extension carpi radialis longue and bravis
- Extensor carpi ulnaris
- Palmaris longus
- Extensor digitorum
- Finger muscles

MUSCLES OF HIP AND LOWER LIMB

- Psoas
- Iliacus
- Quadriceps femoris
- Obturators
- Gluteals
- Sartorius
- Adductor group
- Hamstrings
- Gastrocnemius
- Anterior tibialis
- Soleus

SHORT ESSAY

4. Describe the insertion,action and innervations of extrinsic eye muscles

REFER QUES 3

5. Describe the insertion,action and innervations of extrinsic tongue muscles

INTRODUCTION

TYPES OF EXTRINSIC EYE MUSCLES

- ❖ Superior rectus
- ❖ Inferior rectus
- ❖ Lateral rectus
- ❖ Medial rectus
- ❖ Superior oblique
- ❖ Inferior oblique

INSERTION

ACTION

INNERVATION

CLINICAL SIGNIFICANCE

6. Describe the insertion,action and innervations of respiratory muscles

INTRODUCTION

RESPIRATORY MUSCLES

- ❖ Diaphragm
- ❖ External intercostal muscles
- ❖ Internal intercostals muscles

INSERTION

ACTION

INNERVATION

CLINICAL SIGNIFICANCE

7. Explain the properties and narrate the types of muscle tissue

INTRODUCTION

MUSCLE TISSUE

PROPERTIES

- ❖ Electrical excitability
- ❖ Contractility
- ❖ Extensibility
- ❖ Elasticity

TYPES

- ❖ Skeletal
- ❖ Smooth
- ❖ Cardiac

8. Name muscles of gluteal region. Describe gluteus maximus in detail

INTRODUCTION

TYPES OF MUSCLES IN GLUTEAL REGION

- Gluteus maximus
- Gluteus medius
- Gluteus minimus

GLUTEUS MAXIMUS

STRUCTURE

ORIGIN

NERVES OF GLUTEAL REGION

ARTERIES OF GLUTEAL REGION

Superior gluteal artery

9. Describe rectus sheath and rectus abdominal muscle

INTRODUCTION

DEFINITION OF RECTUS SHEATH

This is an aponeurotic sheath covering the rectus abdominis. It has 2 walls

i. Anterior

ii. Posterior

FEATURES

FORMATION OF WALLS

CONTENTS

- ✓ Muscles (rectus abdominis, pyramidalis)
- ✓ Arteries (superior and inferior epigastric artery)
- ✓ Veins (superior and inferior epigastric vein)
- ✓ Nerves (terminal parts of lower six thoracic vertebrae)

FUNCTIONS

RECTUS ABDOMINIS MUSCLE

- ❖ Origin
- ❖ Insertion
- ❖ Blood supply

SHORT ANSWERS

2. Muscular dystrophy

INTRODUCTION

DEFINITION-It refers to a group of inherited muscle destroying diseases causing progressive degeneration of skeletal muscle fibres

INCIDENCE

CAUSES

CLINICAL FEATURES

DIAGNOSTIC STUDIES

MANAGEMENT

3. Fibromyalgia

DEFINITION- Fibromyalgia is a painful, nonarticular rheumatic disorder that affects fibrous connective tissue.

CAUSES

CLINICAL FEATURES

(Striking sign)

Other symptoms and signs

DIAGNOSTIC STUDIES

MANAGEMENT

4. Gluteus maximus

REFER QUE 8

5. Myasthenia gravis

DEFINITION- Myasthenia Gravis is an autoimmune disorder which causes weakness and fatigue of skeletal muscles caused by antibodies directed against ACH receptors.

CAUSES

SYMPTOMS

DIAGNOSTIC SYMPTOMS

MANAGEMENT

6. Femoral hernia

DEFINITION

CAUSES

SYMPTOMS

DIAGNOSTIC SYMPTOMS

MANAGEMENT

7. Muscles of mastication

TYPES

- ❖ Masseter
- ❖ Temporalis

MOVEMENTS

BLOOD, NERVE, LYMPHATIC SUPPLY

8. Sternocleidomastoid muscle

INTRODUCTION

FUNCTIONS

MOVEMENTS

BLOOD SUPPLY

NERVE SUPPLY

LYMPHATIC SUPPLY

QUESTIONS

UNIT VIII

RENAL SYSTEM

MCQ

1. An organ or structure that is not a component of the urinary system is the:
 - A. Urethra.
 - B. Urinary bladder.
 - C. Ureter.
 - D. Adrenal gland.
 - E. Kidney.
2. The kidneys are:
 - A. Help regulate blood volume.
 - B. Help control blood pressure.
 - C. Help control PH.
 - D. All of the above are correct.
3. All of the following belong to the urinary system except:
 - A. Urethra.
 - B. Ureter.

- C. Bladder.
D. Prostate.
4. The urinary system is the principal system responsible for:
A. Removal of carbon dioxide.
B. Water and electrolyte balance.
C. Excretion of toxic nitrogenous compounds.
D. A and B
E. B and C
F. A, B, and C
5. The structure that connects a kidney to the urinary bladder is the
A. Ureter.
B. Urethra.
C. Renal pelvis.
D. Collecting duct.
6. What portion of the nephron extends into the medulla?
A. Nephron loop.
B. Proximal convoluted tubule.
C. Distal convoluted tubule.
D. Papillary duct.
7. The functional unit of the kidney is called a
A. Glomerulus.
B. Nephron.
C. Corpuscle.
D. Calyx.
8. Which of the following statements concerning the kidneys is false?
A. They are retroperitoneal.
B. They each contain 8 to 15 renal pyramids.
C. They each have a distinct renal cortex and renal medulla region.
D. They are positioned between the third and fifth lumbar.
9. Which of the following is the correct sequence in which urine flows through the kidney toward the urinary bladder?
A. Renal pelvis, major calyx, minor calyx, papillary duct, ureter.
B. Papillary duct, minor calyx, major calyx, renal pelvis, ureter.
C. Minor calyx, major calyx, papillary duct, renal pelvis, ureter.
D. Papillary duct, major calyx, minor calyx, ureter, renal pelvis.
10. Which of the following statements concerning the ureters is false?
A. The mucosa of each ureter is composed of transitional epithelium.
B. They are composed of three tunics.
C. Muscular peristaltic waves move urine through the ureters.
D. They are located within the peritoneal cavity.
11. The micturition reflex center is located in the:
A. kidney.
B. cerebrum.
C. sacral segment of the spinal cord.
D. urinary bladder.
E. hypothalamus.
12. All of the following statements are True related to the urethra of a male except:
A. It serves both the urinary and reproductive systems.
B. It contains a single urethral sphincter near the neck of the urinary bladder.
C. It receives secretions from the bulbourethral glands.
D. It consists of three distinct regions.
E. It is approximately 20 cm long in an adult.
13. Which of the following is the proper sequence of structures in the nephron?
A. Glomerulus, proximal convoluted tubule, distal convoluted tubule, nephron loop.
B. Glomerulus, nephron loop, proximal convoluted tubule, distal convoluted tubule.
C. Glomerulus, proximal convoluted tubule, nephron loop, distal convoluted tubule.
D. Proximal convoluted tubule, glomerulus, nephron loop, distal convoluted tubule.
14. Fibrous connective tissue that surrounds each kidney is the:
A. Cortex.
B. Hilum.
C. Medulla.
D. Renal capsule.
E. Renal pyramids.
15. The apex of the renal pyramid is called the:
A. Major calyx.
B. Minor calyx.
C. Renal papilla.
D. Renal pelvis.
E. Ureter.
16. The tuft of capillaries in the renal corpuscle is called the
A. Podocytes.
B. Glomerulus.
C. Calyx.
D. Renal pyramid.
E. Renal sinus.
17. The location of the kidneys in relationship to the peritoneal lining of the abdominal cavity is referred to as
A. Retroperitoneal.
B. Retro abdominal.
C. Posterior.
D. Dorsal.
18. The triangular area of the urinary bladder between the two ureters posteriorly and the urethra anteriorly is the
A. External urinary sphincter.
B. Internal urinary sphincter.
C. Smooth muscle.
D. Transitional epithelium.
E. Trigone.
- ANSWER KEY
- The correct answer is D.
 - The correct answer is D.
 - The correct answer is D.
 - The correct answer is E.
 - The correct answer is A.
 - The correct answer is A.
 - The correct answer is B.

8. The correct answer is D.
9. The correct answer is B.
10. The correct answer is D.
11. The correct answer is C.
12. The correct answer is B.
13. The correct answer is D.
14. The correct answer is C.
15. The correct answer is B.
16. The correct answer is A.
17. The correct answer is D.
18. The correct answer is A.

LONG ESSAY

1. Mention the parts of excretory system. Describe the urinary bladder under the following headings: _ Position _ surfaces and borders _ Interior _ Applied aspects
2. Describe the external and internal gross anatomical features of the kidneys. Trace the path of blood flow through the kidneys

SHORT ESSAYS

3. Describe the Nephron and its histology
4. Describe the histology of urinary bladder
5. Describe the macroscopic structure of kidney

SHORT NOTES

6. The ureter
7. Histology of kidney
8. Urethra
9. Urinary bladder
10. Relations of right kidney
11. Male urethra
12. Trigone of bladder

ANSWER BRIEFLY

13. Name the parts of a nephron
14. Name the different parts of urinary system
15. Mention the coverings of kidney
16. Draw a labeled diagram of nephron
17. Mention the constrictions of ureter
18. Mention 2 important relations of urinary bladder in females
19. Name the parts of male urethra

DIFFERENTIATE BETWEEN

20. Male and female urethra
21. Ureter and urethra
22. Nephron and neuron

ANSWER KEY

UNIT VIII

RENAL SYSTEM

LONG ESSAY

1. Mention the parts of excretory system. Describe the urinary bladder under the following headings: _ Position _ surfaces and borders _ Interior _ Applied aspects

INTRODUCTION

PARTS OF THE EXCRETORY SYSTEM:

KIDNEYS:

- Within each kidney are an estimated one million microscopic nephrons.
- Filtering of the blood takes place within these areas.

- Each nephron contains a cluster of capillaries called a glomerulus.
- A cup-shaped sac called a Bowman's capsule surrounds each glomerulus.
- White blood cells, red blood cells and proteins remain in the blood.
- The cleansed blood goes into veins that carry the blood from the kidneys and back to the heart.

URETERS:

Ureters are muscular ducts that propel urine from the kidneys to the urinary bladder. In the adult, the ureters are usually 25–30 cm (10–12 in) long.

URETHRA:

A tube that carries urine from the bladder to the outside of the body.

BLADDER:

The urinary bladder is the organ that collects urine excreted by the kidneys prior to disposal by urination. It is a hollow muscular, and elastic organ, and sits on the pelvic floor.

Urine enters the bladder via the ureters and exits via the urethra

Position

External features (diagram)

Empty bladder

- Apex
- Base
- Neck

3 Surfaces

4 borders

Full bladder

- Apex
- Neck

2 Surfaces

Relations

Ligaments

Interior of bladder

Arterial supply

Venous supply

Nerve supply

Lymphatic supply

Applied anatomy

SKIN:

Gives off bodily sweat or perspiration. Skin is the largest organ of the human body. It also protects the inner organs and tissues of the body.

LUNGS:

Lungs give off carbon dioxide, a bodily waste, during respiration.

2. Describe the external and internal gross anatomical features of the kidneys. Trace the path of blood flow through the kidneys

INTRODUCTION

EXTERNAL ANATOMY OF KIDNEYS

INTERNAL ANATOMY OF KIDNEYS

PATH OF BLOOD FLOW

- ❖ Renal artery
- ❖ Segmental artery

- ❖ Interlobar artery
- ❖ Arcuate artery
- ❖ Inter-lobar artery
- ❖ Afferent arterioles
- ❖ Glomerular capillaries
- ❖ Interlobar veins
- ❖ Arcuate veins
- ❖ Interlobar veins
- ❖ Renal vein

APPLIED ANATOMY

SHORT ESSAYS

3. Describe the Nephron and its histology

INTRODUCTION

NEPHRON(diagram)

HISTOLOGY OF NEPHRON AND COLLECTING DUCT

Glomerular capsule

Renal tubule and collecting duct

BLOOD SUPPLY,NERVE SUPPLY AND LYMPHATIC SUPPLY

APPLIED ANATOMY

4. Describe the histology of urinary bladder

INTRODUCTION

URINARY BLADDER

HISTOLOGY OF BLADDER

The wall of the bladder wall has three principal tissue layers or coats:

- ❖ Mucosa.
- ❖ Submucosa.
- ❖ Muscularis

BLOOD SUPPLY

NERVE SUPPLY

APPLIED ANATOMY

5. Describe the macroscopic structure of kidney

INTRODUCTION

KIDNEY

PARTS

MACROSCOPIC STUCTURE(diagram)

VASCUALR SUPPLY,NERVOUS SUPPLY AND LYMPHATIC SUPPLY

APPLIED ANATOMY

SHORT NOTES

6. The ureter

STRUCTURE

DIAGRAM

FUNCTION

BLOOD SUPPLY

APPLIED ANA`TOMY

7.Histology of kidney

REFER

8.Urethra

MALE URETHRA

FEATURES

DIMENSIONS

PARTS

Anterior part

Posterior part

VASCULAR SUPPLY

NERVOUS SUPPLY

INNERVATIONS

FEMALE URETHRA

Features

WALLS OF URETHRA

DEVELOPMENT

APPLIED ANATOMY

9. Urinary bladder

REFER QUE 4

10. Relations of right kidney

11. Male urethra

REFER QUE 8

12. Trigone of bladder

QUESTIONS

UNIT IX

REPRODUCTIVE SYSTEM

MCQ

Qus.1 A cell's capacity to divide into two equal daughter cells is known to as

- (a) Mitosis
- (b) Multiple fission
- (c) Binary fission
- (d) None of these

Ans-(c) Binary fission

Explanation

Binary fission refers to "division into two". It is the ordinary and most usual process of asexual reproduction. In the binary fission method, the organism replicates its genetic material, or deoxyribonucleic acid (DNA), before dividing into two sections (cytokinesis), with each new element obtaining a single copy of DNA. Multiple fission is the asexual reproduction procedure, in which many daughter cells are developed from the parent cell instead of two daughter cells.

Qus2 Normally how many pairs of autosomes are there in humans?

- (a) 12
- (b) 24
- (c) 20
- (d) 22

Ans-(d) 22

Explanation

Normally 22 pairs of autosomes are there in humans. All chromosomes except sex hormones are autosomes. Chromosomes are thread-like structures present in the nucleus. It becomes visible during the procedure of cell division. Humans have 23 pairs of chromosomes i.e. the total number of chromosomes (is 46 chromosomes). Humans have 22 pairs of autosomes means a total no of (44 autosomes) and one pair of allosomes means a sex chromosome. An allosome is equal in shape, form, and behaviour. differs from autosomes. The sex Chromosome is represented by the two symbols that are X and Y., X Chromosome for Male and Y Chromosome for Female.

Qus3 Where sperm are temporarily kept?

- (a) Epididymis
- (b) Vas-deferens

(c) Bladder

(d) All of these

Ans-(a) Epididymis

Explanation

The sperm is temporarily kept in the epididymis. There are approximately 300 million sperm cells created every day, and each one takes 65 to 75 days to develop. Inside the testes, sperm are produced in the organ that is called the seminiferous tubules.

The epididymis, which stores sperm, is placed at the top and back of each testicle (testis).

Qus4 Which of the following organ in the human body does the fertilisation process occur?

(a) Uterus

(b) Fallopian tube

(c) Ovary

(d) Vagina

Ans- (b) Fallopian tube

Explanation

The procedure of fertilization in the human body involved human eggs and sperm in the fallopian tube. There is another procedure known as IVF, which stands for “In Vitro Fertilization,” which later results in the production of a zygote cell or fertilizes an egg. The process of fertilization in humans in the first stage begins with ejaculation during copulation and ovulation. The fallopian tube is also referred to as the uterine tube or salpinges.

Qus5 What is a fertilized cell called?

(a) Embryo

(b) Ovary

(c) Zygote

(d) None of the above

Ans- (c) Zygote

The fertilized cell is called the zygote. The result of fertilization is a zygote, which divides frequently and moves down the oviduct shortly after fertilization. Sexual reproduction constitutes the development of male and female gametes. A zygote is generated by the fusion of gametes. A zygote is developed inside the body of the organism.

Q.6 Which hormone is decreased in the menstrual cycle causing menstruation?

(a) Progesterone

(b) luteinizing

(c) Estrogen

(d) None of These

Ans- (a) Progesterone

Explanation

Menstrual cycle:

Menstruation is the monthly passage of blood and mucosal tissue from the inner lining of the uterus via the vagina. Hormone fluctuations that occur during the menstrual cycle are distinctive. Decreasing progesterone levels cause menstruation, which is a sign that pregnancy has not yet occurred. The cycle is repeated every 28 or 29 days.

There are four phases in the menstrual cycle:

1. Menstrual phase
2. Follicular phase

3. Ovulatory phase

4. Luteal phase

Qus7 Which of the following is not a component of the female reproductive system?

(a) A pair of ovaries

(b) Sperm duct

(c) Uterus

(d) Oviducts

Ans- (b) Sperm duct

Explanation

The sperm duct is not a component of the female reproductive system. The male reproductive organs make up of a pair of testes, sperm ducts, and a penis. The sperms are generated by the testes. The sperms are incredibly tiny in size with a head, a middle piece, and a tail. The female reproductive organs make up of a pair of ovaries, oviducts, and the uterus. The ovaries are responsible for producing eggs (ova).

Qus.8 Name the hormone which is at its peak during ovulation.

(a) Progesterone

(b) Estrogen

(c) FSH

(d) LH

Ans-(d) LH

A mature egg is released from one of the ovaries during ovulation. The egg is released and then travels into the fallopian tube, where it may be fertilized by a sperm cell. luteinizing hormone (LH). LH is quickly released when estrogen levels are high enough to cause it. It normally occurs on or around cycle day thirteen. LH at the peak during ovulation. Luteinizing hormone is generated and secretes in the anterior pituitary gland.

Que. 9 Which is not a component of the female reproductive system?

(a) Eggs produce

(b) Uterus

(c) White Discharge

(d) Vas deferens

Ans-(d) Vas deferens

Explanation

Vas deferens are not a component of the female reproductive system. Vas deferens are also referred to as ductus deferens or sperm ducts. It is a long muscular tube, which extends from the epididymis into the pelvic cavity posterior to the bladder and connects with the urethra via a composition referred to as the ejaculatory duct. It helps in the transportation of produced sperm to the urethra preparation ready for ejaculation.

Que.10 The external genitalia of the female are collectively called the

(a) labia

(b) vulva

(c) clitoris

(d) mons pubis

Ans-(b) Vulva

The vulva is the upper part of the female genitalia, involving the labia majora, labia minora, and clitoris.

The vulva is the collective term for external genitalia of the female reproductive system and involved the pudendal cleft, mons pubis, Bartholin glands, clitoris, labia majora, and labia minora, Vagina.

Que.11 Which of the following performs the function of luteinizing hormone?

(a) It stimulates the interstitial (Leydig) cells to produce testosterone.

(b) It encourages the production of sperm by sustentacular (Sertoli) cells.

(c) It stimulates the anterior pituitary to release FSH.

(d) None of these

Ans-(a) It stimulates the interstitial (Leydig) cells to produce testosterone

Explanation

LH stimulates testosterone production from the interstitial cells of the testes. Luteinizing hormones are known as gonadotropic hormones. It is generated and secreted by the group of cells of the anterior pituitary gland. It is the most specific hormone that regulates the reproductive system by controlling the function of female reproductive organs like ovaries and male reproductive organs like testes.

Que.12 Which organ is responsible for transporting sperm from the testicles to the prostate gland?

(a) Vas deferens

(b) Fallopian tube

(c) seminiferous tubule

(d) All of the Above

Ans-(a) Vas deferens

Explanation

The Vas deferens organ is responsible for transporting sperm from the testicles to the prostate gland. It collects the sperm and moves it out of the scrotal sac. The vas deferens are present between the epididymis and the urethra and attach these together.

Que. 13 Which cells of the testes secrete testosterone?

(a) Sertoli cells

(b) Cells of germinal epithelium

(c) Leydig Cells or Interstitial cells

(d) Secondary spermatocytes

Ans-(c) Leydig Cells or Interstitial cells

Explanation

Leydig Cells or Interstitial cells of the testes secrete testosterone. Leydig cells are interstitial cells placed close to the seminiferous tubules in the testes. Leydig cells are to produce the other hormone androgen, testosterone, under the pulsatile control of pituitary luteinizing hormone (LH). Leydig cells are the main source of testosterone or androgens in males.

Que.14 Which of the following is an assisted reproductive gland in male mammals

(a) Fundic gland

(b) Prostate gland

(c) Mushroom-shaped gland

(d) Gastric gland

Ans-(b) Fundic gland

Explanation

The prostate gland is the only exocrine gland that is situated between the bladder and penis in male mammals. The prostate gland developed a large amount of fluid, which makes up the semen when combined with sperm. Now those the gastric glands or fundic glands are present in the body, the gastric of the mucosa and fundus of the stomach (in all mammals), the mushroom-shaped gland is not found in males and the inguinal canal is the passage in the abdominal wall present in men and women body.

Qus15 What is another name for the bulbourethral gland?

(a) Meibomian gland

(b) Prostate gland

(c) Perineal gland

(d) Cowper's gland

Ans- (d) Cowper's gland

Explanation

A bulbourethral gland is another name Cowper gland is a pair of exocrine glands found in the male reproductive system. They are roughly the size of a pea and situated in inferior to the prostate gland and lateral to the urethra in the urogenital diaphragm. The secretions generated by the Cowper's glands help to cover sperm as it passes through the urethra throughout ejaculation.

Qus16 Which glands are responsible for the secretion to neutralize the acidity of the fluid (semen)?

(a) Cowper's glands

(b) Seminal glands

(c) Spermatogonia glands

(d) Prostate glands

Ans- (d) Prostate glands

Explanation

The prostate gland is responsible for secretion to neutralize the acidity of fluid (semen). The prostate gland produced a high amount of fluid, which creates semen when connected with sperm. A white fluid is produced by the prostate gland. The fluid consists of phosphate, lysozymes, spermin, citrate, fibrinolysin, and other chemicals. Its secretions both encourage sperm production and stop sperm coagulation

Qus17 Who are Primary spermatocytes produced?

(a) spermatogonia

(b) oogonia

(c) primary spermatids

(d) secondary spermatids

Ans-(a) spermatogonia

Explanation

Primary spermatocytes are produced by spermatogonia. The spermatozoa (2N) located in the inner part of the seminiferous tubules are deposited by mitosis and increase in number. They stop undergoing division by mitosis and develop or become primary spermatocytes (2N). Each primary spermatocyte undergoes the process of meiosis to form two identical haploid secondary spermatocytes (N). Each secondary spermatocyte then undergoes second meiosis

to form four homologous. The haploid cells are known as spermatozoa (N). Spermatids obtain nutrition to make sperm through a process called spermatogenesis.

Qus 18 The sac of skin in which the testes are located is called

- (a) seminiferous tubules
- (b) vas deferens
- (c) epididymis
- (d) scrotum

Ans- (d) scrotum

Explanation

The sac of skin in which the testes are located is called Scrotum. The scrotum, a thin outer sac of skin that is divided into two sections, is present in the male reproductive system; Each segment contains one of two testes, the glands that produce sperm, and one of the epididymides, where sperm are stored.

Qus.19 Which of the following cells synthesize and secrete testicular hormones in humans?

- (a) Sertoli cells
- (b) Mucus cells
- (c) Spermatogonia
- (d) Leydig cells

Ans-(d) Leydig cells

Explanation

Leydig cells synthesize and secrete testicular hormones (androgens) in humans (man). The male reproductive system has a pair of testes with accessory ducts, glands, and upper genitals. Each testis contains about 250 compartments known as testicular lobules. Each lobule contains one to three highly coiled semicircular tubules in which spermatozoa are produced. Each seminiferous tubule is lined in its interior by two types of cells known as male germ cells (spermatogonia) and Sertoli cells.

Q.20 Which is the primary sex organ in a man?

- (a) Prostrate
- (b) Testis
- (c) Vas deferens
- (d) Seminal vesicles

Ans-(b) Testis

Explanation

The testes are the primary sex organs or reproductive organs in males. The testes are controls for testosterone and sperm production. Each testis is 4–5 cm long, 2–3 cm wide and weighs about 10–14 g. Males have two testicles of the same size. They are attached to the scrotum which is an extension of the abdominal wall. One testicle is more stranded in the scrotum than the other.

LONG ESSAY

1. Describe the location, structure and histology of the uterus and describe the structure of ovaries
2. Mention the different parts of female reproductive system. Describe the uterus under the following headings.
 - Position and parts
 - Relations and supports
 - Blood supply
 - Applied anatomy

SHORT ESSAY

3. Describe the structure of mammary gland

4. Explain the location and structure of ovary
5. Name the parts of the female reproductive tract. Describe the anatomy of uterus
6. Name the parts of the female reproductive tract. Describe the anatomy of ovary
7. Describe the uterus and its ligaments
8. Describe the female reproductive system
9. Describe the structure of Testes and add a note on sperm

SHORT NOTES

10. Structure of the breast
11. Supports of uterus
12. Fallopian tube
13. Prostate

ANSWER BRIEFLY

14. Name the internal organs of female reproductive system
15. Mention the parts of uterus
16. Name 4 ligaments of the uterus
17. Name the layers of wall of uterus
18. Name parts of fallopian tube
19. Mention the boundaries of ovarian fossa
20. Mention the layers of scrotum
21. Name the parts of male urethra.
22. Draw a diagram on the structure of uterus

DIFFERENTIATE BETWEEN.

23. Sperm and ovum
24. Spermatic cord and epididymis
25. Internal and external genitalia
26. Graffian follicle and Ductus deferens

ANSWER KEY

UNIT IX

REPRODUCTIVE SYSTEM

LONG ESSAY

1. Describe the location, structure and histology of the uterus and describe the structure of ovaries

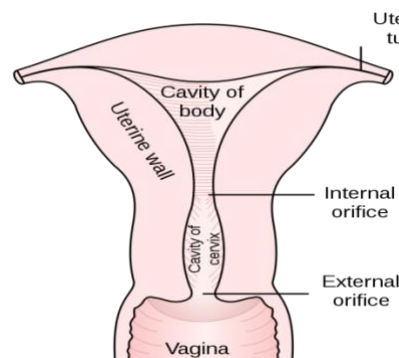
INTRODUCTION

ANATOMY OF UTERUS

DIMENSIONS

BOUNDARIES

ANATOMICAL SUBDIVISIONS OF UTERUS(diagram)



- Fundus
- Body
- Cervix

HISTOLOGY OF UTERUS

- Perimetrium
- Myometrium

- Endometrium

BLOOD SUPPLY

APPLIED ANATOMY

2. Mention the different parts of female reproductive system.
Describe the uterus under the following headings.

- Position and parts □ Relations and supports □ Blood supply
- Applied anatomy

INTRODUCTION

EXTERNAL GENTALIA

- ❖ Mons pubis
- ❖ Labia majora
- ❖ Labia minora
- ❖ Clitoris
- ❖ Vestibule

INTERNAL GENTALIA

- ❖ Ovary
- ❖ Fallopian tube
- ❖ Uterus
- ❖ Vagina

REFER QUE 1

SHORT ESSAY

10.

3. Describe the structure of mammary gland

MAMMARY GLAND

- ❖ Pectoralis muscle
- ❖ Lobules
- ❖ Aerola
- ❖ Laciferous sinus and duct
- ❖ Fatty tissue
- ❖ Nipple

BLOOD SUPPLY

NERVE SUPPLY

APPLIED ANATOMY

4. Explain the location and structure of ovary

OVARY

STRUCTURE

DIMENSIONS

BOUNDARIES

HISTOLOGY

- Germinal epithelium
- Tunica albuginea
- Ovarian cortex
- Ovarian medulla
- Ovarian follicles
- Corpus luteum

MATURE OVUM

LIGAMENTS

VASCULAR SUPPLY, NERVE SUPPLY AND

LYMPHATIC SUPPLY

CLINICAL ANATOMY

5. Name the parts of the female reproductive tract.

Describe the anatomy of uterus

REFER QUE 1

6. Name the parts of the female reproductive tract.

Describe the anatomy of ovary

REFER QUE 1 AND 4

7. Describe the uterus and its ligaments

8. Describe the female reproductive system

9. Describe the structure of Testes and add a note on sperm

INTRODUCTION

STRUCTURE OF TESTIS (diagram)

- ✚ Tunica vaginalis
- ✚ Tunica albuginea
- ✚ Epididymis
- ✚ Semiferous tubules

SPERM

- HEAD
- TAIL
- Neck
- Principal piece
- Middle piece
- End piece

VASCULAR SUPPLY

NERVE SUPPLY

APPLIED ANATOMY

SHORT NOTES

10. Structure of the breast

INTRODUCTION

BREAST

STRUCTURE

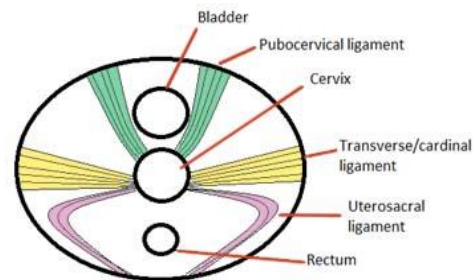
(diagram)

The main parts of the female breast are lobules (milk producing glands), ducts (milk passages that connect the lobules and the nipple), and stroma (fatty tissue and ligaments surrounding the ducts and lobules, blood vessels, and lymphatic vessels).

11. Supports of uterus

UTERUS

MAJOR LIGAMENTS AND SUPPORTING AREAS



INNERVATION

BLOOD SUPPLY

APPLIED ANATOMY

12. Fallopian tube

FEATURES

PARTS

- Interstitial segment – extends from the uterine cavity through the uterine muscle
- Isthmic segment – narrow muscular portion adjacent to the uterus
- Ampullary segment – wider and longer middle part of the tube
- Infundibular segment – funnel shaped segment next to the fimbrial end

- Fimbrial segment – wide opening at the end of the tube facing the ovary

BLOOD SUPPLY

APPLIED ANATOMY

13. Prostate

PROSTATE

FEATURES

AREAS

BOUNDARIES

DIMENSIONS

BLOOD SUPPLY

NERVE SUPPLY

LYMPHATIC SUPPLY

APPLIED ANATOMY

QUESTIONS

UNIT X

NERVOUS SYSTEM

MCQ

1. Which of the following is one of the parts of the hindbrain?

- (a) hypothalamus
- (b) cerebellum
- (c) corpus callosum
- (d) spinal cord

Answer: (b)

2. If there is an injury in the hypothalamus region of the brain, it is most likely to affect

- (a) regulation of body temperature
- (b) decision making
- (c) co-ordination during locomotion
- (d) short-term memory

Answer: (a)

3. A cranial nerve with the highest number of branches

- (a) Facial nerve
- (b) Trigeminal nerve
- (c) Vagus nerve
- (d) Both (b) and (c)

Answer: (b)

4. Afferent neurons carry nerve impulses from

- (a) CNS to muscles
- (b) CNS to receptors
- (c) receptors to CNS
- (d) effector organs to CNS

Answer: (c)

5. The cranial nerve that regulates the heartbeat

- (a) VII
- (b) VIII
- (c) IX
- (d) X

Answer: (d)

6. Which of the following is another name for the third ventricle of the brain?

- (a) diocoel
- (b) paracoel
- (c) rhinocoel
- (d) metacoel

Answer: (a)

7. Nissl's granules present in the neurons are made up of

- (a) protein
- (b) ribosome

(c) RNA

(d) DNA

Answer: (b)

8. Which of the following is a part of the rhombencephalon?

- (a) Medulla
- (b) Substantia nigra
- (c) Thalamus
- (d) Temporal lobe

Answer: (a)

9. A neuron that carries information from the peripheral nervous system to the central nervous system is

- (a) afferent neuron
- (b) efferent neuron
- (c) both
- (d) none

Answer: (a)

10. Which of the following parts of the brain controls the body temperature and urge of eating?

- (a) thalamus
- (b) cerebellum
- (c) pons
- (d) hypothalamus

Answer: (d)

11. System of the body which coordinates and controls its activity is known as _____

- a) Organ system
- b) Muscular system
- c) Nervous tissue
- d) Nervous system

12. Peripheral nervous system consists of the brain and spinal cord.

- a) True
- b) False

13. Which of the following is NOT the component of the PNS?

- a) Elastic connective tissue
- b) Cranial nerves
- c) Spinal nerves
- d) Ganglia

14. Name the basic structural and functional unit of the nervous system.

- a) Neuroglia
- b) Glial cells
- c) Neurons
- d) Perikaryon

15. Which of the following cells supports, nourishes, and protect the neurons?

- a) Nissl bodies
- b) Perikaryon
- c) Ganglia
- d) Glial cells

16. What is the speed of nerve impulse?

- a) 60 m/s
- b) 0.5 – 130 m/s
- c) 10-30 m/s
- d) 50 -150 m/s

17. What are Nissl bodies?

- a) Golgi bodies
- b) Lysosomes

- c) Cluster of rough endoplasmic reticulum
d) Mitochondria
18. Which of the following is NOT found in axon?
a) Endoplasmic reticulum
b) Neurofibrils
c) Mitochondria
d) Microtubules
19. Name the multipolar neuron which is located entirely within the central nervous system.
a) Motor neuron
b) Efferent neuron
c) Afferent neuron
d) Interneuron
20. Out of the following, which one does not affect the speed of conduction of nerve impulse.
a) No. of ganglia
b) Myelin sheath
c) Axon diameter
d) Temperature
21. The term central nervous system refers to the:
A) autonomic and peripheral nervous systems
B) brain, spinal cord, and cranial nerves
C) brain and cranial nerves
D) spinal cord and spinal nerves
E) brain and spinal cord
22. The peripheral nervous system consists of:
A) spinal nerves only
B) the brain only
C) cranial nerves only
D) the brain and spinal cord
E) the spinal and cranial nerves
23. Which of these cells are not a type of neuroglia found in the CNS:
A) astrocytes
B) microglia
C) Schwann cells
D) ependymal cells
E) oligodendrocytes
24. The Schwann cells form a myelin sheath around the:
A) dendrites
B) cell body
C) nucleus
D) axon
E) nodes of Ranvier
25. The neuron processes that normally receives incoming stimuli are called:
A) axons
B) dendrites
C) neurolemmas
D) Schwann cells
E) satellite cells
26. Collections of nerve cell bodies inside the PNS are called:
A) ganglia
B) tracts
C) nerves
D) nuclei
E) tracts or ganglia
27. Which of the following best describes the waxy-appearing material called myelin:
A) an outer membrane on a neuroglial cell
B) a lipid-protein (lipoprotein) cell membrane on the outside of axons
C) a mass of white lipid material that surrounds the cell body of a neuron
D) a mass of white lipid material that insulates the axon of a neuron
E) a mass of white lipid material that surrounds the dendrites of a neuron
28. A neuron with a cell body located in the CNS whose primary function is connecting other neurons is called a(n):
A) efferent neuron
B) afferent neuron
C) interneuron
D) glial cell
E) satellite cell
29. White matter refers to myelinated fibers in the:
A) CNS
B) PNS
C) ANS
D) SNS
E) both ANS and SNS
30. Impulse conduction is fastest in neurons that are:
A) myelinated
B) unmyelinated
C) sensory
D) motor
E) cerebral
31. An action potential:
A) is essential for nerve impulse propagation
B) involves the influx of negative ions to depolarize the membrane
C) involves the outflux of negative ions to depolarize the membrane
D) involves the outflux of positive ions to depolarize the membrane
E) is initiated by potassium ion movements
32. Immediately after an action potential is propagated, which one of the following ions rapidly diffuses out of the cell into the tissue fluid:
A) sodium
B) chloride
C) calcium
D) potassium
E) magnesium
33. The ability to respond to a stimulus is termed:

- A) polarized
 - B) irritability
 - C) depolarized
 - D) conductivity
 - E) all-or-none response
34. Which one of the following is the correct sequence of events that follows a threshold potential:
1. the membrane becomes depolarized
 2. sodium channels open and sodium ions diffuse inward
 3. the membrane becomes repolarized
 4. potassium channels open and potassium ions diffuse outward while sodium is actively transported out of the cell
- A) 3,2,4,1 B) 2,1,4,3 C) 2,1,3,4 D) 1,2,4,3 E) 4,1,3,2
35. Which one of the following describes saltatory conduction:
- A) occurs only if the myelin sheath is continuous
 - B) occurs only if the nodes of Ranvier are lacking
 - C) occurs only in the absence of axon hillocks
 - D) is faster than conduction on an unmyelinated fiber
 - E) is slower than conduction on an unmyelinated fiber
36. The substance that is released at axonal endings to propagate a nervous impulse is called:
- A) an ion
 - B) nerve glue
 - C) a neurotransmitter
 - D) the sodium-potassium pump
 - E) an action potential
37. Which of the following is the correct sequence in a typical reflex arc:
- A) effector, afferent neuron, integration centre, efferent neuron, receptor
 - B) receptor, afferent neuron, integration centre, efferent neuron, effector
 - C) effector, efferent neuron, integration centre, afferent neuron, receptor
 - D) receptor, efferent neuron, integration centre, afferent neuron, effector
 - E) receptor, afferent neuron, efferent neuron, integration centre, effector
38. The elevated ridges of tissue on the surface of the cerebral hemispheres are known as _____ while the shallow grooves are termed _____.
- A) sulci; gyri
 - B) gyri; sulci
 - C) ganglia; gyri
 - D) tracts; ganglia
 - E) receptors; effectors
39. The olfactory area is found within the:
- A) occipital lobe
 - B) temporal lobe
 - C) frontal lobe
 - D) parietal lobe
 - E) pyramidal tract
40. Sally has a brain injury; she knows what she wants to say but can't vocalize the words. The part of her brain that deals with the ability to speak is the:
- A) longitudinal fissure
 - B) gyrus
 - C) central sulcus
 - D) Broca's area
 - E) primary motor control
41. The midbrain, pons and medulla oblongata are housed in the:
- A) diencephalon
 - B) hypothalamus
 - C) brain stem
 - D) pineal gland
 - E) cerebellum
42. Lobe that contains the primary motor area that enables voluntary control of skeletal muscle movements:
- A) parietal lobe
 - B) temporal lobe
 - C) occipital lobe
 - D) frontal lobe
 - E) diencephalon
43. Afferent nerves are called _____, and motor nerves are called _____.
- A) motor nerves; sensory nerves
 - B) peripheral nerves; cranial nerves
 - C) mixed nerves; motor nerves
 - D) sensory nerves; efferent nerves
 - E) cranial nerves; peripheral nerves
44. The area of the brain stem that plays a role in consciousness and the awake/sleep cycle is the:
- A) thalamus
 - B) reticular activating system (RAS)
 - C) pineal gland
 - D) limbic system
 - E) cerebellum
45. Control of temperature, endocrine activity, metabolism, and thirst are functions associated with the:
- A) medulla oblongata
 - B) cerebellum
 - C) hypothalamus
 - D) thalamus
 - E) cerebrum
46. The vital centers for the control of visceral activities such as heart rate, breathing, blood pressure, swallowing, and vomiting are located in the:
- A) pons
 - B) medulla oblongata
 - C) midbrain
 - D) cerebrum
 - E) hypothalamus

47. Loss of muscle coordination results from damage to the:

- A) cerebrum
- B) hypothalamus
- C) cerebellum
- D) thalamus
- E) midbrain

48. Which of the following is a traumatic brain injury:

- A) cerebrovascular accident (CVA)
- B) Alzheimer's disease
- C) aphasia
- D) cerebral edema
- E) Parkinson's disease

49. Which one of the following is the correct sequence of nerves that exit the spinal cord, going from superior to inferior:

- A) thoracic spinal nerves, cervical spinal nerves, lumbar spinal nerves, sacral spinal nerves
- B) cervical spinal nerves, lumbar spinal nerves, thoracic spinal nerves, sacral spinal nerves
- C) thoracic spinal nerves, cervical spinal nerves, sacral spinal nerves, lumbar spinal nerves
- D) cervical spinal nerves, thoracic spinal nerves, sacral spinal nerves, lumbar spinal nerves
- E) cervical spinal nerves, thoracic spinal nerves, lumbar spinal nerves, sacral spinal nerves

50. The nerve that contains sensory fibers that are involved in hearing is:

- A) cranial nerve II
- B) cranial nerve III
- C) cranial nerve V
- D) cranial nerve VIII
- E) cranial nerve IX

51. Preparing the body for "fight-or-flight" response during threatening situations is the role of the:

- A) sympathetic nervous system
- B) cerebrum
- C) parasympathetic nervous system
- D) somatic nervous system
- E) afferent nervous

Solutions 1. E 2. E 3. C 4. D 5. B 6. A 7. D 8. C 9. A 10. A 11. A 12. D 13. B 14. B 15. D 16. C 17. B 18. B 19. B 20. D 21. C 22. D 23. D 24. C 25. C 26. B 27. C 28. D 29. E 30. D 31. A

LONG ESSAY

1. Mention the different parts of cerebral hemisphere and describe the supero-lateral surface of right cerebral hemisphere under the following headings:

_ sulci _ gyri _ functional areas

SHORT ESSAY

1. Name the parts of central nervous system and describe the cerebral hemisphere
2. Name the various lobes of cerebrum. Describe various functional areas of cerebrum
3. Explain the structure of cerebellum in detail
4. Describe brachial plexus of nerves
5. Spinal cord
6. Draw a neat labeled diagram of circle of Willis
7. Trochlear nerve
8. Explain cerebellum with its anatomical significance.
9. Classify nervous system and write a note on neurons
10. Ulnar nerve and its applied anatomy
11. Radial nerve and its applied anatomy

SHORT ANSWERS

12. Structure of a neuron
13. Spinal ganglion
14. What is wry neck
15. Erb's palsy
16. Carpal tunnel syndrome
17. Neuralgia
18. Bell's palsy
19. Lumbar puncture

20. Draw a labeled diagram of neurons

ANSWER BRIEFLY

21. Name the parts of the brain stem
22. Name ventricles of brain
23. Mention the function of hypoglossal nerve
24. Explain the deformity in klumpke's paralysis
25. Special features of Medulla oblongata
26. Name major parts of central nervous system
27. Mention the functions of CSF
28. Name 4 cranial nerves
29. Mention the two functions of cerebellum
30. Mention 2 parts of sciatic nerve
31. Name the branches of the trigeminal nerve
32. Mention the parts of the brain
33. List the cranial nerves.
34. Name the parts of central nervous system
35. Name the meningeal coverings of brain in order.

DIFFERENTIATE BETWEEN

36. Sympathetic and parasympathetic nervous system
37. Autonomic and somatic nervous system.
38. Cranial nerves and spinal nerves
39. Synapse and stimulus.
40. Microglia and neuroglia
41. Sulci and gyri

ANSWER KEY

UNIT X

NERVOUS SYSTEM

LONG ESSAY

1. Mention the different parts of cerebral hemisphere and describe the supero-lateral surface of right cerebral hemisphere under the following headings:

_ sulci _ gyri _ functional areas

INTRODUCTION

CEREBRAL hemispheres

EXTERNAL FEATURES

- 3 Surfaces
- 4 Borders
- 3 Poles

LOBES- 4 LOBES

- ✓ FRONTAL
- ✓ PARIETAL
- ✓ OCCIPITAL
- ✓ TEMPORAL

CEREBRAL SULCI AND GYRI

FUNCTIONAL OR CORTICAL AREAS OF CERBRAL CORTEX.

- Motor areas
- Sensory areas
- Association areas.

FUNCTIONS OF CEREBRAL CORTEX.

- ❖ Cerebral dominance
- ❖ Discriminatory aspects
- ❖ Associative functions
- ❖ Controlling of Voluntary movements

BLOOD SUPPLY

CLINICAL SIGNIFICANCE

SHORT ESSAY

2. Name the parts of central nervous system and describe the cerebral hemisphere

INTRODUCTION

CENTRAL NERVOUS SYSTEM

PARTS

- Brain
 - ❖ Fore brain
 - Telencephalon(cerebrum)
 - Diencephalon
 - Thalamus
 - Hypothalamus
 - Metathalamus
 - Epithalamus
 - Subthalamus
 - ❖ Mid brain
 - Crus cerbri
 - Substantia nigra
 - Tegmentum
 - Tectum
 - ❖ Hind brain
 - Metencephalon(cerebellum and pons)
 - Myelencephelon or medulla oblongata.
 - Spinal cord
 - ❖ External features
 - ❖ Internal structure
 - ❖ SPINAL NERVES
 - PAIRS OF
 - 8 CERVICAL
 - 12 THORACIC
 - 5 LUMBAR
 - 5 SACRAL
 - 1 COCCYGEAL
 - ❖ TRACTS OF SPINAL CORD
 - Descending tract

- Ascending tract

❖ CLINICAL SIGNIFICANCE

2. Name the various lobes of cerebrum. Describe various functional areas of cerebrum
REFER QUES 1

3. Explain the structure of cerebellum in detail
INTRODUCTION

EXTERNAL FEATURES with diagram

PARTS--seperates by FISSURES

- ✓ Horizontal fissure
- ✓ Primary fissure
- ✓ Posterolateral fissure
- ANTERIOR LOBE
- MIDDLE LOBE
- FLOCCULONODULAR LOBE.
Functional division:-
 - LATERAL
 - INTERMEDIATE
 - VERMIS
 - FLOCCULONODULAR LOBE

CONNECTIONS

FUNCTIONS OF CEREBELLUM

DEVELOPMENT

ARTERIAL, VENOUS SUPPLY

CLINICAL SIGNIFICANCE

4. Describe brachial plexus of nerves
INTRODUCTION

Formed by ventral rami of C5,C6,C7,C8 and T1 nerves.

BOUNDARIES

BRANCHES FROM RAMI

BRANCHES OF UPPER TRUNK

BRANCHES OF LATERAL CORD

CLINICAL ANATOMY

5. Spinal cord
REFER QUE 2
6. Draw a neat labeled diagram of circle of Willis
INTRODUCTION
DIAGRAM WITH LABELS

7. Trochlear nerve
INTRODUCTION
FOURTH CRANIAL NERVE
FUNCTIONAL COMPONENTS
COURSE AND DISTRIBUTION

- Intraneural course
- Cavernous sinus
- Orbit
- Lavetor palpebrae superioris
- Superior oblique muscle
- CLINICAL SIGNIFICANCE

8. Explain cerebellum with its anatomical significance.
REFER QUES NO-3

9. Classify nervous system and write a note on neurons
INTRODUCTION

DIVISIONS OF NERVOUS SYSTEM

Anatomical

- ❖ CENTRAL NERVOUS SYSTEM
 - BRAIN

- SPINAL CORD
- ❖ PERIPHERAL NERVOUS SYSTEM
- 12 PAIRS OF CRANIAL NERVES
- 31 PAIRS OF SPINAL NERVES
- ❖ Autonomic nervous system
 - Sympathetic nervous system
 - Parasympathetic nervous system
- ❖ Somatic nervous system

NEURON

- ❖ CLASSIFICATION
- Multipolar neurons
- Bipolar neurons
- Pseudounipolar neurons
- Unipolar neurons
- SENSORY NEURONS
- Primary sensory neurons
- Secondary sensory neurons
- Tertiary sensory neurons

MOTOR NEURONS

- Somatic motor neuron
- Autonomic neuron
- STRUCTURE OF NEURON WITH DIAGRAM
- CHARACTERISTICS
- CLINICAL SIGNIFICANCE

10. Ulnar nerve and its applied anatomy

INTRODUCTION

In human anatomy, the **ulnar nerve** is a nerve that runs near the ulna bone. The ulnar collateral ligament of elbow joint is in relation with the ulnar nerve.

The nerve is the largest unprotected nerve in the human body (meaning unprotected by muscle or bone), so injury is common.

This nerve is directly connected to the little finger, and the adjacent half of the ring finger, innervating the palmar aspect of these fingers, including both front and back of the tips, perhaps as far back as the fingernail beds.

STRUCTURE

- ARM
- FOREARM
- HAND
- FUNCTION
- SENSORY
- MOTOR
- CLINICAL SIGNIFICANCE

11. Radial nerve and its applied anatomy

INTRODUCTION

The **radial nerve** is a nerve in the human body that supplies the posterior portion of the upper limb.

It innervates the medial and lateral heads of the triceps brachii muscle of the arm, as well as all 12 muscles in the posterior osteofascial compartment of the forearm and the associated joints and overlying skin.

It originates from the brachial plexus, carrying fibers from the ventral roots of spinal nerves C5, C6, C7, C8 & T1.

The radial nerve and its branches provide motor innervation to the dorsal arm muscles (the triceps brachii and the anconeus) and the extrinsic extensors of the wrists and hands; it also

provides cutaneous sensory innervation to most of the back of the hand, except for the back of the little finger and adjacent half of the ring finger (which are innervated by the ulnar nerve).

The radial nerve divides into a deep branch, which becomes the posterior interosseous nerve, and a superficial branch, which goes on to innervate the dorsum (back) of the hand.



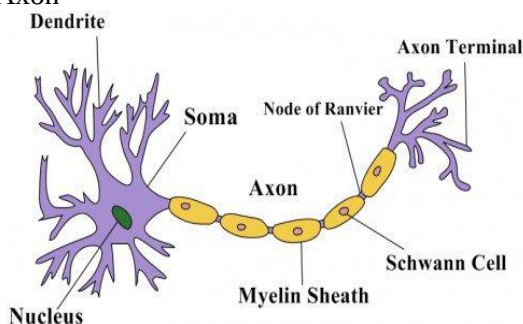
- STRUCTURE
 - ARM
 - FOREARM
 - VARIATION
- FUNCTION
 - CUTANEOUS
 - MOTOR
- CLINICAL SIGNIFICANCE
 - INJURY

SHORT ANSWERS

12. Structure of a neuron

3 PARTS

- Cell body
- Dendrites
- Axon



13. Spinal ganglion

INTRODUCTION

A **dorsal root ganglion** (or **spinal ganglion**) (also known as a **posterior root ganglion**), is a cluster of neurons (a ganglion) in a dorsal root of a spinal nerve.

The cell bodies of sensory neurons known as **first-order neurons** are located in the dorsal root ganglia.



- STRUCTURE
 - DISTAL SECTION
 - LOCATION
 - DEVELOPMENT
- FUNCTION
 - NOCICEPTION
 - MECHANOSENSITIVE CHANNELS
 - HIGH-THRESHOLD MECHANOSENSITIVE CHANNELS
 - PRESYNAPTIC CONTROL

14. Wry neck

INTRODUCTION

DEFINITION

Wry neck, or torticollis, is a painfully twisted and tilted neck. The top of the head generally tilts to one side while the chin tilts to the other side

CAUSES

- CONGENITAL (PRESENT AT BIRTH) OR ACQUIRED.
- DAMAGE TO THE NECK MUSCLES OR BLOOD SUPPLY

TYPES

- ❖ Temporary torticollis
- ❖ Fixed torticollis
- ❖ Muscular torticollis
- ❖ Klippel-Feil syndrome
- ❖ Cervical dystonia

SYMPTOMS

DIAGNOSIS

TREATMENT

15. Erb's palsy

INTRODUCTION

DEFINITION

Erb's palsy or **Erb–Duchenne palsy** is a paralysis of the arm caused by injury to the upper group of the arm's main nerves, specifically the severing of the upper trunk C5–C6 nerves.

CAUSES

- ❖ Dystocia, an abnormal or difficult childbirth or labor.
- ❖ Erb's palsy can also affect neonates affected by a clavicle fracture unrelated to dystocia.
- ❖ Trauma to the head and shoulder
- ❖ Injury may also occur as the result of direct violence, including gunshot wounds and traction on the arm, or attempting to diminish shoulder joint dislocation.

SIGNS AND SYMPTOMS

DIAGNOSIS

TREATMENT

16. Carpal tunnel syndrome

INTRODUCTION

DEFINITION

Carpal tunnel syndrome is a common condition that causes pain, numbness, and tingling in the hand and arm. The condition occurs when one of the major nerves to the hand — the median nerve — is squeezed or compressed as it travels through the wrist.

CAUSES

CLINICAL FEATURES

- Numbness, tingling, burning, and pain
- Occasional shock-like sensations that radiate to the thumb and index, middle, and ring fingers
- Pain or tingling that may travel up the forearm toward the shoulder
- Weakness and clumsiness in the hand
- Dropping things

DIAGNOSIS

MANAGEMENT

17. Neuralgia

INTRODUCTION

DEFINITION

Neuralgia is a stabbing, burning, and often severe pain due to an irritated or damaged nerve.

CAUSES

DIAGNOSIS

MANAGEMENT

18. Bell's palsy

INTRODUCTION

DEFINITION

Bell's palsy is a type of facial **paralysis** that results in an inability to control the facial muscles on the affected side.

CAUSES

DIAGNOSIS

MANAGEMENT

19. Lumbar puncture

INTRODUCTION

DEFINITION

Lumbar puncture is a procedure that is often performed in the emergency department to obtain information about the cerebrospinal fluid (CSF).

A **lumbar puncture** (**spinal** tap) is performed in your lower back, in the **lumbar** region. During a **lumbar puncture**, a needle is inserted between two **lumbar** bones (vertebrae) to remove a sample of cerebrospinal fluid. This is the fluid that surrounds your brain and **spinal** cord to protect them from injury.

INDICATION

CONTRAINDICATION

PURPOSES

PRE-PROCEDURE CARE

INTRA-PROCEDURE CARE

POST-PROCEDURE CARE

COMPLICATIONS

20. Draw a labeled diagram of neurons

REFER QUES-12